

Project Data Page:





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DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\A150NRL14CM001\\Let down vessel7%.pvdb

Date of Analysis : Sep 18, 2026 1:55pm

PV Elite 28 U2, April 2026



Vessel Design Summary:

ASME Code, Section VIII Division 1, 2025

Diameter Spec : 2100.000 mm ID	
Vessel Design Length, Tangent to Tangent	9800.00 mm
Specified Datum Line Distance	0.00 mm
Internal Design Temperature	85 °C
Internal Design Pressure	2.500 MPa
External Design Temperature	85 °C
Maximum Allowable Working Pressure	2.66 MPa
Shop Test Pressure	3.457 MPa
Field Test Pressure	3.457 MPa
Required Minimum Design Metal Temperature	-40.0 °C
Warmest Computed Minimum Design Metal Temperature	-29.0 °C

Warning: Computed overall MDMT was higher than the required value !

Wind Design Code	ASCE/SEI 7-22
Earthquake Design Code	ASCE/SEI 7-22

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-516 70	...	...	K02700	+N+IT
Head	SA-516 70	...	...	K02700	+N+IT
Nozzle	SA-350 LF1	1	...	K03009	+IT
Nozzle Flg	SA-350 LF1	1	...	K03009	+IT
Nozzle Flg	SA-105	...	...	K03504	
Hrz Bolting	SA-36	...	...	K02600	
Saddles	SA-285 C	...	...	K02801	

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized
 +Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (MPa & mm):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
Head-1	2.500	0.00	3.480	1.5000	No	No
Shell-1	2.500	0.00	2.660	1.5000	N/A	No
Shell-2	2.500	0.00	3.170	1.5000	N/A	No
Shell-3	2.500	0.00	2.660	1.5000	N/A	No
Shell-4	2.500	0.00	2.660	1.5000	N/A	No
Head-2	2.500	0.00	2.690	1.5000	No	No

Element Types and Properties:

Element Type	"To" Elev mm	Element Length mm	Nominal Thickness mm	Finished Thickness mm	Reqd Thk Internal mm	Reqd Thk External mm	Long Eff	Circ Eff
Ellipse	50.0	50.0	34.0	28.0	20.5	...	1.00	1.00
Cylinder	3050.0	3000.0	22.0	22.0	20.8	...	1.00	1.00
Cylinder	5550.0	2500.0	26.0	26.0	20.8	...	1.00	1.00
Cylinder	8050.0	2500.0	22.0	22.0	20.8	...	1.00	1.00
Cylinder	9750.0	1700.0	22.0	22.0	20.8	...	1.00	1.00
Ellipse	9800.0	50.0	26.0	22.0	20.5	...	1.00	1.00

Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
A1_6"	WRC-297	0.993	Shell	Passed
S1_12"	WRC-297	0.833	Shell	Passed
V1_3"	WRC-297	0.993	Shell	Passed
B2_6"	WRC-297	0.993	Shell	Passed
B1_2"	WRC-297	0.993	Shell	Passed

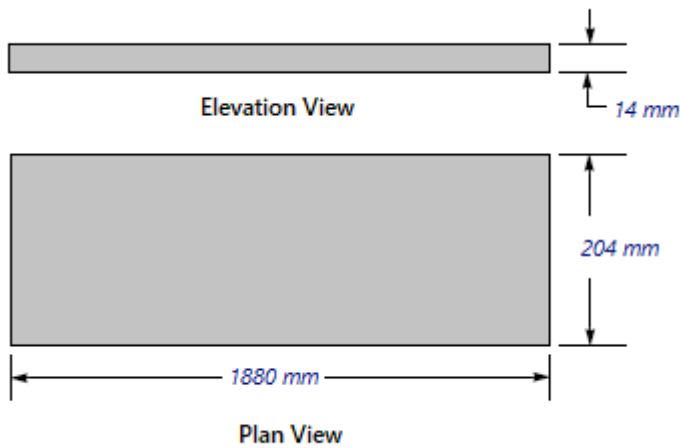
Foundation/Support Summary

Saddle Parameters:

Saddle Width	178.000 mm
Saddle Bearing Angle	120.000 deg.
Centerline Dimension	1600.000 mm
Wear Pad Width	230.000 mm
Wear Pad Thickness	10.000 mm
Wear Pad Bearing Angle	132.000 deg.
Distance from Saddle to Tangent	1500.000 mm

Baseplate Length	1880.000 mm
Baseplate Thickness	14.000 mm
Baseplate Width	204.000 mm
Number of Ribs (including outside ribs)	3
Rib Thickness	14.000 mm
Web Thickness	14.000 mm
Height of Center Web	492.000 mm
Number of Bolts in Baseplate	4

Baseplate Sketch



Maximum Tensile Bolt Load 984. N

Summary of Maximum Saddle Loads, Operating Case, Un-Factored:

Maximum Vertical Saddle Load	196856.11 N
Maximum Transverse Saddle Shear Load	31663.33 N
Maximum Longitudinal Saddle Shear Load	19506.10 N

Summary of Maximum Saddle Loads, Pressure Test Case:

Maximum Vertical Saddle Load	325368.09 N
Maximum Transverse Saddle Shear Load	10448.90 N
Maximum Longitudinal Saddle Shear Load	5940.22 N

Weights:

Fabricated - Bare w/o removable internals	17804.3 kgm
Shop Test - Fabricated + water (full)	54241.5 kgm



Shipping	- Fab. + rem. intls.+ shipping app.	21891.4	kgm
Erected	- Fab. + rem. intls.+ insul. (etc)	21976.4	kgm
Empty	- Fab. + intls. + details + wghts.	21976.4	kgm
Operating	- Empty + operating liquid (no ca)	21976.4	kgm
Field Test	- Empty weight + test medium (full)	58328.6	kgm

Nozzle Calculation Summary (MPa & mm):

Description	MAWP	Ext	MAPNC	UG-45	[tr]	Weld Path	Areas or Stresses
M1_24"	2.72	...	...	...		OK	Passed
A1_6"	2.66	...	...	OK	7.72	OK	Passed
P1_2"	2.66	...	...	OK	4.92	OK	No Calc[*]
L7_2"	2.66	...	...	OK	4.92	OK	No Calc[*]
L8_2"	2.66	...	...	OK	4.92	OK	No Calc[*]
P2_2"	3.17	...	...	OK	4.92	OK	No Calc[*]
S1_12"	2.85	...	...	OK	9.83	OK	Passed
L3_2"	3.17	...	...	OK	4.92	OK	No Calc[*]
L4_2"	3.17	...	...	OK	4.92	OK	No Calc[*]
L5_2"	3.17	...	...	OK	4.92	OK	No Calc[*]
L6_2"	3.17	...	...	OK	4.92	OK	No Calc[*]
V1_3"	2.66	...	...	OK	6.30	OK	Passed
L1_2"	2.66	...	...	OK	4.92	OK	No Calc[*]
L2_2"	2.66	...	...	OK	4.92	OK	No Calc[*]
B2_6"	2.66	...	...	OK	7.72	OK	Passed
B1_2"	2.66	...	...	OK	6.30	OK	No Calc[*]

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 2.66 Nozzle : B12"
 Minimum MAWP Shells/Flanges : 2.66 Element : Shell-4
 Minimum MAPnc Shells/Flanges : 2.86 Element : Shell-4

Computed Vessel M.A.W.P. : 2.66 MPa

[*] - This was a small opening and the areas were not computed.

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	X Coordinate mm	Layout Angle deg	Dia. Limit mm
10	M1_24"	0.000	0.000	1121.200
20	A1_6"	1550.000	90.000	304.145
20	P1_2"	2550.000	90.000	101.325
20	L7_2"	2750.000	30.000	101.325
20	L8_2"	2750.000	330.000	101.325
30	P2_2"	3550.000	90.000	109.325
30	S1_12"	5050.000	90.000	615.600
30	L3_2"	5050.000	30.000	109.325
30	L4_2"	5050.000	330.000	109.325
30	L5_2"	4050.000	30.000	109.325
30	L6_2"	4050.000	330.000	109.325
40	V1_3"	7550.000	90.000	144.862
40	L1_2"	6550.000	30.000	101.325
40	L2_2"	6550.000	330.000	101.325
50	B2_6"	9250.000	270.000	304.145
50	B1_2"	9049.999	90.000	125.328

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.

lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!



Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia in	Wall Thk mm	Reinforcing Pad Diameter	Pad Thk mm	Cut Length mm	Flg Class
P1_2"	2.000 in	160	LWN	2.375	8.738	...	...	223.03	300
L7_2"	2.000 in	160	LWN	2.375	8.738	...	...	223.03	300
L8_2"	2.000 in	160	LWN	2.375	8.738	...	...	223.03	300
P2_2"	2.000 in	160	LWN	2.375	8.738	...	...	227.03	300
L3_2"	2.000 in	160	LWN	2.375	8.738	...	...	227.03	300
L4_2"	2.000 in	160	LWN	2.375	8.738	...	...	227.03	300
L5_2"	2.000 in	160	LWN	2.375	8.738	...	...	227.03	300
L6_2"	2.000 in	160	LWN	2.375	8.738	...	...	227.03	300
L1_2"	2.000 in	160	LWN	2.375	8.738	...	...	223.03	300
L2_2"	2.000 in	160	LWN	2.375	8.738	...	...	223.03	300
B1_2"	2.000 in Actual		LWN	3.320	16.764	...	...	219.70	600
V1_3"	3.000 in	160	WNF	3.500	11.125	...	...	223.87	600
A1_6"	6.000 in	80	WNF	6.625	10.973	...	...	228.71	300
B2_6"	6.000 in	80	WNF	6.625	10.973	...	...	228.71	300
S1_12"	12.000 in Actual		WNF	12.945	12.000	...	...	248.96	300
M1_24"	24.000 in Actual		WNF	24.000	26.000	...	...	352.45	300

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
P1_2"	SA-350 LF1	22.000	10.000	...	...	...
L7_2"	SA-350 LF1	22.000	10.000	...	...	...
L8_2"	SA-350 LF1	22.000	10.000	...	...	...
P2_2"	SA-350 LF1	22.000	10.000	...	...	...
L3_2"	SA-350 LF1	22.000	10.000	...	...	...
L4_2"	SA-350 LF1	22.000	10.000	...	...	...
L5_2"	SA-350 LF1	22.000	10.000	...	...	...
L6_2"	SA-350 LF1	22.000	10.000	...	...	...
L1_2"	SA-350 LF1	22.000	10.000	...	...	...
L2_2"	SA-350 LF1	22.000	10.000	...	...	...
B1_2"	SA-350 LF1	22.000	10.000	...	...	...
V1_3"	SA-350 LF1	22.000	10.000	...	...	...
A1_6"	SA-350 LF1	22.000	10.000	...	...	...
B2_6"	SA-350 LF1	22.000	10.000	...	...	...
S1_12"	SA-350 LF1	22.000	10.000	...	...	...
M1_24"	SA-350 LF1	22.000	10.000	...	...	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum mm	Layout Angle deg	Proj Outside mm	Proj Inside mm	Installed in Component
P1_2"	2550.000	90.0	200.00	0.00	Shell-1
L7_2"	2750.000	30.0	200.00	0.00	Shell-1
L8_2"	2750.000	330.0	200.00	0.00	Shell-1
P2_2"	3550.000	90.0	200.00	0.00	Shell-2
L3_2"	5050.000	30.0	200.00	0.00	Shell-2



L4_2"	5050.000	330.0	200.00	0.00	Shell-2
L5_2"	4050.000	30.0	200.00	0.00	Shell-2
L6_2"	4050.000	330.0	200.00	0.00	Shell-2
L1_2"	6550.000	30.0	200.00	0.00	Shell-3
L2_2"	6550.000	330.0	200.00	0.00	Shell-3
B1_2"	9049.999	90.0	196.85	0.00	Shell-4
V1_3"	7550.000	90.0	200.00	0.00	Shell-3
A1_6"	1550.000	90.0	200.00	0.00	Shell-1
B2_6"	9249.999	270.0	200.00	0.00	Shell-4
S1_12"	5050.000	90.0	200.00	0.00	Shell-2
M1_24"	...	0.0	300.00	0.00	Head-1



Bill of Materials:

QTY	DESCRIPTION	MATERIAL
1	ELLIPTICAL HEAD: 2.0 X 1, 34.0mm THK X 2100.0mm ID X 50.0mm	SA-516 70
1	ELLIPTICAL HEAD: 2.0 X 1, 26.0mm THK X 2100.0mm ID X 50.0mm	SA-516 70
2	INSULATION: 575mm X 40mm THK	CC
2	SADDLE: 178.0mm X 120 DEG	SA-285 C
1	PLATFORM: 1100mm X 2144mm WIDE	
1	LADDER: 2500.0mm	...
1	INSULATION: 3000mm X 40mm THK	CC
2	INSULATION: 2500mm X 40mm THK	CC
1	INSULATION: 1700mm X 40mm THK	CC
1	CLASS 300 GR 1.4, 24.0" BLIND FLANGE(S)	SA-350 LF1
2	CYLINDRICAL SEGMENT 22.0mm THK X 3048.0mm X 3000.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 22.0mm THK X 501.3mm X 3000.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 26.0mm THK X 3048.0mm X 2500.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 26.0mm THK X 3549.3mm X 2500.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 26.0mm THK X 0.0mm X 2500.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 22.0mm THK X 3048.0mm X 2500.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 22.0mm THK X 3549.3mm X 2500.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 22.0mm THK X 0.0mm X 2500.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 22.0mm THK X 3048.0mm X 1700.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 22.0mm THK X 3549.3mm X 1700.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 22.0mm THK X 0.0mm X 1700.0mm	SA-516 70
1	NAMEPLATE	...



Minimum Design Metal Temperature Results Summary (°C):

Description	Notes	Curve	Basic MDMT	Reduced MDMT	UG-20(f) MDMT	Thickness ratio	Gov Thk mm	E*	PWHT reqd
Head-1	[10]	!	-40	-53		0.765	28.000	1.00	No
Head-1	[7]	!	-40	-61		0.631	34.000	1.00	No
Shell-1	[8]	!	-40	-40		1.000	22.000	1.00	No
Shell-2	[8]	!	-40	-49		0.837	26.000	1.00	No
Shell-3	[8]	!	-40	-40		1.000	22.000	1.00	No
Shell-4	[8]	!	-40	-40		1.000	22.000	1.00	No
Head-2	[10]	!	-40	-41		0.988	22.000	1.00	No
Head-2	[7]	!	-40	-49		0.837	26.000	1.00	No
M124"	[1]	!	-29	-29	-29	0.689	28.000	1.00	No
Nozzle Flg	[5]	!	-29	-50					
A16"	[1]	!	-29	-29	-29	1.000	22.000	1.00	No
Nozzle Flg	[5]	!	-29	-50					
P12"	[1]	!	-29	-29	-29	1.000	22.000	1.00	No
Nozzle Flg	[5]	!	-40	-40					
L72"	[1]	!	-29	-29	-29	1.000	22.000	1.00	No
Nozzle Flg	[5]	!	-40	-40					
L82"	[1]	!	-29	-29	-29	1.000	22.000	1.00	No
Nozzle Flg	[5]	!	-40	-40					
P22"	[1]	!	-29	-104	-29	0.024	24.000	1.00	No
Nozzle Flg	[5]	!	-40	-40					
S112"	[1]	!	-29	-29	-29	0.837	26.000	1.00	No
Nozzle Flg	[5]	!	-29	-50					
L32"	[1]	!	-29	-104	-29	0.024	24.000	1.00	No
Nozzle Flg	[5]	A	-18	-48					
L42"	[1]	!	-29	-104	-29	0.024	24.000	1.00	No
Nozzle Flg	[5]	A	-18	-48					
L52"	[1]	!	-29	-104	-29	0.024	24.000	1.00	No
Nozzle Flg	[5]	A	-18	-48					
L62"	[1]	!	-29	-104	-29	0.024	24.000	1.00	No
Nozzle Flg	[5]	A	-18	-48					
V13"	[1]	!	-29	-29	-29	1.000	22.000	1.00	No
Nozzle Flg	[5]	!	-29	-104					
L12"	[1]	!	-29	-29	-29	1.000	22.000	1.00	No
Nozzle Flg	[5]	!	-40	-40					
L22"	[1]	!	-29	-29	-29	1.000	22.000	1.00	No
Nozzle Flg	[5]	!	-40	-40					
B26"	[1]	!	-29	-29	-29	1.000	22.000	1.00	No
Nozzle Flg	[5]	!	-29	-50					
B12"	[1]	!	-29	-104	-29	0.040	16.764	1.00	No
Nozzle Flg	[5]	!	-40	-40					
Nozl. Bolting	[22]		-48			1.000	64.000	1.00	No
Warmest MDMT:			-18	-29					

Warmest MDMT: -18 -29

Required Minimum Design Metal Temperature -40.0 °C
 Warmest Computed Minimum Design Metal Temperature -29.0 °C

Warning: Computed overall MDMT was higher than the required value !

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable



- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)
- [20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld

UG-84(b)(2) was not considered.

UCS-66(g) was not considered.

UCS-66(i) was not considered.

Notes:

Impact test temps were not entered in and not considered in the analysis.

UCS-66(i) applies to impact tested materials not by specification and

UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).

The Basic MDMT includes the (30F) PWHT credit if applicable.

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Class From To : Basic Element Checks.

Class From To: Check of Additional Element Data

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



One or more nozzles in this model was specified using the actual thickness basis and a standard flange is also specified. In many cases a matching nominal diameter cannot be determined and the weight of the flange and optional blind flange cannot be determined. When actual thickness basis is specified, you must calculate the weight of the assembly (by hand) and enter it into the 'Overriding Weight' field in the nozzle dialog when there is a standard flange specified. However, for FVC nozzles that have their nominal diameter specified in the FVC dialog, entering the overriding weight for that nozzle is not required.



The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.

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Units used in this Analysis (SIASME):

Name	System Unit	Constant	User Unit
Length	Feet	304.8000	mm
Force	Pounds	4.4480	N
Mass	Pounds	0.4536	kgm
Area	sq. inches	645.1600	mm ²
Moment	ft. lbs.	1356.3000	N-mm
Stress	lbs./sq.in.	0.0069	MPa
Temperature	Degrees F	0.5556	°C
Pressure	psig	0.0069	MPa
Elast. Modulus	lbs./sq.in.	0.0069	MPa
Pipe Density	lbs./cu.in.	0.0277	kg/cm ³
Ins. Density	lbs./cu.ft.	0.1600E-04	kg/cm ³
Fluid Density	lbs./cu.ft.	0.1600E-04	kg/cm ³
Wind Speed	miles/hr	1.6093	km/hr
Tray Weight	lbs./sq.ft.	0.0005	kg/cm ²
Inertia	in.4	416231.0000	mm ⁴
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	0.0479	kPa
Elevation	Feet	304.8000	mm
Volume	in.3	0.0164	ltr
Diameter	inches	25.4000	mm
Thickness	inches	25.4000	mm

PV Elite Vessel Analysis Program: Input Data

Design Internal Pressure (for Hydrotest)	2.5	MPa
Design Internal Temperature	85.0	°C
Projection of Nozzle from Vessel Top	0	mm
Projection of Nozzle from Vessel Bottom	0	mm
Minimum Design Metal Temperature	-40.0	°C
Type of Construction	Welded	
Special Service	None	
Degree of Radiography	RT-1	
Use Higher Longitudinal Stresses (Flag)	Y	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	
Consider Vortex Shedding	N	

Shop Pressure Test:

Type of Pressure Test	UG-99(b)
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	No

Field Pressure Test:

Type of Pressure Test	UG-99(b)
Pressure Test Position	Vertical
Test Performed in Corroded Condition	Yes

Load Case 1	NP+EW+WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW



Load Case 16	IP+VE+EW	
Load Case 17	NP+VO+OW	
Load Case 18	FS+BS+IP+OW	
Load Case 19	FS+BS+EP+OW	
Load Case 20	BL+IP+OW	
Wind Design Code	ASCE/SEI 7-22	
Wind Load Reduction Scale Factor	1.100	
Basic Wind Speed	[V] 180	km/hr
Surface Roughness Category	D: Flat, unobstructed	
Type of Surface	Moderately Smooth	
Base Elevation	300	mm
Percent Wind for Hydrotest	33.0	
Using User defined Wind Press. Vs Elev.	N	
Height of Hill or Escarpment	H or Hh	0 mm
Distance Upwind of Crest	Lh	0 mm
Distance from Crest to the Vessel	x	0 mm
Type of Terrain (Hill, Escarpment)	Flat	
Damping Factor (Beta) for Wind (Ope)	0.0100	
Damping Factor (Beta) for Wind (Empty)	0.0000	
Damping Factor (Beta) for Wind (Filled)	0.0000	
Site Elevation above Sea Level	0	mm
Seismic Design Code	ASCE/SEI 7-22	
Importance Factor	1.500	
Force Modification Factor R	3.000	
Site Class	D	
Component Elevation Ratio	z/h	0.000
Amplification Factor	Ap	1.000
Force Factor		0.000
Consider Vertical Acceleration	No	
Minimum Acceleration Multiplier		0.000
User Value of Sds (used if > 0)		0.000
User Value of Sd1 (used if > 0)		0.000
Component resonance duct. fact.	[CAR]	0.000
Component strength factor	[Rpo]	0.000
Overstrength factor	[Omega0]	0.000
Response Acceleration Factor	[SMS]	0.326
Response Acceleration Factor	[SM1]	0.139
Building Importance Factor	[Ie]	1.000
Site Latitude		0.000 deg
Site Longitude		0.000 deg
M.A.W.P + Static Head to Nozzle	Y	
Consider MAP New and Cold in Noz. Design	N	
Consider External Loads for Nozzle Des.	Y	
Use ASME VIII-1 Appendix 1-9	N	
Perform Blast Load Analysis	No	
Material Database Year	Current w/Addenda or Code Year	

Configuration Directives:

g0 is taken as min(g0, attached shell thk)	Yes
Do not use Nozzle MDMT Interpretation VIII-1 01-37	No
Use Table G instead of exact equation for "A"	Yes
Shell Head Joints are Tapered	Yes
Compute "K" in corroded condition	Yes
Use Code Case 2286	No
Use the MAWP to compute the MDMT	Yes
For thickness ratios <= 0.35, MDMT will be -155F (-104C)	Yes
For PWHT & P1 Materials the MDMT can be < -55F (-48C)	No
Using Metric Material Databases, ASME II D	Yes
Calculate B31.3 type stress for Nozzles with Loads	Yes
Reduce the MDMT due to lower membrane stress	Yes
Consider Longitudinal Stress in MDMT Calculations	Yes

Complete Listing of Vessel Elements and Details:



Element#	1/6
Element From Node	10
Element To Node	20
Element Type	Elliptical
Description	Head-1
Distance "FROM" to "TO"	50 mm
Inside Diameter	2100 mm
Element Thickness	28 mm
Internal Corrosion Allowance	1.5 mm
Nominal Thickness	34 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	2.5 MPa
Design Temperature Internal Pressure	85 °C
Design External Pressure	0 MPa
Design Temperature External Pressure	85 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70 +N+IT
Allowable Stress, Ambient	138 MPa
Allowable Stress, Operating	138 MPa
Allowable Stress, Hydrotest	234 MPa
Material Density	0.00775 kg/cm³
P Number Thickness	32.004 mm
Yield Stress, Operating	242 MPa
UCS-66 Chart Curve Designation	Impact Tested
External Pressure Chart Name	CS-2
UNS Number	K02700
Product Form	Plate
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Elliptical Head Factor	2.0
Weld is pre-Heated	No
Element From Node	10
Detail Type	Insulation
Detail ID	Ins: 10
Dist. from "FROM" Node / Offset dist	-525 mm
Height/Length of Insulation	575 mm
Thickness of Insulation	40 mm
Density	0.40000E-04 kg/cm³
Element From Node	10
Detail Type	Nozzle
Detail ID	M124"
Dist. from "FROM" Node / Offset dist	0 mm
Nozzle Diameter	24 in.
Nozzle Schedule	None
Nozzle Class	300
Layout Angle	0.0
Blind Flange (Y/N)	Y
Weight of Nozzle (Used if > 0)	7901.6 N
Grade of Attached Flange	GR 1.4
Nozzle Matl	SA-350 LF1 +IT

Element#	2/6
Element From Node	20
Element To Node	30
Element Type	Cylinder
Description	Shell-1
Distance "FROM" to "TO"	3000 mm
Inside Diameter	2100 mm
Element Thickness	22 mm
Internal Corrosion Allowance	1.5 mm
Nominal Thickness	22 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	2.5 MPa
Design Temperature Internal Pressure	85 °C
Design External Pressure	0 MPa
Design Temperature External Pressure	85 °C



Effective Diameter Multiplier 1.2
 Material Name SA-516 70 +N+IT
 Efficiency, Longitudinal Seam 1.0
 Efficiency, Circumferential Seam 1.0
 Weld is pre-Heated No

Element From Node 20
 Detail Type Saddle
 Detail ID Left Saddle
 Dist. from "FROM" Node / Offset dist 1450 mm
 Width of Saddle 178 mm
 Height of Saddle at Bottom 1600 mm
 Saddle Contact Angle 120.0
 Height of Composite Ring Stiffener 0 mm
 Width of Wear Plate 230 mm
 Thickness of Wear Plate 10 mm
 Contact Angle, Wear Plate (degrees) 132.0
 Friction coefficient 0.60000002
 Moment Factor 3.0
 Dimension E at base (optional) 0 mm
 Circumferential Eff. over Saddle 1.0
 Circumferential Eff. at Midspan 1.0
 Tangent to Tangent dist. (optional) 0 mm

Element From Node 20
 Detail Type Platform
 Detail ID PLAT:[1 OF 1]
 Dist. from "FROM" Node / Offset dist 0 mm
 Platform Start Angle (degrees) 0.0
 Platform End Angle (degrees) 0.0
 Platform Wind Area 22779200 mm²
 Platform Weight 36269 N
 Platform Railing Weight 0 N/mm
 Platform Grating Weight 0.017 kg/cm²
 Platform Width 2144 mm
 Platform Height 1100 mm
 Platform Clearance or End Offset 125 mm
 Platform Force Coefficient 2.0
 Ladder Layout Angle 0.0
 Ladder Start Elevation 0 mm
 Ladder End Elevation 2500 mm
 Unit Weight of Ladder 0.49 N/mm
 Platform Length (top head platform) 9800 mm

Element From Node 20
 Detail Type Insulation
 Detail ID Ins: 10
 Dist. from "FROM" Node / Offset dist 0 mm
 Height/Length of Insulation 3000 mm
 Thickness of Insulation 40 mm
 Density 0.40000E-04 kg/cm³

Element From Node 20
 Detail Type Nozzle
 Detail ID A16"
 Dist. from "FROM" Node / Offset dist 1500 mm
 Nozzle Diameter 6 in.
 Nozzle Schedule 80
 Nozzle Class 300
 Layout Angle 90.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 498.32 N
 Grade of Attached Flange GR 1.4
 Nozzle Matl SA-350 LF1 +IT

Element From Node 20
 Detail Type Nozzle
 Detail ID P12"
 Dist. from "FROM" Node / Offset dist 2500 mm
 Nozzle Diameter 2 in.



Nozzle Schedule 160
 Nozzle Class 300
 Layout Angle 90.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 85.981 N
 Grade of Attached Flange GR 1.1
 Nozzle Matl SA-350 LF1 +IT

Element From Node 20
 Detail Type Nozzle
 Detail ID L72"
 Dist. from "FROM" Node / Offset dist 2700 mm
 Nozzle Diameter 2 in.
 Nozzle Schedule 160
 Nozzle Class 300
 Layout Angle 30.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 85.981 N
 Grade of Attached Flange GR 1.1
 Nozzle Matl SA-350 LF1 +IT

Element From Node 20
 Detail Type Nozzle
 Detail ID L82"
 Dist. from "FROM" Node / Offset dist 2700 mm
 Nozzle Diameter 2 in.
 Nozzle Schedule 160
 Nozzle Class 300
 Layout Angle 330.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 85.981 N
 Grade of Attached Flange GR 1.1
 Nozzle Matl SA-350 LF1 +IT

Element From Node 20
 Detail Type Weight
 Detail ID Lifting lug
 Dist. from "FROM" Node / Offset dist 2000 mm
 Miscellaneous Weight 4903 N
 Offset from Element Centerline 0 mm

Element From Node 20
 Detail Type Weight
 Detail ID Half ope pipeA
 Dist. from "FROM" Node / Offset dist 1500 mm
 Miscellaneous Weight 833.51 N
 Offset from Element Centerline 0 mm

Element# 3/6
 Element From Node 30
 Element To Node 40
 Element Type Cylinder
 Description Shell-2
 Distance "FROM" to "TO" 2500 mm
 Inside Diameter 2100 mm
 Element Thickness 26 mm
 Internal Corrosion Allowance 1.5 mm
 Nominal Thickness 26 mm
 External Corrosion Allowance 0 mm
 Design Internal Pressure 2.5 MPa
 Design Temperature Internal Pressure 85 °C
 Design External Pressure 0 MPa
 Design Temperature External Pressure 85 °C
 Effective Diameter Multiplier 1.2
 Material Name SA-516 70 +N+IT
 Efficiency, Longitudinal Seam 1.0
 Efficiency, Circumferential Seam 1.0
 Weld is pre-Heated No



Element From Node	30
Detail Type	Insulation
Detail ID	Ins: 10
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	2500 mm
Thickness of Insulation	40 mm
Density	0.40000E-04 kg/cm ³

Element From Node	30
Detail Type	Nozzle
Detail ID	P22"
Dist. from "FROM" Node / Offset dist	500 mm
Nozzle Diameter	2 in.
Nozzle Schedule	160
Nozzle Class	300
Layout Angle	90.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	85.981 N
Grade of Attached Flange	GR 1.4
Nozzle Matl	SA-350 LF1 +IT

Element From Node	30
Detail Type	Nozzle
Detail ID	S112"
Dist. from "FROM" Node / Offset dist	2000 mm
Nozzle Diameter	12 in.
Nozzle Schedule	None
Nozzle Class	300
Layout Angle	90.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	1359.8 N
Grade of Attached Flange	GR 1.4
Nozzle Matl	SA-350 LF1 +IT

Element From Node	30
Detail Type	Nozzle
Detail ID	L32"
Dist. from "FROM" Node / Offset dist	2000 mm
Nozzle Diameter	2 in.
Nozzle Schedule	160
Nozzle Class	300
Layout Angle	30.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	66.114 N
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-350 LF1 +IT

Element From Node	30
Detail Type	Nozzle
Detail ID	L42"
Dist. from "FROM" Node / Offset dist	2000 mm
Nozzle Diameter	2 in.
Nozzle Schedule	160
Nozzle Class	300
Layout Angle	330.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	85.981 N
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-350 LF1 +IT

Element From Node	30
Detail Type	Nozzle
Detail ID	L52"
Dist. from "FROM" Node / Offset dist	1000 mm
Nozzle Diameter	2 in.
Nozzle Schedule	160
Nozzle Class	300
Layout Angle	30.0
Blind Flange (Y/N)	N



Weight of Nozzle (Used if > 0)	85.981	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-350 LF1	+IT
Element From Node	30	
Detail Type	Nozzle	
Detail ID	L62"	
Dist. from "FROM" Node / Offset dist	1000	mm
Nozzle Diameter	2	in.
Nozzle Schedule	160	
Nozzle Class	300	
Layout Angle	330.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	85.981	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-350 LF1	+IT
Element From Node	30	
Detail Type	Weight	
Detail ID	Clipwt	
Dist. from "FROM" Node / Offset dist	0	mm
Miscellaneous Weight	1372.8	N
Offset from Element Centerline	0	mm

Element#	4/6
Element From Node	40
Element To Node	50
Element Type	Cylinder
Description	Shell-3
Distance "FROM" to "TO"	2500 mm
Inside Diameter	2100 mm
Element Thickness	22 mm
Internal Corrosion Allowance	1.5 mm
Nominal Thickness	22 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	2.5 MPa
Design Temperature Internal Pressure	85 °C
Design External Pressure	0 MPa
Design Temperature External Pressure	85 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70 +N+IT
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	40
Detail Type	Insulation
Detail ID	Ins: 10
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	2500 mm
Thickness of Insulation	40 mm
Density	0.40000E-04 kg/cm³

Element From Node	40
Detail Type	Nozzle
Detail ID	V13"
Dist. from "FROM" Node / Offset dist	2000 mm
Nozzle Diameter	3 in.
Nozzle Schedule	160
Nozzle Class	600
Layout Angle	90.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	159.5 N
Grade of Attached Flange	GR 1.4
Nozzle Matl	SA-350 LF1
	+IT

Element From Node	40
Detail Type	Nozzle



Detail ID	L12"
Dist. from "FROM" Node / Offset dist	1000 mm
Nozzle Diameter	2 in.
Nozzle Schedule	160
Nozzle Class	300
Layout Angle	30.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	85.981 N
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-350 LF1 +IT

Element From Node	40
Detail Type	Nozzle
Detail ID	L22"
Dist. from "FROM" Node / Offset dist	1000 mm
Nozzle Diameter	2 in.
Nozzle Schedule	160
Nozzle Class	300
Layout Angle	330.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	85.981 N
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-350 LF1 +IT

Element#	5/6
Element From Node	50
Element To Node	60
Element Type	Cylinder
Description	Shell-4
Distance "FROM" to "TO"	1700 mm
Inside Diameter	2100 mm
Element Thickness	22 mm
Internal Corrosion Allowance	1.5 mm
Nominal Thickness	22 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	2.5 MPa
Design Temperature Internal Pressure	85 °C
Design External Pressure	0 MPa
Design Temperature External Pressure	85 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70 +N+IT
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	50
Detail Type	Saddle
Detail ID	New Saddle
Dist. from "FROM" Node / Offset dist	250 mm
Width of Saddle	178 mm
Height of Saddle at Bottom	1600 mm
Saddle Contact Angle	120.0
Height of Composite Ring Stiffener	0 mm
Width of Wear Plate	230 mm
Thickness of Wear Plate	10 mm
Contact Angle, Wear Plate (degrees)	132.0
Friction coefficient	0.0
Moment Factor	3.0
Dimension E at base (optional)	0 mm
Circumferential Eff. over Saddle	1.0
Circumferential Eff. at Midspan	1.0
Tangent to Tangent dist. (optional)	0 mm

Element From Node	50
Detail Type	Insulation
Detail ID	Ins: 10
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	1700 mm



Thickness of Insulation	40	mm
Density	0.40000E-04	kg/cm ³
Element From Node	50	
Detail Type	Nozzle	
Detail ID	B26"	
Dist. from "FROM" Node / Offset dist	1200	mm
Nozzle Diameter	6	in.
Nozzle Schedule	80	
Nozzle Class	300	
Layout Angle	270.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	498.32	N
Grade of Attached Flange	GR 1.4	
Nozzle Matl	SA-350 LF1 +IT	
Element From Node	50	
Detail Type	Nozzle	
Detail ID	B12"	
Dist. from "FROM" Node / Offset dist	1000	mm
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	600	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	108.25	N
Grade of Attached Flange	GR 1.4	
Nozzle Matl	SA-350 LF1 +IT	

Element#	6/6
Element From Node	60
Element To Node	70
Element Type	Elliptical
Description	Head-2
Distance "FROM" to "TO"	50 mm
Inside Diameter	2100 mm
Element Thickness	22 mm
Internal Corrosion Allowance	1.5 mm
Nominal Thickness	26 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	2.5 MPa
Design Temperature Internal Pressure	85 °C
Design External Pressure	0 MPa
Design Temperature External Pressure	85 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70 +N+IT
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Elliptical Head Factor	2.0
Weld is pre-Heated	No

Element From Node	60
Detail Type	Insulation
Detail ID	Ins: 10
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	575 mm
Thickness of Insulation	40 mm
Density	0.40000E-04 kg/cm ³



XY Coordinate Calculations:

From	To	X (Horiz.) mm	Y (Vert.) mm	DX (Horiz.) mm	DY (Vert.) mm
	Head-1	50	...	50	...
	Shell-1	3050	...	3000	...
	Shell-2	5550	...	2500	...
	Shell-3	8050	...	2500	...
	Shell-4	9750	...	1700	...
	Head-2	9800	...	50	...

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd MPa	Nominal Thickness mm	Total Corr Allowance mm	Element Diameter mm	Allowable Stress(SE) MPa	Element#
	Head-1	2.5	34	1.5	2100	138	1
	Shell-1	2.5	22	1.5	2100	138	2
	Shell-2	2.5	26	1.5	2100	138	3
	Shell-3	2.5	22	1.5	2100	138	4
	Shell-4	2.5	22	1.5	2100	138	5
	Head-2	2.5	26	1.5	2100	138	6

Element Required Thickness and MAWP :

From	To	Design Pressure MPa	M.A.W.P. Corroded MPa	M.A.P. New & Cold MPa	Minimum Thickness mm	Required Thickness mm	Element#
	Head-1	2.5	3.48	3.67	28	20.5477	1
	Shell-1	2.5	2.66	2.86	22	20.7586	2
	Shell-2	2.5	3.17	3.37	26	20.7586	3
	Shell-3	2.5	2.66	2.86	22	20.7586	4
	Shell-4	2.5	2.66	2.86	22	20.7586	5
	Head-2	2.5	2.69	2.89	22	20.5476	6

Minimum 2.66 2.86

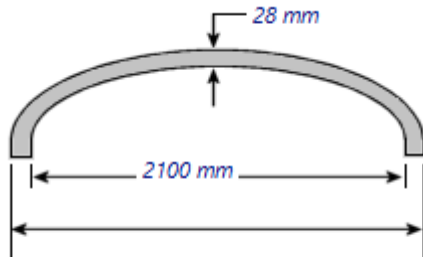
MAWP: 2.66 MPa, limited by: Shell-4.

Elements Suitable for Design Internal Pressure.

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Head-1

Elliptical Head Design Information:					
Design Pressure	P	2.500 MPa	Int. Design Temperature		85 °C
Material		SA-516 70	Ext. Design Temperature		85 °C
Length		50.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.5000 mm	Allowable Stress (ope)	S	138.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	2100.000 mm	Actual Stress		99.3 MPa
Spec. Min. Thk t _e , or s _a		28.0000 mm	Spec. Nominal Thk.		34.000 mm
Surface Area		5435522.5 mm ²	SG of Contents		0.000
Element Volume		1385.7 ltr	Weight of Contents		0.0 N
Empty Weight		14950.3 N	Operating Weight		14950.3 N
			Comp. All. Ope. Stress		134.8 MPa
			Aspect Ratio		2.00

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 26.250 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 13.125 mm

Material treatment options: +N+IT

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (2.5 \cdot 2103 \cdot 1) / (2 \cdot 138 \cdot 1 - 0.2 \cdot 2.5)$$

$$= 19.0477 + 1.5000 = 20.5477 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138 \cdot 1 \cdot 26.5) / (1 \cdot 2103 + 0.2 \cdot 26.5)$$

$$= 3.48 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138 \cdot 1 \cdot 28) / (1 \cdot 2100 + 0.2 \cdot 28)$$

$$= 3.67 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (2.5 \cdot (1 \cdot 2103 + 0.2 \cdot 26.5)) / (2 \cdot 1 \cdot 26.5)$$

$$= 99.260 \text{ MPa}$$



Straight Flange Required Thickness:

$$\begin{aligned} &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \quad \text{per UG-27 (c)(1)} \\ &= (2.5 \cdot 1052) / (138 \cdot 1 - 0.6 \cdot 2.5) + 1.5 \\ &= 20.759 \text{ mm} \end{aligned}$$

Straight Flange Maximum Allowable Working Pressure:

$$\begin{aligned} &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \quad \text{per UG-27 (c)(1)} \\ &= (138 \cdot 1 \cdot 32.5) / (1052 + 0.6 \cdot 32.5) \\ &= 4.19 \text{ MPa} \end{aligned}$$

Factor K, corroded condition [Kcor]:

$$\begin{aligned} &= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth}))^2) / 6 \\ &= (2 + (2103 / (2 \cdot 526))^2) / 6 \\ &= 0.998102 \end{aligned}$$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 6.818 %

Note: Please Check Requirements of UCS-79 as Elongation is > 5%.

MDMT Calculations in the Knuckle Portion:

Note: This Element/Detail was specified as being Impact Tested.

MDMT Calculations in the Head Straight Flange:

Note: This Element/Detail was specified as being Impact Tested.

Tolerance for Formed Heads per UG-81(a):

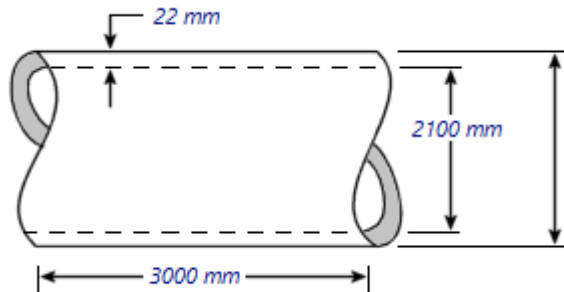
Head inner surface maximum deviation outside the specified shape, 1.25% of D: 26.250 mm
Head inner surface maximum deviation inside the specified shape, 0.625% of D: 13.125 mm

Material treatment options: +N+IT

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Shell-1

Cylindrical Shell Design Information:					
Design Pressure	P	2.500 MPa	Int. Design Temperature	85 °C	
Material		SA-516 70	Ext. Design Temperature	85 °C	
Length		3000.000 mm	External Pressure Chart	CS-2	
			UNS Number	K02700	
Int. Corr. All.	c	1.5000 mm	Allowable Stress (ope)	S	138.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	2100.000 mm	Actual Stress	129.7 MPa	
Spec. Min. Thk t _e , or s _a		22.0000 mm	Spec. Nominal Thk.	22.000 mm	
Surface Area		20206724.0 mm ²	SG of Contents	0.000	
Element Volume		10392.7 ltr	Weight of Contents	0.0 N	
Empty Weight		33439.7 N	Operating Weight	33439.7 N	
			Comp. All. Ope. Stress	129.1 MPa	

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +N+IT

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (2.5 \cdot 1052) / (138 \cdot 1 - 0.6 \cdot 2.5)$$

$$= 19.2586 + 1.5000 = 20.7586 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 20.5) / (1052 + 0.6 \cdot 20.5)$$

$$= 2.66 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 22) / (1050 + 0.6 \cdot 22)$$

$$= 2.86 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (2.5 \cdot (1052 + 0.6 \cdot 20.5)) / (1 \cdot 20.5)$$

$$= 129.732 \text{ MPa}$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 1.037 %

Minimum Design Metal Temperature Results:



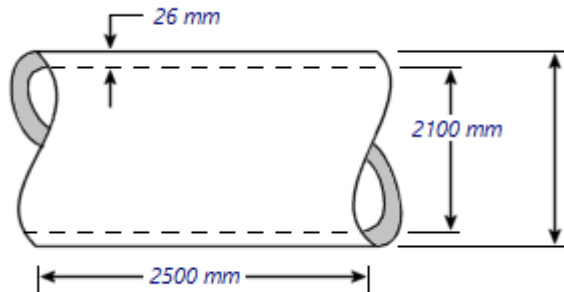
Note: This Element/Detail was specified as being Impact Tested.

Material treatment options: +N+IT

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Shell-2

Cylindrical Shell Design Information:					
Design Pressure	P	2.500 MPa	Int. Design Temperature		85 °C
Material		SA-516 70	Ext. Design Temperature		85 °C
Length		2500.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.5000 mm	Allowable Stress (ope)	S	138.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	2100.000 mm	Actual Stress		108.8 MPa
Spec. Min. Thk t _e , or s _a		26.0000 mm	Spec. Nominal Thk.		26.000 mm
Surface Area		16901768.0 mm ²	SG of Contents		0.000
Element Volume		8660.6 ltr	Weight of Contents		0.0 N
Empty Weight		32994.9 N	Operating Weight		32994.9 N
			Comp. All. Ope. Stress		133.1 MPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +N+IT

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (2.5 \cdot 1052) / (138 \cdot 1 - 0.6 \cdot 2.5)$$

$$= 19.2586 + 1.5000 = 20.7586 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 24.5) / (1052 + 0.6 \cdot 24.5)$$

$$= 3.17 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 26) / (1050 + 0.6 \cdot 26)$$

$$= 3.37 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (2.5 \cdot (1052 + 0.6 \cdot 24.5)) / (1 \cdot 24.5)$$

$$= 108.796 \text{ MPa}$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 1.223 %

Minimum Design Metal Temperature Results:



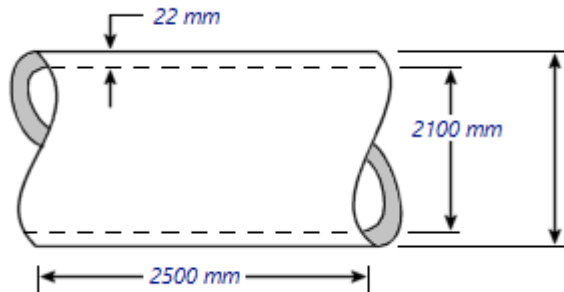
Note: This Element/Detail was specified as being Impact Tested.

Material treatment options: +N+IT

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Shell-3

Cylindrical Shell Design Information:					
Design Pressure	P	2.500 MPa	Int. Design Temperature		85 °C
Material		SA-516 70	Ext. Design Temperature		85 °C
Length		2500.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.5000 mm	Allowable Stress (ope)	S	138.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	2100.000 mm	Actual Stress		129.7 MPa
Spec. Min. Thk t _e , or s _a		22.0000 mm	Spec. Nominal Thk.		22.000 mm
Surface Area		16838938.0 mm ²	SG of Contents		0.000
Element Volume		8660.6 ltr	Weight of Contents		0.0 N
Empty Weight		27866.3 N	Operating Weight		27866.3 N
			Comp. All. Ope. Stress		129.1 MPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +N+IT

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (2.5 \cdot 1052) / (138 \cdot 1 - 0.6 \cdot 2.5)$$

$$= 19.2586 + 1.5000 = 20.7586 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 20.5) / (1052 + 0.6 \cdot 20.5)$$

$$= 2.66 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 22) / (1050 + 0.6 \cdot 22)$$

$$= 2.86 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (2.5 \cdot (1052 + 0.6 \cdot 20.5)) / (1 \cdot 20.5)$$

$$= 129.732 \text{ MPa}$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 1.037 %

Minimum Design Metal Temperature Results:



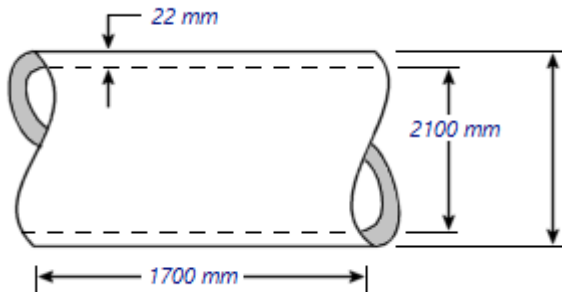
Note: This Element/Detail was specified as being Impact Tested.

Material treatment options: +N+IT

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Shell-4

Cylindrical Shell Design Information:					
Design Pressure	P	2.500 MPa	Int. Design Temperature		85 °C
Material		SA-516 70	Ext. Design Temperature		85 °C
Length		1700.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.5000 mm	Allowable Stress (ope)	S	138.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	2100.000 mm	Actual Stress		129.7 MPa
Spec. Min. Thk t _e , or s _a		22.0000 mm	Spec. Nominal Thk.		22.000 mm
Surface Area		11450478.0 mm ²	SG of Contents		0.000
Element Volume		5889.2 ltr	Weight of Contents		0.0 N
Empty Weight		18949.1 N	Operating Weight		18949.1 N
			Comp. All. Ope. Stress		129.1 MPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +N+IT

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (2.5 \cdot 1052) / (138 \cdot 1 - 0.6 \cdot 2.5)$$

$$= 19.2586 + 1.5000 = 20.7586 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 20.5) / (1052 + 0.6 \cdot 20.5)$$

$$= 2.66 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 22) / (1050 + 0.6 \cdot 22)$$

$$= 2.86 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (2.5 \cdot (1052 + 0.6 \cdot 20.5)) / (1 \cdot 20.5)$$

$$= 129.732 \text{ MPa}$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 1.037 %

Minimum Design Metal Temperature Results:



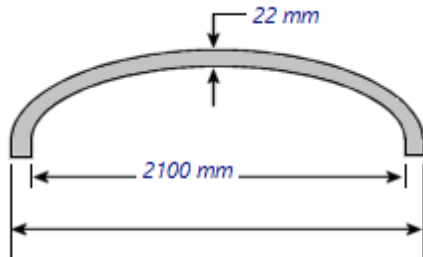
Note: This Element/Detail was specified as being Impact Tested.

Material treatment options: +N+IT

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Head-2

Elliptical Head Design Information:					
Design Pressure	P	2.500 MPa	Int. Design Temperature	85 °C	
Material		SA-516 70	Ext. Design Temperature	85 °C	
Length		50.000 mm	External Pressure Chart	CS-2	
			UNS Number	K02700	
Int. Corr. All.	c	1.5000 mm	Allowable Stress (ope)	S	138.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	2100.000 mm	Actual Stress	128.2 MPa	
Spec. Min. Thk t _e , or s _a		22.0000 mm	Spec. Nominal Thk.	26.000 mm	
Surface Area		5358084.0 mm ²	SG of Contents	0.000	
Element Volume		1385.7 ltr	Weight of Contents	0.0 N	
Empty Weight		11349.5 N	Operating Weight	11349.5 N	
			Comp. All. Ope. Stress	129.1 MPa	
			Aspect Ratio	2.00	

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 26.250 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 13.125 mm

Material treatment options: +N+IT

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (2.5 \cdot 2103 \cdot 1) / (2 \cdot 138 \cdot 1 - 0.2 \cdot 2.5)$$

$$= 19.0476 + 1.5000 = 20.5476 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138 \cdot 1 \cdot 20.5) / (1 \cdot 2103 + 0.2 \cdot 20.5)$$

$$= 2.69 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138 \cdot 1 \cdot 22) / (1 \cdot 2100 + 0.2 \cdot 22)$$

$$= 2.89 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (2.5 \cdot (1 \cdot 2103 + 0.2 \cdot 20.5)) / (2 \cdot 1 \cdot 20.5)$$

$$= 128.238 \text{ MPa}$$



Straight Flange Required Thickness:

$$\begin{aligned} &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \quad \text{per UG-27 (c)(1)} \\ &= (2.5 \cdot 1052) / (138 \cdot 1 - 0.6 \cdot 2.5) + 1.5 \\ &= 20.759 \text{ mm} \end{aligned}$$

Straight Flange Maximum Allowable Working Pressure:

$$\begin{aligned} &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \quad \text{per UG-27 (c)(1)} \\ &= (138 \cdot 1 \cdot 24.5) / (1052 + 0.6 \cdot 24.5) \\ &= 3.17 \text{ MPa} \end{aligned}$$

Factor K, corroded condition [Kcor]:

$$\begin{aligned} &= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth}))^2) / 6 \\ &= (2 + (2103 / (2 \cdot 526))^2) / 6 \\ &= 0.998102 \end{aligned}$$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 5.270 %

Note: Please Check Requirements of UCS-79 as Elongation is > 5%.

MDMT Calculations in the Knuckle Portion:

Note: This Element/Detail was specified as being Impact Tested.

MDMT Calculations in the Head Straight Flange:

Note: This Element/Detail was specified as being Impact Tested.

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 26.250 mm
Head inner surface maximum deviation inside the specified shape, 0.625% of D: 13.125 mm

Material treatment options: +N+IT

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Hydrostatic Test Pressure Results:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	3.457 MPa
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	3.250 MPa
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	3.692 MPa
UG100	= 1.10 * M.A.W.P. * Sa/S	2.925 MPa
PED	= max(1.43*DP, 1.25*DP*ratio)	3.575 MPa
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	3.575 MPa
App 27-4	= M.A.W.P.	2.659 MPa

UG-99(b), Test Pressure Calculation:
 = Test Factor • MAWP • Stress Ratio
 = 1.3 • 2.66 • 1
 = 3.457 MPa

Horizontal Test performed per: UG-99b

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (MPa):

From To	Stress	Allowable	Ratio	Pressure
Head-1	130.8	234.0	0.559	3.48
Shell-1	168.1	234.0	0.718	3.48
Shell-2	142.5	234.0	0.609	3.48
Shell-3	168.1	234.0	0.718	3.48
Shell-4	168.1	234.0	0.718	3.48
Head-2	166.3	234.0	0.711	3.48

Theoretical Maximum Allowable test pressure is : 4.821 MPa

Stress ratios for Nozzle and Pad Materials (MPa):

Description	Pad/Nozzle	Ambient	Operating	Ratio
M1_24"	Nozzle	118.00	118.00	1.000
A1_6"	Nozzle	118.00	118.00	1.000
P1_2"	Nozzle	118.00	118.00	1.000
L7_2"	Nozzle	118.00	118.00	1.000
L8_2"	Nozzle	118.00	118.00	1.000
P2_2"	Nozzle	118.00	118.00	1.000
S1_12"	Nozzle	118.00	118.00	1.000
L3_2"	Nozzle	118.00	118.00	1.000
L4_2"	Nozzle	118.00	118.00	1.000
L5_2"	Nozzle	118.00	118.00	1.000
L6_2"	Nozzle	118.00	118.00	1.000
V1_3"	Nozzle	118.00	118.00	1.000
L1_2"	Nozzle	118.00	118.00	1.000
L2_2"	Nozzle	118.00	118.00	1.000
B2_6"	Nozzle	118.00	118.00	1.000
B1_2"	Nozzle	118.00	118.00	1.000

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (MPa):

Description	Ambient	Operating	Ratio
Head-1	138.00	138.00	1.000
Shell-1	138.00	138.00	1.000
Shell-2	138.00	138.00	1.000
Shell-3	138.00	138.00	1.000
Shell-4	138.00	138.00	1.000
Head-2	138.00	138.00	1.000

Minimum 1.000

Hoop Stress in Nozzle Wall during Pressure Test (MPa):

Description	Ambient	Operating	Ratio
M1_24"	39.38	184.50	0.213
A1_6"	29.08	184.50	0.158
P1_2"	12.33	184.50	0.067
L7_2"	12.33	184.50	0.067
L8_2"	12.33	184.50	0.067
P2_2"	12.33	184.50	0.067
S1_12"	46.25	184.50	0.251
L3_2"	12.33	184.50	0.067
L4_2"	12.33	184.50	0.067
L5_2"	12.33	184.50	0.067
L6_2"	12.33	184.50	0.067
V1_3"	14.49	184.50	0.079
L1_2"	12.33	184.50	0.067
L2_2"	12.33	184.50	0.067
B2_6"	29.08	184.50	0.158
B1_2"	7.36	184.50	0.040

Elements Suitable for Test Pressure.

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Hydrostatic Test Pressure Results:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	3.457 MPa
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	3.250 MPa
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	3.692 MPa
UG100	= 1.10 * M.A.W.P. * Sa/S	2.925 MPa
PED	= max(1.43*DP, 1.25*DP*ratio)	3.575 MPa
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	3.575 MPa
App 27-4	= M.A.W.P.	2.659 MPa

UG-99(b), Test Pressure Calculation:
 = Test Factor • MAWP • Stress Ratio
 = 1.3 • 2.66 • 1
 = 3.457 MPa

Vertical Test performed per: UG-99b

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (MPa):

From To	Stress	Allowable	Ratio	Pressure
Head-1	138.1	234.0	0.590	3.48
Shell-1	180.5	234.0	0.771	3.48
Shell-2	151.3	234.0	0.647	3.48
Shell-3	180.5	234.0	0.771	3.48
Shell-4	180.5	234.0	0.771	3.48
Head-2	178.4	234.0	0.762	3.48

Theoretical Maximum Allowable test pressure is : 4.489 MPa

Stress ratios for Nozzle and Pad Materials (MPa):

Description	Pad/Nozzle	Ambient	Operating	Ratio
M1_24"	Nozzle	118.00	118.00	1.000
A1_6"	Nozzle	118.00	118.00	1.000
P1_2"	Nozzle	118.00	118.00	1.000
L7_2"	Nozzle	118.00	118.00	1.000
L8_2"	Nozzle	118.00	118.00	1.000
P2_2"	Nozzle	118.00	118.00	1.000
S1_12"	Nozzle	118.00	118.00	1.000
L3_2"	Nozzle	118.00	118.00	1.000
L4_2"	Nozzle	118.00	118.00	1.000
L5_2"	Nozzle	118.00	118.00	1.000
L6_2"	Nozzle	118.00	118.00	1.000
V1_3"	Nozzle	118.00	118.00	1.000
L1_2"	Nozzle	118.00	118.00	1.000
L2_2"	Nozzle	118.00	118.00	1.000
B2_6"	Nozzle	118.00	118.00	1.000
B1_2"	Nozzle	118.00	118.00	1.000

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (MPa):

Description	Ambient	Operating	Ratio
Head-1	138.00	138.00	1.000
Shell-1	138.00	138.00	1.000
Shell-2	138.00	138.00	1.000
Shell-3	138.00	138.00	1.000
Shell-4	138.00	138.00	1.000
Head-2	138.00	138.00	1.000

Minimum 1.000

Hoop Stress in Nozzle Wall during Pressure Test (MPa):

Description	Ambient	Operating	Ratio
M1_24"	39.38	184.50	0.213
A1_6"	29.08	184.50	0.158
P1_2"	12.33	184.50	0.067
L7_2"	12.33	184.50	0.067
L8_2"	12.33	184.50	0.067
P2_2"	12.33	184.50	0.067
S1_12"	46.25	184.50	0.251
L3_2"	12.33	184.50	0.067
L4_2"	12.33	184.50	0.067
L5_2"	12.33	184.50	0.067
L6_2"	12.33	184.50	0.067
V1_3"	14.49	184.50	0.079
L1_2"	12.33	184.50	0.067
L2_2"	12.33	184.50	0.067
B2_6"	29.08	184.50	0.158
B1_2"	7.36	184.50	0.040

Elements Suitable for Test Pressure.

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External Pressure Calculation Results Summary:

ASME Code, Section VIII Division 1, 2025

External Pressure Calculations:

From	To	Section Length mm	Outside Diameter mm	Corroded Thickness mm	Factor A	Factor B MPa
10	20	No Calc	2156	26.5	0.0017071	100.043
20	30	10150	2144	20.5	0.00023878	23.873
30	40	10150	2152	24.5	0.0003105	31.0426
40	50	10150	2144	20.5	0.00023878	23.873
50	60	10150	2144	20.5	0.00023878	23.873
60	70	No Calc	2144	20.5	0.001328	93.5433

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure MPa	External M.A.W.P. MPa
10	20	28	No Calc	...	1.37
20	30	22	No Calc	...	0.3
30	40	26	No Calc	...	0.47
40	50	22	No Calc	...	0.3
50	60	22	No Calc	...	0.3
60	70	22	No Calc	...	0.99

Minimum

0.3

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm	Allowable Length Bet. Stiffeners mm	Ring Inertia Required mm^4	Ring Inertia Available mm^4
10	20	No Calc	No Calc	No Calc	No Calc
20	30	10150	No Calc	No Calc	No Calc
30	40	10150	No Calc	No Calc	No Calc
40	50	10150	No Calc	No Calc	No Calc
50	60	10150	No Calc	No Calc	No Calc
60	70	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

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Element and Detail Weights:

From	To	Element Metal Wgt. kgm	Element ID Volume ltr	Corroded Metal Wgt. kgm	Corroded ID Volume ltr	Extra due Misc % kgm
10	20	1524.61	1385.69	1457.35	1391.39	106.723
20	30	3410.12	10392.7	3179.86	10422.4	238.709
30	40	3364.77	8660.57	3172.88	8685.33	235.534
40	50	2841.76	8660.57	2649.88	8685.33	198.924
50	60	1932.4	5889.19	1801.92	5906.02	135.268
60	70	1157.41	1385.69	1090.63	1391.39	81.0184
Total		14231	36374.39	13352	36481.86	996

Weight of Details:

From	Type	Weight of Detail kgm	X Offset, Dtl. Cent. mm	Y Offset, Dtl. Cent. mm	Z Offset Dtl. Cent. mm	Description
10	Ins1	11.6134	-237.5	...	...	Ins: 10
10	Noz1	862.197	-525	...	...	M1_24"
20	Sad1	348.434	1450	1309	...	Left Saddle
20	Plat	3957.57	4900	1148.2	...	PLAT:[1 OF 1]
20	Ins1	32.9725	1500	...	...	Ins: 10
20	Noz1	54.3748	1500	1124.54	...	A1_6"
20	Noz1	9.382	2500	1072.52	...	P1_2"
20	Noz1	9.382	2700	536.259	928.827	L7_2"
20	Noz1	9.382	2700	-536.259	928.827	L8_2"
20	Wght	500	2000	...	...	Lifting lug
20	Wght	85	1500	...	...	Half ope pipe_A
30	Ins1	27.5778	1250	...	...	Ins: 10
30	Noz1	9.382	500	1072.52	...	P2_2"
30	Noz1	148.378	2000	1202.4	...	S1_12"
30	Noz1	7.21412	2000	536.259	928.827	L3_2"
30	Noz1	9.382	2000	-536.259	928.827	L4_2"
30	Noz1	9.382	1000	536.259	928.827	L5_2"
30	Noz1	9.382	1000	-536.259	928.827	L6_2"
30	Wght	140	...	...	...	Clip_wt
40	Ins1	27.4771	1250	...	...	Ins: 10
40	Noz1	17.4042	2000	1084.72	...	V1_3"
40	Noz1	9.382	1000	536.259	928.827	L1_2"
40	Noz1	9.382	1000	-536.259	928.827	L2_2"
50	Sad1	348.434	250	1309	...	New Saddle
50	Ins1	18.6844	850	...	...	Ins: 10
50	Noz1	54.3748	1200	-1124.54	...	B2_6"
50	Noz1	11.8115	1000	1075.4	...	B1_2"
60	Ins1	11.191	287.5	...	...	Ins: 10

Total Weight of Each Detail Type:

Saddles	696.9
Platforms	3957.6
Insulation	129.5
Nozzles	1240.2
Weights	725.0

Sum of the Detail Weights 6749.1 kgm

Weight Summation Results: (kgm)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	15227.3	15227.3	15227.3	15227.3	15227.3	15227.3
Saddles	696.9	696.9	696.9	696.9	696.9	696.9
Nozzles	1240.2	1240.2	1240.2	1240.2	1240.2	1240.2
Wld Weights	640.0	640.0	640.0	640.0	640.0	640.0

Platforms	...	...	3957.6	3957.6	3957.6	3957.6
Insulation	...	...	129.5	129.5	129.5	129.5
Empty Weights	...	...	...	85.0	85.0	...
Test Weights	...	85.0	...	...	...	...
Ope Weights	...	...	...	...	...	85.0
Test Liquid	...	36352.2	...	...	...	...
Totals	17804.3	54241.5	21891.4	21976.4	21976.4	21976.4

Field Installation Options:

Miscellaneous Weight Percent: 7.0 %

Note that the above value for the miscellaneous weight percent has been applied to the shells/heads/flange/tubesheets/tubes etc. in the weight calculations for metallic components.

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	17804.3 kgm
Shop Test Wt.	- Fabricated weight + water (full)	54241.5 kgm
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	21891.4 kgm
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	21976.4 kgm
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	21976.4 kgm
Operating Wt.	- Empty weight + operating liq. uncorroded	21976.4 kgm
Oper. Wt. + CA	- Corr wt. + operating liquid	21036.3 kgm
Field Test Wt.	- Empty weight + test medium (full)	58328.6 kgm

The Corroded Weight and thickness are used in the Horizontal Vessel Analysis (Ope Case) and Earthquake Load Calculations.

Surface Areas of Elements:

From	To	Outside Surface Area mm ²	Inside Surface Area mm ²
10	20	5435522	5110243
20	30	20206724	19792034
30	40	16901768	16493364
40	50	16838938	16493364
50	60	11450478	11215487
60	70	5358084	5110243

Total	76191520.0	74214736.0 mm ²
Total	76.2	74.2 m ²

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Nozzle Flange MAWP Results: (MPa & °C)

Nozzle Description	Flange Rating		Des. Temp	Class	Grade/Group	Equiv. Press	Max Pressure		
	Ope.	Ambient					UG-44	50%	DNV
M1_24"	3.97	4.26	85	300	GR 1.4	...	...	...	...
A1_6"	3.97	4.26	85	300	GR 1.4	3.191	2.766	3.971	2.766
P1_2"	4.76	5.10	85	300	GR 1.1	...	...	...	...
L7_2"	4.76	5.10	85	300	GR 1.1	...	...	...	...
L8_2"	4.76	5.10	85	300	GR 1.1	...	...	...	...
P2_2"	3.97	4.26	85	300	GR 1.4	...	...	...	...
S1_12"	3.97	4.26	85	300	GR 1.4	2.111	3.846	3.971	3.846
L3_2"	4.76	5.10	85	300	GR 1.1	...	...	...	...
L4_2"	4.76	5.10	85	300	GR 1.1	...	...	...	...
L5_2"	4.76	5.10	85	300	GR 1.1	...	...	...	...
L6_2"	4.76	5.10	85	300	GR 1.1	...	...	...	...
V1_3"	7.94	8.52	85	600	GR 1.4	4.383	7.531	7.943	7.531
L1_2"	4.76	5.10	85	300	GR 1.1	...	...	...	...
L2_2"	4.76	5.10	85	300	GR 1.1	...	...	...	...
B2_6"	3.97	4.26	85	300	GR 1.4	3.191	2.766	3.971	2.766
B1_2"	7.94	8.52	85	600	GR 1.4	5.665	6.249	7.943	6.249
Min Rating	2.766	4.260, [for Core Elements]					...	...	...

Selected Method for Derating ASME B16.5 Flange MAWP: ASME UG-44(b)

The UG-44(b) Method is based on the paper PVP 2013-97814. PV Elite uses the maximum loads from each load category to determine ME and FE. In many cases, the computed maximum allowable pressure will be greater than the flange rating. In these cases, the minimum of the rating from the table and the UG-44(b) method will be used. SA-193 B8 Cl. 2 bolts or ones with higher allowable stresses at the specified bolt size shall be used.

The axial force FE must put the nozzle in tension (-P for 107/297; +Fr for PD 5500 AnnexG) to be considered.

Nozzle Description	Gasket Dimension G mm	Fm Value	Resolved Moment ME N-mm	Axial Force FE N	MAWP PD MPa	Maximum Pressure Pmax MPa	Pass Fail
A1_6"	196.088	0.50	4725218	0	2.659	2.766	Pass
S1_12"	357.251	0.50	18900894	0	2.659	3.846	Pass
V1_3"	111.125	0.50	1181304	0	2.659	7.531	Pass
B2_6"	196.088	0.50	4725218	0	2.659	2.766	Pass
B1_2"	77.851	0.50	525024	0	2.659	6.249	Pass

Evaluating Flange on Nozzle [A16"]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot ((PR - PD) + FM \cdot PR)$$

$$= 16(4725218) + 4 \cdot \text{abs}(0) \cdot 196.088 \leq \pi \cdot 196.088^3 \cdot ((3.971 - 2.659) + .500 \cdot 3.971)$$

$$= 0.75603E+08 \leq 0.78135E+08 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle A16" [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(3.971, 0.500 \cdot 3.971 + 3.971 - 75603448.0 / (\pi \cdot 196.088^3))$$

$$= \min(3.971, 2.766)$$

$$= 2.766 \text{ MPa}$$

Evaluating Flange on Nozzle [S112"]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot ((PR - PD) + FM \cdot PR)$$

$$= 16(18900894) + 4 \cdot \text{abs}(0) \cdot 357.251 \leq \pi \cdot 357.251^3 \cdot ((3.971 - 2.659) + .500 \cdot 3.971)$$

$$= 0.30241E+09 \leq 0.47251E+09 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle S112" [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(3.971, 0.500 \cdot 3.971 + 3.971 - 302414208.0 / (\pi \cdot 357.251^3))$$

$$= \min(3.971, 3.846)$$

$$= 3.846 \text{ MPa}$$



Evaluating Flange on Nozzle [V13"]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR]$$

$$= 16(1181304) + 4 \cdot \text{abs}(0) 111.125 \leq \pi \cdot 111.125^3 [(7.943 - 2.659) + .500 \cdot 7.943]$$

$$= 0.18901E+08 \leq 0.39910E+08 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle V13" [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(7.943, 0.500 \cdot 7.943 + 7.943 - 18900834.0 / (\pi \cdot 111.125^3))$$

$$= \min(7.943, 7.531)$$

$$= 7.531 \text{ MPa}$$

Evaluating Flange on Nozzle [B26"]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR]$$

$$= 16(4725218) + 4 \cdot \text{abs}(0) 196.088 \leq \pi \cdot 196.088^3 [(3.971 - 2.659) + .500 \cdot 3.971]$$

$$= 0.75603E+08 \leq 0.78135E+08 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle B26" [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(3.971, 0.500 \cdot 3.971 + 3.971 - 75603448.0 / (\pi \cdot 196.088^3))$$

$$= \min(3.971, 2.766)$$

$$= 2.766 \text{ MPa}$$

Evaluating Flange on Nozzle [B12"]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR]$$

$$= 16(525024) + 4 \cdot \text{abs}(0) 77.851 \leq \pi \cdot 77.851^3 [(7.943 - 2.659) + .500 \cdot 7.943]$$

$$= 0.84004E+07 \leq 0.13723E+08 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle B12" [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(7.943, 0.500 \cdot 7.943 + 7.943 - 8400381.0 / (\pi \cdot 77.851^3))$$

$$= \min(7.943, 6.249)$$

$$= 6.249 \text{ MPa}$$

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

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Wind Load Calculation Results:

Input Values:

Wind Design Code	ASCE/SEI 7-22
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed	[V] 180 km/hr
Surface Roughness Category	D: Flat, unobstructed
Type of Surface	Moderately Smooth
Base Elevation	300 mm
Percent Wind for Hydrotest	33.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm
Distance Upwind of Crest Lh	0 mm
Distance from Crest to the Vessel x	0 mm
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm

Wind Analysis Results

The importance factor is not used in the calculation of wind pressure vs. elevation for the selected wind code.

Static Gust-Effect Factor, Operating Case [G]:
 $= \min(0.85, 0.925((1 + 1.7 \cdot g_Q \cdot I_{zbar} \cdot Q) / (1 + 1.7 \cdot g_V \cdot I_{zbar})))$
 $= \min(0.85, 0.925((1 + 1.7 \cdot 3.4 \cdot 0.19 \cdot 0.98) / (1 + 1.7 \cdot 3.4 \cdot 0.19)))$
 $= \min(0.85, 0.91)$
 $= 0.850$

Natural Frequency of Vessel (Operating)	33.000 Hz
Natural Frequency of Vessel (Empty)	33.000 Hz
Natural Frequency of Vessel (Test)	33.000 Hz

User Entered Importance Factor is	1.000
Force Coefficient	[Cf] 0.562
Structure Height to Diameter ratio	4.723

This is classified as a rigid structure. Static analysis performed.

Sample Calculation for the First Element:

The ASCE code performs calculations in Imperial Units. The wind pressure is therefore computed in these units.

Value of $[\alpha]$ and $[Z_g]$:
 Exposure Category: D from Table 26.11-1
 $\alpha = 11.5 : Z_g = 589788 \text{ mm}$

Effective Height $[z]$:
 $= \text{Centroid Height} + \text{Vessel Base Elevation}$
 $= 1600 + 300 = 1900 \text{ mm}$
 $= 6.23 \text{ ft. Imperial Units}$

Velocity Pressure coefficient evaluated at height z $[K_z]$:
 Because z (6.23 ft.) $< 15 \text{ ft.}$
 $= 2.41(15 / Z_g)^{2/\alpha}$
 $= 2.41(15/1935)^{2/11.5}$
 $= 1.04$

Type of Hill: No Hill

Wind Directionality Factor $[K_d]$:
 $= 1, \text{ per Table 26.6-1}$

As there is No Hill Present: $[K_{zt}]$:
 $K_1 = 0, K_2 = 0, K_3 = 0$



Topographical Factor [Kzt]:

$$= (1 + K1 \cdot K2 \cdot K3)^2$$

$$= (1 + 0 \cdot 0 \cdot 0)^2$$

$$= 1$$

Velocity Pressure evaluated at height z, Imperial Units [qz]:

$$= \max(16, 0.00256 \cdot Kz \cdot Kzt \cdot Kd \cdot Ke \cdot V(\text{mph})^2)$$

$$= \max(16, 0.00256 \cdot 1.04 \cdot 1 \cdot 1 \cdot 1 \cdot 112^2)$$

$$= 33.1 \text{ psf [1.59] kPa}$$

Force on the first element [F]:

$$= qz \cdot G \cdot Cf \cdot \text{WindArea}$$

$$= 33.1 \cdot 0.85 \cdot 0.56 \cdot 12.7$$

$$= 200 \text{ lbs. [891] N}$$

Element	Hgt (z) mm	K1	K2	K3	Kz	Kzt	qz kPa
Head-1	1900.0	0.000	0.000	0.000	1.035	1.000	1.587
Shell-1	1900.0	0.000	0.000	0.000	1.035	1.000	1.587
Shell-2	1900.0	0.000	0.000	0.000	1.035	1.000	1.587
Shell-3	1900.0	0.000	0.000	0.000	1.035	1.000	1.587
Shell-4	1900.0	0.000	0.000	0.000	1.035	1.000	1.587
Head-2	1900.0	0.000	0.000	0.000	1.035	1.000	1.587

Platform Load Calculations

ID	Wind Area mm ²	Elevation mm	Pressure kPa	Force N	Cf
PLAT:[1 OF 1]	22779200.00	1900.00	1.59	36152.83	2.00

Wind Loads on Masses/Equipment/Piping

ID	Wind Area mm ²	Elevation mm	Pressure kPa	Force N
Lifting lug	0.00	1900.00	1.59	0.00
Half ope pipe	0.00	1900.00	1.59	0.00
Clipwt	0.00	1900.00	1.59	0.00

Wind Load Calculation:

From	To	Wind Height mm	Wind Diameter mm	Wind Area mm ²	Wind Pressure kPa	Element Wind Load N
10	20	1900	2683.2	1175557	1.58714	980.484
20	30	1900	2668.8	8006400	1.58714	46445.9
30	40	1900	2678.4	6696000	1.58714	5584.86
40	50	1900	2668.8	6672000	1.58714	5564.84
50	60	1900	2668.8	4536960	1.58714	3784.09
60	70	1900	2668.8	1158768	1.58714	966.481

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

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Earthquake Load Calculation Results:

Earthquake Load Calculation:

Input Values:

Seismic Design Code		ASCE/SEI 7-22	
Importance Factor		1.500	
Force Modification Factor R		3.000	
Site Class		D	
Component Elevation Ratio	z/h	0.000	
Amplification Factor	Ap	1.000	
Force Factor		0.000	
Consider Vertical Acceleration		No	
Minimum Acceleration Multiplier		0.000	
User Value of Sds (used if > 0)		0.000	
User Value of Sd1 (used if > 0)		0.000	
Component resonance duct. fact.	[CAR]	0.000	
Component strength factor	[Rpo]	0.000	
Overstrength factor	[Omega0]	0.000	
Response Acceleration Factor	[SMS]	0.326	
Response Acceleration Factor	[SM1]	0.139	
Building Importance Factor	[Ie]	1.000	
Site Latitude		0.000	deg
Site Longitude		0.000	deg

Seismic Analysis Results:

$$SDS = 2/3 * Sms = 2/3 * 0.33 = 0.22$$

$$SD1 = 2/3 * Sm1 = 2/3 * 0.14 = 0.093$$

Check Approximate Fundamental Period from 12.8-7 [Ta]:
 = $C_t \cdot h_{nx}$ where $C_t = 0.020$, $x = 0.75$ and h_n = Structural Height (ft.)
 = $0.020(8.690.75)$
 = 0.101 seconds

The Coefficient C_u from Table 12.8-1 is : 1.700

Fundamental Period (1/Frequency) [T]:
 = (1/Natural Frequency) = (1/33)
 = 0.030

Check the Value of T which is the smaller of $C_u \cdot T_a$ and T:
 = Minimum Value of ($1.7 \cdot 0.1$, 0.03) per 12.8.2
 = 0.030
 As the time period is < 0.06 second, use section 15.4.2.

Base Shear per equation 15.4-5, [V]:
 = $0.3 \cdot Sds \cdot W \cdot I$
 = $0.3 \cdot 0.22 \cdot 199449 \cdot 1.5$
 = 19506.100 N

Earthquake Load Calculation:

From	To	Earthquake Height mm	Earthquake Weight N	Element Ope Load N
10	20	1050	24931.1	2438.26
20	30	1050	24931.1	2438.26
20	30	1050	24931.1	2438.26
30	40	1050	24931.1	2438.26
40	50	1050	24931.1	2438.26
50	0	1050	24931.1	2438.26
50	60	1050	24931.1	2438.26
60	70	1050	24931.1	2438.26

The Earthquake Weight does not include the weight of the saddles.





Shop/Field Installation Options :

Platform(s) installed in the Shop.
Insulation is installed in the Shop.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Saddles	4899.999 mm
Center of Gravity of the Platforms	4950.000 mm
Center of Gravity of the Insulation	4882.777 mm
Center of Gravity of the Nozzles	1220.567 mm
Center of Gravity of the Added Weights (Operating)	2184.482 mm
Center of Gravity of the Added Weights (Empty)	2184.482 mm
Center of Gravity of Bare Shell New and Cold	4745.190 mm
Center of Gravity of Bare Shell Corroded	4735.185 mm
Vessel CG in the Operating Condition	4486.855 mm
Vessel CG in the Fabricated (Shop/Empty) Condition	4504.411 mm
Vessel CG in the Test Condition	4751.000 mm

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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Operating Case

Note:

Wear Pad Width (230.00) is less than $1.56 \cdot \sqrt{r_m \cdot t}$
 and less than $2a$. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

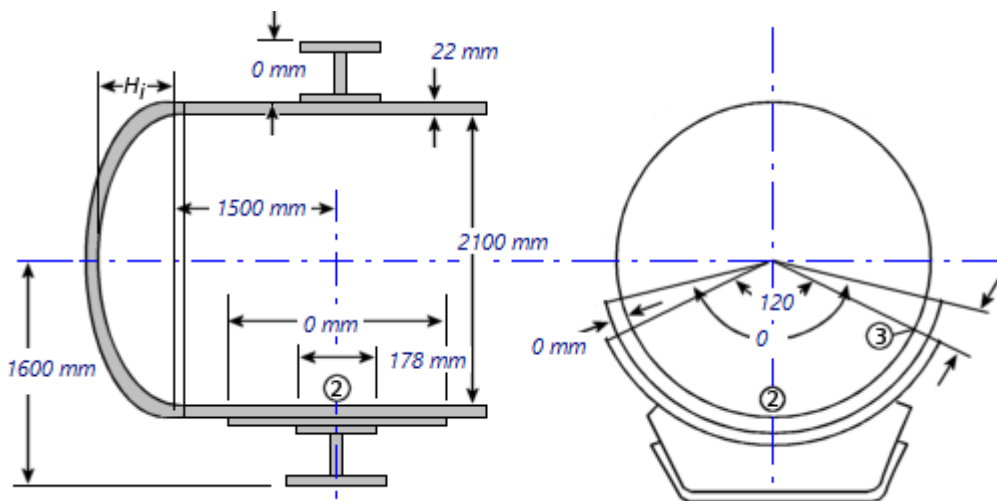
$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(178 + 1.56 \cdot \sqrt{1062 \cdot 20.5}, 2 \cdot 1500)$$

$$= 408.1509 \text{ mm}$$

Input and Calculated Values:

Vessel Mean Radius	Rm	1061.75	mm
Shell Thickness used in this Case	t	22.000	mm
Stiffened Vessel Length per 4.15.6	L	9800.00	mm
Distance from Saddle to Vessel tangent	a1 or a	1500.00	mm
Saddle Width	b1 or b	178.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Inside or Mean Depth of Head	Hi or hm	539.75	mm
Shell Allowable Stress used in Calculation		138.00	MPa
Head Allowable Stress used in Calculation		138.00	MPa
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1600.00	mm
Coefficient of Friction	mu	0.60	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case	193439.36	N
Pressure used in Analysis	P	2.500 MPa

Horizontal Vessel Analysis Results:	Actual MPa	Allowable MPa
Long. Stress at Top of Midspan	62.55	138.00
Long. Stress at Bottom of Midspan	66.93	138.00
Long. Stress at Top of Saddles	71.66	138.00
Long. Stress at Bottom of Saddles	60.91	138.00
Tangential Shear in Shell	6.73	110.40
Circ. Stress at Horn of Saddle	42.27	172.50

Circ. Compressive Stress in Shell	1.76	138.00
-----------------------------------	------	--------

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= Ftr(Ft/Num\ of\ Saddles + Z\ Force\ Load) \cdot B / E$
 $= 3 (63327/2 + 0) \cdot 1600/1874$
 $= 81098.0\ N$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(Fl,wind; Sum\ of\ X\ Forces) \cdot B / Ls$
 $= \max(18001, 0) \cdot 1600/6800$
 $= 4235.5\ N$

Saddle Reaction Force due to Earthquake Fl [Fsl]:
 $= \max(Fl,eq; Sum\ of\ X\ Forces) \cdot B / Ls$
 $= \max(19506, 0) \cdot 1600/6800$
 $= 4589.7\ N$

Saddle Reaction Force due to Earthquake Ft [Fst]:
 $= Ftr(Ft/Num\ of\ Saddles + Z\ Force\ Load) \cdot B / E$
 $= 3 (19506/2 + 0) \cdot 1600/1874$
 $= 24980.1\ N$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= Saddle\ Load + \max(Fwl, Fwt, Fsl, Fst)$
 $= 112341 + \max(4235, 81098, 4590, 24980)$
 $= 193439.4\ N$

Longitudinal Wind Force [Fl]:
 $= WindScalar \cdot WindPress(Platform\ Area + (\pi/4(OD \cdot WindDiaMult)^2))$
 $= 1.1 \cdot 1587 (4.72 + \pi/4(2.22 \cdot 1.2)^2)$
 $= 18001.596\ N$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	196856.11	N
Transverse Shear Load	31663.33	N
Longitudinal Shear Load	19506.10	N

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:
 $= \min(2(ShellID/2 + t + WearPadThickness)\sin(\theta/2), 2 \cdot Rm)$
 $= \min(2(2100/2 + 22 + 10)\sin(120/2), 2 \cdot 1062)$
 $= 1874.079\ mm$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	

Note: Dimension a is greater than or equal to $Rm/2$.

Moment per Equation 4.15.1 [M1]:
 $= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L))/(1 + (4hm)/3L)]$
 $= -193439 \cdot 1500 [1 - (1 - 1500/9800 + (1062^2 - 540^2)/(2 \cdot 1500 \cdot 9800))/(1 + (4 \cdot 540)/(3 \cdot 9800))]$
 $= -53559572.0\ N \cdot mm$

Moment per Equation 4.15.2 [M2]:
 $= Q \cdot L/4 [(1 + 2(Rm^2 - hm^2)/(L^2))/(1 + (4hm)/(3L)) - 4a/L]$
 $= 193439 \cdot 9800/4 [(1 + 2(1062^2 - 540^2)/(9800^2))/(1 + (4 \cdot 540)/(3 \cdot 9800)) - 4 \cdot 1500/9800]$
 $= 159095968.0\ N \cdot mm$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:
 $= P \cdot Rm/(2t) - M2/(\pi \cdot Rm^2 t)$
 $= 2.5 \cdot 1062/(2 \cdot 20.5) - 159095968/(\pi \cdot 1062^2 \cdot 20.5)$
 $= 62.55\ MPa$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t)$$

$$= 2.5 \cdot 1062 / (2 \cdot 20.5) + 159095968 / (\pi \cdot 1062^2 \cdot 20.5)$$

$$= 66.93 \text{ MPa}$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_3]:

$$= P \cdot R_m / (2t) - M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 2.5 \cdot 1062 / (2 \cdot 20.5) - 53559572 / (0.11 \cdot \pi \cdot 1062^2 \cdot 20.5)$$

$$= 71.66 \text{ MPa}$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_4]:

$$= P \cdot R_m / (2t) + M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 2.5 \cdot 1062 / (2 \cdot 20.5) + 53559572 / (0.19 \cdot \pi \cdot 1062^2 \cdot 20.5)$$

$$= 60.91 \text{ MPa}$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 193439 (9800 - 2 \cdot 1500) / (9800 + (4 \cdot 540 / 3))$$

$$= 125040.8 \text{ N}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot 125041 / (1062 \cdot 20.5)$$

$$= 6.73 \text{ MPa}$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{R_m \cdot t}$$

$$= 0.78 \cdot \sqrt{1062 \cdot 20.5}$$

$$= 115.075 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K_5 \cdot Q \cdot k / (t(b + X_1 + X_2))$$

$$= -0.76 \cdot 193439 \cdot 0.1 / (20.5 (178 + 115 + 115))$$

$$= -1.76 \text{ MPa}$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.22) [σ_7]:

$$= -Q / (4 \cdot t(b + X_1 + X_2)) - 3 \cdot K_7 \cdot Q / (2 \cdot t^2)$$

$$= -193439 / (4 \cdot 20.5 (178 + 115 + 115)) - 3 \cdot 0.053 \cdot 193439 / (2 \cdot 20.5^2)$$

$$= -42.27 \text{ MPa}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(178 + 1.56 \cdot \sqrt{1062 \cdot 20.5}, 2 \cdot 1500)$$

$$= 408.15 \text{ mm}$$

Distance between Saddle Supports [Ls]:

$$= 6800.0 \text{ mm}$$

Free Un-Restrained Thermal Expansion between the Saddles [Exp]:

$$= \alpha \cdot L_s (\text{Design Temperature} - \text{Ambient Temperature})$$

$$= 0.12005\text{E-}04 \cdot 6800 (85 - 21.1)$$

$$= 5.215 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1880.0000	mm
Baseplate Thickness	Bpthk	14.0000	mm
Baseplate Width	Bpwid	204.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	14.0000	mm
Web Thickness	Webtk	14.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	207.0	MPa
Height of Web at Center	Hw,c	492.0	mm
Friction Coefficient	mu	0.600	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth}) t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(1052 \cdot 20.5)} + 230) 20.5^3) / 12$$

$$= 329554.938 \text{ mm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$$

$$= 329555 + 9410 (171 - 10.2)^2$$

$$= 243151888.000 \text{ mm}^4$$

	B mm	D mm	Y mm	A mm ²	AY mm ³	I + AD ² mm ⁴
Shell	459.0	20.5	10.2	9410.3	96455.2	0.243E+09
Wearplate	230.0	10.0	25.5	2300.0	58650.0	0.486E+08
Web	14.0	504.0	282.5	7056.0	1993319.6	0.237E+09
BasePlate	204.0	14.0	541.5	2856.0	1546523.9	0.392E+09
Totals	...	...	...	21622.3	3694948.5	0.921E+09

Distance to Centroid [C1]:

$$= AY / A$$

$$= 3694948 / 21622$$

$$= 170.886 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 193439$$

$$= 39369.1055 \text{ N}$$

Tension Stress, St = (Fh/As) = 3.2241 MPa
 Allowed Stress, Sa = 0.6 * Yield Str = 124.1978 MPa

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\text{theta}/2) / (\text{theta}/2 \text{ in radians})$$

$$= 1600 - 1052 \cdot \sin(120/2) / 1.05$$

$$= 730.416 \text{ mm}$$

Bending Moment, M = Fh * d = 28767500.0000 N-mm

Bending Stress, Sb = (M * C1 / I) = 5.3337 MPa
 Allowed Stress, Sa = 2/3 * Yield Str = 137.9975 MPa

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(193439 + 3417) 204) / (4 \cdot 1880 \cdot 138)}$$

$$= 10.775 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle} / 2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120/2) (1050 + 22 + 10)$$

$$= 1874.079 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 1874 / (3 - 1)$$

$$= 937.039 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot Bpwid / 2$$

$$= 937 \cdot 204/2$$

$$= 95578.023 \text{ mm}^2$$

Bearing Pressure [Bp]:

$$= (Q + SaddleWt) / (BasePlateLength \cdot BasePlateWidth)$$

$$= 196856 / (1880 \cdot 204)$$

$$= 0.513 \text{ N/mm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 95578 \cdot 0.51$$

$$= 49059.031 \text{ N}$$

Area of the Rib and Web [Ar]:

$$= Rib \text{ Area} + Web \text{ Area}$$

$$= 2296 + 6559$$

$$= 8855.276 \text{ mm}^2$$

Compressive Stress [Sc]:

$$= P/Ar$$

$$= 49059/8855$$

$$= 5.541 \text{ MPa}$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	14.0	178.0	...	2492.0	...	0.658E+07

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 178 - 14$$

$$= 164.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/8855$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 178/2$$

$$= 89.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(6579704/8855)}$$

$$= 27.259 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\text{theta}/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1600 - \cos(120/2) (1050 + 22 + 10) - 14$$

$$= 1045.000 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 202713/207)}$$

$$= 139.035$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 1045/27.3$$

$$= 38.337$$

Bending Moment [Rm]:

$$= FI / (2 \cdot Bplen) \cdot e \cdot L / 2$$

$$= 19506 / (2 \cdot 1880) \cdot 937 \cdot 1045/2$$

= 2540989.000 N-mm

Compressive Allowable, $KL/r < C_c$ (38.3 < 139) per AISC E2-1 [Sca]:

$$= \left(1 - \frac{(KL/r)^2}{(2 \cdot C_c^2)} \right) F_y / \left(\frac{5}{3} + 3 \cdot \frac{(KL/r)}{(8 \cdot C_c)} - \frac{(KL/r)^3}{(8 \cdot C_c^3)} \right)$$

$$= \left(1 - \frac{(38.3)^2}{(2 \cdot 139^2)} \right) 207 / \left(\frac{5}{3} + 3 \cdot \frac{(38.3)}{(8 \cdot 139)} - \frac{(38.3)^3}{(8 \cdot 139^3)} \right)$$

$$= 112.7 \text{ MPa}$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 5.54 / 113 + (2540989 \cdot 89 / 6579711) / 138$$

$$= 0.298$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	14.0	164.0	0.0	2296.0	0.0	0.658E+07
Web	937.0	14.0	0.0	13118.6	0.0	0.214E+06
Totals	...	...	...	15414.6	...	0.679E+07

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0 / 15415$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 178 / 2$$

$$= 89.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - B_{pthk}$$

$$= 1600 - \cos(1082/2) (14 + 0 + 0) - 0$$

$$= 504.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(6790772 / 15415)}$$

$$= 20.989 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 504 / 21$$

$$= 24.012$$

Unit Force [Force,u]:

$$= F / (2 \cdot \text{Baseplate Length})$$

$$= 19506 / (2 \cdot 1880)$$

$$= 5.188 \text{ N/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 5.19 \cdot 937 \cdot 504$$

$$= 2451020.750 \text{ N-mm}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 2451020 / 76301$$

$$= 32.113 \text{ MPa}$$

Compressive Allowable, $KL/r < C_c$ (24 < 139) per AISC E2-1 [Sca]:

$$= \left(1 - \frac{(KL/r)^2}{(2 \cdot C_c^2)} \right) F_y / \left(\frac{5}{3} + 3 \cdot \frac{(KL/r)}{(8 \cdot C_c)} - \frac{(KL/r)^3}{(8 \cdot C_c^3)} \right)$$

$$= \left(1 - \frac{(24)^2}{(2 \cdot 139^2)} \right) 207 / \left(\frac{5}{3} + 3 \cdot \frac{(24)}{(8 \cdot 139)} - \frac{(24)^3}{(8 \cdot 139^3)} \right)$$

$$= 117.8 \text{ MPa}$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C_1 / I) / S_{ba}$$

$$= 6.36/118 + (2451020 \cdot 89/6790772)/138$$

$$= 0.287$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-36
Bolt Allowable Stress	Stba	149.94 MPa
Bolt Corrosion Allowance	Bca	3.0 mm
Distance from Bolts to Edge	Edgedis	50.0 mm
Nominal Bolt Diameter	Bnd	30.0000 mm
Thread Series	Series	TEMA Metric
BasePlate Allowable Stress	S	108.00 MPa
Area Available in a Single Bolt	BlArea	292.9056 mm ²
Saddle Load QO (Weight)	QO	115758.1 N
Saddle Load QL (Wind/Seismic contribution)	QL	4589.7 N
Maximum Transverse Force	Ft	31663.3 N
Maximum Longitudinal Force	Fl	19506.1 N
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [τ_{b,l}]:

$$= Fl / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 19506 / (293 \cdot 4)$$

$$= 16.7 \text{ MPa. Must be less than } 99.9 \text{ MPa.}$$

Shear Stress in a Single Bolt, Transverse Direction [τ_{b,t}]:

$$= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 31663 / (293 \cdot 4)$$

$$= 27.0 \text{ MPa. Must be less than } 99.9 \text{ MPa.}$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (QO > QL --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= Fl / (\text{BoltShearAllowable} \cdot \text{Nbolts})$$

$$= 19506 / (99.9 \cdot 4)$$

$$= 48.8040 \text{ mm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot Ft + \text{Sum of X Moments}$$

$$= 1600 \cdot 31663 + 0$$

$$= 50681872.00 \text{ N-mm}$$

Eccentricity (e):

$$= Rmom / QO$$

$$= 50681872 / 115758$$

$$= 437.65 \text{ mm > Bplen/6 --> Uplift in Transverse direction}$$

f = Bplen / 2 - Edgedis

$$= 1880 / 2 - 50$$

$$= 889.99 \text{ mm}$$

Modular Ratio Of Steel/Concrete (n1):

$$= ES / EC$$

$$= 203402 / 21526$$

$$= 9.45$$

K1 = 3 (e - 0.5 • Bplen)

$$= 3 (438 - 0.5 \cdot 1880)$$

$$= -1507.06 \text{ mm}$$

K2 = 6 • n1 • At / Bpwid • (f + e)

$$= 6 \cdot 9.45 \cdot 586 / 204 \cdot (890 + 438)$$

$$= 216144.46 \text{ mm}^2$$

$$\begin{aligned} K3 &= -K2 \cdot (0.5 \cdot Bplen + f) \\ &= -216144 \cdot (0.5 \cdot 1880 + 890) \\ &= -395541603.85 \text{ mm}^3 \end{aligned}$$

Iteratively Solving for the Effective Bearing Length:

$$\begin{aligned} Y^3 + K1 \cdot Y^2 + K2 \cdot Y + K3 &= 0 \\ Y^3 + -1507 \cdot Y^2 + 216144 \cdot Y + -395541600 &= 0 \\ Y &= 1534.22 \text{ mm} \end{aligned}$$

$$\begin{aligned} \text{Num} &= (Bplen / 2 - Y / 3 - e) \\ &= (1880/2 - 1534/3 - 438) \\ &= -9.05 \end{aligned}$$

$$\begin{aligned} \text{Denom} &= (Bplen / 2 - Y / 3 + f) \\ &= (1880/2 - 1534/3 + 890) \\ &= 1318.58 \end{aligned}$$

$$\begin{aligned} \text{Total Bolt Tension Force [Tforce]:} \\ &= -QO \cdot \text{Num} / \text{Denom} \\ &= -115758 \cdot -9.05/1319 \\ &= 794.79 \text{ N} \end{aligned}$$

$$\begin{aligned} \text{Bolt Area Required due to Transverse Load [Bltareart]:} \\ &= \text{Tforce} / (\text{Stba} \cdot \text{Nbt}) \\ &= 795 / (150 \cdot 2) \\ &= 2.6505 \text{ mm}^2 \end{aligned}$$

$$\begin{aligned} \text{Required Area of a Single Bolt [Bltarear]:} \\ &= \max[\text{Bltarearl}, \text{Bltarears}, \text{Bltareart}] \\ &= \max[0, 48.8, 2.65] \\ &= 48.8040 \text{ mm}^2 \end{aligned}$$

Baseplate Thickness Calculation per D. Moss:

$$\begin{aligned} \text{Bearing Pressure (fc)} \\ &= 2(QO + \text{Tforce}) / (Y \cdot \text{Bpwid}) \\ &= 2(115758 + 795) / (1534 \cdot 204) \\ &= 0.74 \text{ MPa} \end{aligned}$$

$$\begin{aligned} \text{Distance from Baseplate Edge to the Web [ADIST]:} \\ &= (Bplen - \text{WebIngh}) / 2 \\ &= (1880 - 1829) / 2 \\ &= 25.4000 \text{ mm} \end{aligned}$$

$$\begin{aligned} \text{Overturning Moment due To Bolt Tension [Mt]:} \\ &= \text{Tforce} \cdot \text{Adist} \\ &= 795 \cdot 25.4 \\ &= 20195.85 \text{ N-mm} \end{aligned}$$

$$\begin{aligned} \text{Equivalent Bearing Pressure (f1):} \\ &= fc \cdot (Y - \text{Adist}) / Y \\ &= 0.74 \cdot (1534 - 25.4) / 1534 \\ &= 0.73 \text{ MPa} \end{aligned}$$

$$\begin{aligned} \text{Overturning Moment due to Bearing Pressure [Mc]:} \\ &= (\text{Adist}^2 \cdot \text{Bpwid} / 6) \cdot (f1 + 2 \cdot fc) \\ &= (25.4^2 \cdot 204 / 6) \cdot (0.73 + 2 \cdot 0.74) \\ &= 48761.50 \text{ N-mm} \end{aligned}$$

$$\begin{aligned} \text{Baseplate Required Thickness [Treq]:} \\ &= \sqrt{(6 \cdot \max(\text{Mt}, \text{Mc}) / (\text{Bpwid} \cdot \text{Sba}))} \\ &= \sqrt{(6 \cdot \max(20196, 48761) / (204 \cdot 162))} \\ &= 2.9749 \text{ mm} \end{aligned}$$

Note: Saddle and wear plate material are both specified as impact tested.

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Note:

Wear Pad Width (230.00) is less than $1.56 \cdot \sqrt{R_m \cdot t}$
 and less than $2a$. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

$$= \min(b + 1.56 \sqrt{ R_m \cdot t }, 2a)$$

$$= \min(178 + 1.56 \sqrt{ 1062 \cdot 20.5 }, 2 \cdot 1500)$$

$$= 408.1509 \text{ mm}$$

Input and Calculated Values:

Vessel Mean Radius	Rm	1061.75	mm
Shell Thickness used in this Case	t	22.000	mm
Stiffened Vessel Length per 4.15.6	L	9800.00	mm
Distance from Saddle to Vessel tangent	a1 or a	1500.00	mm
Saddle Width	b1 or b	178.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Inside or Mean Depth of Head	Hi or hm	536.75	mm
Shell Allowable Stress used in Calculation		138.00	MPa
Head Allowable Stress used in Calculation		138.00	MPa
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1600.00	mm
Coefficient of Friction	mu	0.00	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case		168205.42	N
Pressure used in Analysis	P	2.500	MPa

Horizontal Vessel Analysis Results:		Actual MPa	Allowable MPa
Long. Stress at Top of Midspan		62.83	138.00
Long. Stress at Bottom of Midspan		66.65	138.00
Long. Stress at Top of Saddles		70.74	138.00
Long. Stress at Bottom of Saddles		61.41	138.00
Tangential Shear in Shell		5.85	110.40
Circ. Stress at Horn of Saddle		36.76	172.50
Circ. Compressive Stress in Shell		1.53	138.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$$

$$= 3 (63327 / 2 + 0) \cdot 1600 / 1874$$

$$= 81098.0 \text{ N}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(F_{l, \text{wind}}, \text{Sum of X Forces}) \cdot B / L_s$$

$$= \max(18001, 0) \cdot 1600 / 6800$$

$$= 4235.5 \text{ N}$$

Saddle Reaction Force due to Earthquake Fl [Fsl]:

$$= \max(F_{l, \text{eq}}, \text{Sum of X Forces}) \cdot B / L_s$$

$$= \max(19506, 0) \cdot 1600 / 6800$$

$$= 4589.7 \text{ N}$$

Saddle Reaction Force due to Earthquake Ft [Fst]:

$$= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$$

$$= 3 (19506 / 2 + 0) \cdot 1600 / 1874$$

$$= 24980.1 \text{ N}$$

Load Combination Results for Q + Wind or Seismic [Q]:

= Saddle Load + max(Fwl, Fwt, Fsl, Fst)
 = 87107 + max(4235, 81098, 4590, 24980)
 = 168205.4 N

Longitudinal Wind Force [FI]:

= WindScalar • WindPress(Platform Area + ($\pi/4$ (OD • WindDiaMult)²))
 = 1.1 • 1587 (4.72 + $\pi/4$ (2.22 • 1.2)²)
 = 18001.596 N

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	171622.17	N
Transverse Shear Load	31663.33	N
Longitudinal Shear Load	19506.10	N

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

= min(2(ShellID/2 + t + WearPadThickness)sin($\theta/2$), 2•Rm)
 = min(2(2100/2 + 22 + 10)sin(120/2), 2•1062)
 = 1874.079 mm

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

= -Q•a [1 - (1 - a/L + (Rm²-hm²)/(2a•L))/(1+(4hm)/3L)]
 = -168205•1500 [1 - (1 - 1500/9800 + (1062²-537²)/(2•1500•9800))/(1+(4•537)/(3•9800))]
 = -46468648.0 N-mm

Moment per Equation 4.15.2 [M2]:

= Q•L/4((1+2(Rm²-hm²)/(L²))/(1+(4hm)/(3L)) - 4a/L)
 = 168205•9800/4((1+2(1062²-537²)/(9800²))/(1+(4•537)/(3•9800)) - 4•1500/9800)
 = 138516560.0 N-mm

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

= P • Rm/(2t) - M2/(π •Rm²t)
 = 2.5 • 1062/(2•20.5) - 138516560/(π •1062²•20.5)
 = 62.83 MPa

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

= P • Rm/(2t) + M2/(π • Rm² • t)
 = 2.5 • 1062/(2 • 20.5) + 138516560/(π • 1062² • 20.5)
 = 66.65 MPa

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_3]:

= P • Rm/(2t) - M1/(K1• π •Rm²t)
 = 2.5•1062/(2•20.5) - 46468648/(0.11• π •1062²•20.5)
 = 70.74 MPa

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_4]:

= P • Rm/(2t) + M1/(K1* • π • Rm² • t)
 = 2.5•1062/(2•20.5) + 46468648/(0.19• π •1062²•20.5)
 = 61.41 MPa

Maximum Shear Force in the Saddle (4.15.3) [T]:

= Q(L-2a)/(L+(4•hm/3))
 = 168205 (9800 - 2 • 1500)/(9800 + (4 • 537/3))
 = 108770.8 N

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

= K2 • T / (Rm • t)
 = 1.17 • 108771/(1062 • 20.5)
 = 5.85 MPa

Decay Length (4.15.20) [x1,x2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(1062 \cdot 20.5)}$$

$$= 115.075 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K_5 \cdot Q \cdot k / (t(b + X_1 + X_2))$$

$$= -0.76 \cdot 168205 \cdot 0.1 / (20.5 (178 + 115 + 115))$$

$$= -1.53 \text{ MPa}$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.22) [σ_7]:

$$= -Q / (4 \cdot t(b + X_1 + X_2)) - 3 \cdot K_7 \cdot Q / (2 \cdot t^2)$$

$$= -168205 / (4 \cdot 20.5 (178 + 115 + 115)) -$$

$$3 \cdot 0.053 \cdot 168205 / (2 \cdot 20.5^2)$$

$$= -36.76 \text{ MPa}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(178 + 1.56 \cdot \sqrt{(1062 \cdot 20.5)}, 2 \cdot 1500)$$

$$= 408.15 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1880.0000	mm
Baseplate Thickness	Bpthk	14.0000	mm
Baseplate Width	Bpwid	204.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	14.0000	mm
Web Thickness	Webtk	14.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	207.0	MPa
Height of Web at Center	Hw,c	492.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth}) t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(1052 \cdot 20.5)} + 230) 20.5^3) / 12$$

$$= 329554.938 \text{ mm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}} (C_1 - Y)^2$$

$$= 329555 + 9410 (171 - 10.2)^2$$

$$= 243151888.000 \text{ mm}^4$$

	B mm	D mm	Y mm	A mm ²	AY mm ³	I + AD ² mm ⁴
Shell	459.0	20.5	10.2	9410.3	96455.2	0.243E+09
Wearplate	230.0	10.0	25.5	2300.0	58650.0	0.486E+08
Web	14.0	504.0	282.5	7056.0	1993319.6	0.237E+09
BasePlate	204.0	14.0	541.5	2856.0	1546523.9	0.392E+09
Totals	...	...	...	21622.3	3694948.5	0.921E+09

Distance to Centroid [C1]:

$$= AY / A$$

$$= 3694948 / 21622$$

$$= 170.886 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

= 0.2035

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 168205$$

$$= 34233.4531 \text{ N}$$

$$\text{Tension Stress, } St = (Fh/As) = 2.8035 \text{ MPa}$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 124.1978 \text{ MPa}$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 1600 - 1052 \cdot \sin(120/2) / 1.05$$

$$= 730.416 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 25014814.0000 \text{ N-mm}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 4.6380 \text{ MPa}$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 137.9975 \text{ MPa}$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{3(168205 + 3417) \cdot 204 / (4 \cdot 1880 \cdot 138)}$$

$$= 10.061 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120/2) (1050 + 22 + 10)$$

$$= 1874.079 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 1874 / (3 - 1)$$

$$= 937.039 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 937 \cdot 204 / 2$$

$$= 95578.023 \text{ mm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 171622 / (1880 \cdot 204)$$

$$= 0.447 \text{ N/mm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 95578 \cdot 0.45$$

$$= 42770.414 \text{ N}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 2296 + 6559$$

$$= 8855.276 \text{ mm}^2$$

Compressive Stress [Sc]:

$$= P/Ar$$

$$= 42770/8855$$

$$= 4.830 \text{ MPa}$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	14.0	178.0	...	2492.0	...	0.658E+07

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 178 - 14$$

$$= 164.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/8855$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 178/2$$

$$= 89.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{6579704/8855}$$

$$= 27.259 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\text{theta}/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1600 - \cos(120/2) (1050 + 22 + 10) - 14$$

$$= 1045.000 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 202713/207)}$$

$$= 139.035$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 1045/27.3$$

$$= 38.337$$

Bending Moment [Rm]:

$$= FI / (2 \cdot \text{Bplen}) \cdot e \cdot L / 2$$

$$= 19506 / (2 \cdot 1880) \cdot 937 \cdot 1045/2$$

$$= 2540989.000 \text{ N-mm}$$

Compressive Allowable, $KL/r < Cc$ ($38.3 < 139$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (38.3)^2 / (2 \cdot 139^2)) 207 /$$

$$(5/3 + 3 \cdot (38.3) / (8 \cdot 139) - (38.3)^3 / (8 \cdot 139^3))$$

$$= 112.7 \text{ MPa}$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 4.83/113 + (2540989 \cdot 89/6579711) / 138$$

$$= 0.292$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	14.0	164.0	0.0	2296.0	0.0	0.658E+07
Web	937.0	14.0	0.0	13118.6	0.0	0.214E+06
Totals	...	...	...	15414.6	...	0.679E+07

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/15415$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 178/2$$

$$= 89.000 \text{ mm}$$

Length of Inner Rib [L]:
 = Saddle Height - Outside Radius - Bpthk
 = $1600 - \cos(1082/2)(14 + 0 + 0) - 0$
 = 504.000 mm

Radius of Gyration [r]:
 = $\sqrt{\text{Total Inertia} / \text{Total Area}}$
 = $\sqrt{6790772/15415}$
 = 20.989 mm

Slenderness ratio [KL/r]:
 = KL/r
 = $1 \cdot 504/21$
 = 24.012

Unit Force [Force,u]:
 = FI / (2 • Baseplate Length)
 = $19506 / (2 \cdot 1880)$
 = 5.188 N/mm

Moment at base of inner Rib [Mbase,c]:
 = Unit Force • e • L
 = $5.19 \cdot 937 \cdot 504$
 = 2451020.750 N-mm

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:
 = Bending Moment / Section Modulus
 = $2451020/76301$
 = 32.113 MPa

Compressive Allowable, KL/r < Cc (24 < 139) per AISC E2-1 [Sca]:
 = $(1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$
 = $(1 - (24)^2 / (2 \cdot 139^2)) 207 / (5/3 + 3 \cdot (24) / (8 \cdot 139) - (24^3) / (8 \cdot 139^3))$
 = 117.8 MPa

AISC Unity Check of Inside Ribs (must be ≤ 1)
 = $Sc/Sca + (Mbase,c \cdot C1/I) / Sba$
 = $5.55/118 + (2451020 \cdot 89/6790772) / 138$
 = 0.280

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-36
Bolt Allowable Stress	Stba	149.94 MPa
Bolt Corrosion Allowance	Bca	3.0 mm
Distance from Bolts to Edge	Edgedis	50.0 mm
Nominal Bolt Diameter	Bnd	30.0000 mm
Thread Series	Series	TEMA Metric
BasePlate Allowable Stress	S	108.00 MPa
Area Available in a Single Bolt	BlArea	292.9056 mm ²
Saddle Load Q0 (Weight)	Q0	90524.2 N
Saddle Load QL (Wind/Seismic contribution)	QL	4589.7 N
Maximum Transverse Force	Ft	31663.3 N
Maximum Longitudinal Force	Fl	19506.1 N
Saddle Bolted to Steel Foundation		No

Friction force due to thermal growth, [Ff]:
 = 69454.9 N

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b ,l]:
 = $\max(Ff, Fl) / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 = $69455 / (293 \cdot 4)$
 = 59.3 MPa. Must be less than 99.9 MPa.

Shear Stress in a Single Bolt, Transverse Direction [τ_b ,t]:
 = $Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$



$$= 31663 / (293 \cdot 4)$$

$$= 27.0 \text{ MPa. Must be less than } 99.9 \text{ MPa.}$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
 = 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
 = $F_l / (\text{BoltShearAllowable} \cdot N_{\text{bolts}})$
 = $19506 / (99.9 \cdot 4)$
 = 48.8040 mm^2

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
 = $B \cdot F_t + \text{Sum of X Moments}$
 = $1600 \cdot 31663 + 0$
 = 50681872.00 N-mm

Eccentricity (e):
 = R_{mom} / QO
 = $50681872 / 90524$
 = $559.64 \text{ mm} > B_{\text{plen}} / 6$ --> Uplift in Transverse direction

$f = B_{\text{plen}} / 2 - \text{Edgedis}$
 = $1880 / 2 - 50$
 = 889.99 mm

Modular Ratio Of Steel/Concrete (n1):
 = E_S / E_C
 = $203402 / 21526$
 = 9.45

$K1 = 3 (e - 0.5 \cdot B_{\text{plen}})$
 = $3 (560 - 0.5 \cdot 1880)$
 = -1141.07 mm

$K2 = 6 \cdot n1 \cdot A_t / B_{\text{pwid}} \cdot (f + e)$
 = $6 \cdot 9.45 \cdot 586 / 204 \cdot (890 + 560)$
 = 236005.89 mm^2

$K3 = -K2 \cdot (0.5 \cdot B_{\text{plen}} + f)$
 = $-236006 \cdot (0.5 \cdot 1880 + 890)$
 = $-431887766.08 \text{ mm}^3$

Iteratively Solving for the Effective Bearing Length:
 $Y^3 + K1 \cdot Y^2 + K2 \cdot Y + K3 = 0$
 $Y^3 + -1141 \cdot Y^2 + 236006 \cdot Y + -431887744 = 0$
 $Y = 1233.57 \text{ mm}$

$\text{Num} = (B_{\text{plen}} / 2 - Y / 3 - e)$
 = $(1880 / 2 - 1234 / 3 - 560)$
 = -30.83

$\text{Denom} = (B_{\text{plen}} / 2 - Y / 3 + f)$
 = $(1880 / 2 - 1234 / 3 + 890)$
 = 1418.80

Total Bolt Tension Force [Tforce]:
 = $-QO \cdot \text{Num} / \text{Denom}$
 = $-90524 \cdot -30.8 / 1419$
 = 1967.30 N

Bolt Area Required due to Transverse Load [Bltareart]:
 = $T_{\text{force}} / (Stba \cdot N_{\text{bt}})$
 = $1967 / (150 \cdot 2)$
 = 6.5607 mm^2

Required Area of a Single Bolt [Bltarear]:



= max[Bltarearl, Bltarears, Bltareart]
= max[0, 48.8, 6.56]
= 48.8040 mm²

Baseplate Thickness Calculation per D. Moss:

Bearing Pressure (fc)
= $2(QO + Tforce) / (Y \cdot Bpwid)$
= $2(90524 + 1967) / (1234 \cdot 204)$
= 0.74 MPa

Distance from Baseplate Edge to the Web [ADIST]:
= $(Bplen - WebLength) / 2$
= $(1880 - 1829) / 2$
= 25.4000 mm

Overturning Moment due To Bolt Tension [Mt]:
= $Tforce \cdot Adist$
= $1967 \cdot 25.4$
= 49989.64 N-mm

Equivalent Bearing Pressure (f1):
= $fc \cdot (Y - Adist) / Y$
= $0.74 \cdot (1234 - 25.4) / 1234$
= 0.72 MPa

Overturning Moment due to Bearing Pressure [Mc]:
= $(Adist^2 \cdot Bpwid / 6) \cdot (f1 + 2 \cdot fc)$
= $(25.4^2 \cdot 204 / 6) \cdot (0.72 + 2 \cdot 0.74)$
= 48060.78 N-mm

Baseplate Required Thickness [Treq]:
= $\sqrt{(6 \cdot \max(Mt, Mc) / (Bpwid \cdot Sba))}$
= $\sqrt{(6 \cdot \max(49990, 48061) / (204 \cdot 162))}$
= 3.0122 mm

Note: Saddle and wear plate material are both specified as impact tested.

Review notes about nozzle loadings and supports in the Warnings report.

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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Test Case

Note:

Wear Pad Width (230.00) is less than $1.56 \cdot \sqrt{r_m \cdot t}$
 and less than $2a$. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

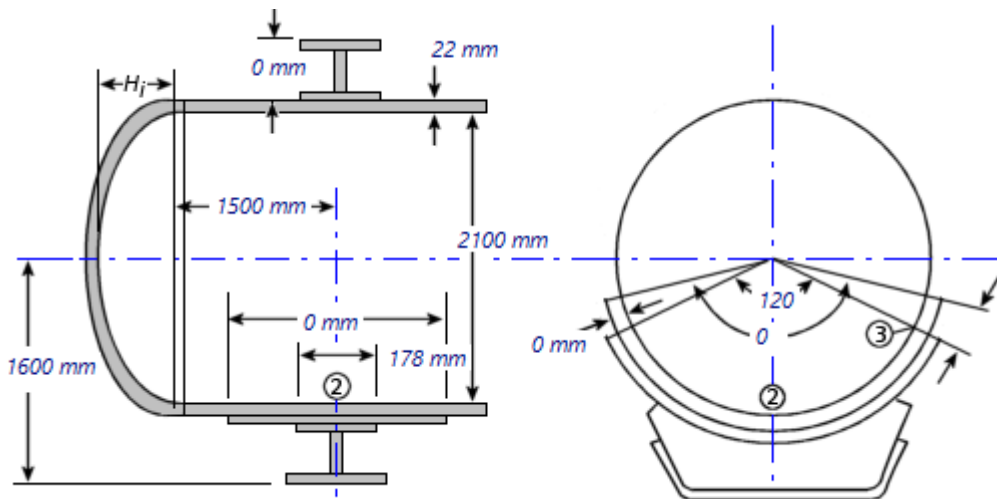
$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(178 + 1.56 \cdot \sqrt{1061 \cdot 22}, 2 \cdot 1500)$$

$$= 416.3382 \text{ mm}$$

Input and Calculated Values:

Vessel Mean Radius	Rm	1061.00	mm
Shell Thickness used in this Case	t	22.000	mm
Stiffened Vessel Length per 4.15.6	L	9800.00	mm
Distance from Saddle to Vessel tangent	a1 or a	1500.00	mm
Saddle Width	b1 or b	178.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Inside or Mean Depth of Head	Hi or hm	539.00	mm
Shell Allowable Stress used in Calculation		234.00	MPa
Head Allowable Stress used in Calculation		234.00	MPa
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1600.00	mm
Coefficient of Friction	mu	0.60	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces	321951.38	N
Pressure used in Analysis	P	3.467 MPa

Horizontal Vessel Analysis Results:	Actual MPa	Allowable MPa
Long. Stress at Top of Midspan	80.21	234.00
Long. Stress at Bottom of Midspan	87.02	234.00
Long. Stress at Top of Saddles	94.35	234.00
Long. Stress at Bottom of Saddles	77.66	234.00
Tangential Shear in Shell	10.44	138.58
Circ. Stress at Horn of Saddle	61.53	350.99

Circ. Compressive Stress in Shell	2.67	234.00
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Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3 (20898/2 + 0) \cdot 1600/1874$
 $= 26762.3 \text{ N}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(F_{l,wind}; \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(5940, 0) \cdot 1600/6800$
 $= 1397.7 \text{ N}$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$
 $= 295189 + \max(1398, 26762, 0, 0)$
 $= 321951.4 \text{ N}$

Longitudinal Wind Force [FI]:
 $= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4 (\text{OD} \cdot \text{WindDiaMult})^2))$
 $= 1.1 \cdot 1587 (4.72 + \pi/4 (2.22 \cdot 1.2)^2)$
 $= 18001.596 \text{ N}$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	325368.09	N
Transverse Shear Load	10448.90	N
Longitudinal Shear Load	5940.22	N

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:
 $= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot R_m)$
 $= \min(2(2100/2 + 22 + 10)\sin(120/2), 2 \cdot 1061)$
 $= 1874.079 \text{ mm}$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:
 $= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L))/(1 + (4h_m)/3L)]$
 $= -321951 \cdot 1500 [1 - (1 - 1500/9800 + (1061^2 - 539^2)/(2 \cdot 1500 \cdot 9800))/(1 + (4 \cdot 539)/(3 \cdot 9800))]$
 $= -89116576.0 \text{ N-mm}$

Moment per Equation 4.15.2 [M2]:
 $= Q \cdot L/4 ((1 + 2(R_m^2 - h_m^2)/(L^2))/(1 + (4h_m^2)/(3L)) - 4a/L)$
 $= 321951 \cdot 9800/4 ((1 + 2(1061^2 - 539^2)/(9800^2))/(1 + (4 \cdot 539)/(3 \cdot 9800)) - 4 \cdot 1500/9800)$
 $= 264850912.0 \text{ N-mm}$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:
 $= P \cdot R_m/(2t) - M_2/(\pi \cdot R_m^2 t)$
 $= 3.47 \cdot 1061/(2 \cdot 22) - 264850912/(\pi \cdot 1061^2 \cdot 22)$
 $= 80.21 \text{ MPa}$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:
 $= P \cdot R_m/(2t) + M_2/(\pi \cdot R_m^2 \cdot t)$
 $= 3.47 \cdot 1061/(2 \cdot 22) + 264850912/(\pi \cdot 1061^2 \cdot 22)$
 $= 87.02 \text{ MPa}$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:
 $= P \cdot R_m/(2t) - M_1/(K_1 \cdot \pi \cdot R_m^2 t)$
 $= 3.47 \cdot 1061/(2 \cdot 22) - 89116576/(0.11 \cdot \pi \cdot 1061^2 \cdot 22)$
 $= 94.35 \text{ MPa}$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot R_m / (2t) + M1 / (K1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 3.47 \cdot 1061 / (2 \cdot 22) + 89116576 / (0.19 \cdot \pi \cdot 1061^2 \cdot 22)$$

$$= 77.66 \text{ MPa}$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 321951 (9800 - 2 \cdot 1500) / (9800 + (4 \cdot 539 / 3))$$

$$= 208131.8 \text{ N}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot 208132 / (1061 \cdot 22)$$

$$= 10.44 \text{ MPa}$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(1061 \cdot 22)}$$

$$= 119.169 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K5 \cdot Q \cdot k / (t(b + X1 + X2))$$

$$= -0.76 \cdot 321951 \cdot 0.1 / (22(178 + 119 + 119))$$

$$= -2.67 \text{ MPa}$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.22) [σ_7]:

$$= -Q / (4 \cdot t(b + X1 + X2)) - 3 \cdot K7 \cdot Q / (2 \cdot t^2)$$

$$= -321951 / (4 \cdot 22(178 + 119 + 119)) - 3 \cdot 0.053 \cdot 321951 / (2 \cdot 22^2)$$

$$= -61.53 \text{ MPa}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(178 + 1.56 \cdot \sqrt{(1061 \cdot 22)}, 2 \cdot 1500)$$

$$= 416.34 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1880.0000	mm
Baseplate Thickness	Bpthk	14.0000	mm
Baseplate Width	Bpwid	204.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	14.0000	mm
Web Thickness	Webtk	14.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	205.0	MPa
Height of Web at Center	Hw,c	492.0	mm
Friction Coefficient	mu	0.600	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(1050 \cdot 22)} + 230)^3) / 12$$

$$= 414472.531 \text{ mm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 414473 + 10276(166 - 11)^2$$

$$= 246848000.000 \text{ mm}^4$$

	B mm	D mm	Y mm	A mm ²	AY mm ³	I + AD ⁽²⁾ mm ⁴
Shell	467.1	22.0	11.0	10276.2	113038.1	0.247E+09
Wearplate	230.0	10.0	27.0	2300.0	62100.0	0.444E+08
Web	14.0	504.0	284.0	7056.0	2003903.8	0.248E+09
BasePlate	204.0	14.0	543.0	2856.0	1550807.9	0.406E+09
Totals	...	...	...	22488.2	3729849.5	0.945E+09

Distance to Centroid [C1]:

$$= AY / A$$

$$= 3729849/22488$$

$$= 165.858 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 321951$$

$$= 65524.0859 \text{ N}$$

$$\text{Tension Stress, } St = (Fh / As) = 5.3660 \text{ MPa}$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 122.9978 \text{ MPa}$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 1600 - 1050 \cdot \sin(120/2) / 1.05$$

$$= 731.657 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 47960592.0000 \text{ N-mm}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 8.4120 \text{ MPa}$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 136.6642 \text{ MPa}$$

Minimum Thickness of Baseplate per Moss:

The maximum of 2/3*Yield and the Saddle Allowable Stress value was used to determine the baseplate thickness.

$$= \sqrt{3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{3(321951 + 3417)204 / (4 \cdot 1880 \cdot 138)}$$

$$= 13.853 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120/2) (1050 + 22 + 10)$$

$$= 1874.079 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nrifs} - 1)$$

$$= 1874 / (3 - 1)$$

$$= 937.039 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 937 \cdot 204/2$$

$$= 95578.023 \text{ mm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 325368 / (1880 \cdot 204)$$

$$= 0.848 \text{ N/mm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 95578 \cdot 0.85$$

$$= 81085.852 \text{ N}$$

Area of the Rib and Web [Ar]:



$$\begin{aligned}
 &= \text{Rib Area} + \text{Web Area} \\
 &= 2296 + 6559 \\
 &= 8855.276 \text{ mm}^2
 \end{aligned}$$

$$\begin{aligned}
 &\text{Compressive Stress [Sc]:} \\
 &= P/Ar \\
 &= 81086/8855 \\
 &= 9.158 \text{ MPa}
 \end{aligned}$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	14.0	178.0	...	2492.0	...	0.658E+07

$$\begin{aligned}
 &\text{Rib dimension [D]:} \\
 &= \text{Saddle Width} - \text{Web Thickness} \\
 &= 178 - 14 \\
 &= 164.000 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 &\text{Distance to Centroid from Datum [ytot]:} \\
 &= AY / A \\
 &= 0/8855 \\
 &= 0.000 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 &\text{Distance to Centroid [C1]:} \\
 &= \text{Saddle Width} / 2 \\
 &= 178/2 \\
 &= 89.000 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 &\text{Radius of Gyration [r]:} \\
 &= \sqrt{(\text{Total Inertia} / \text{Total Area})} \\
 &= \sqrt{(6579704/8855)} \\
 &= 27.259 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 &\text{Length of Outer Rib [L]:} \\
 &= \text{Saddle Height} - \cos(\text{theta}/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk} \\
 &= 1600 - \cos(120/2) (1050 + 22 + 10) - 14 \\
 &= 1045.000 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 &\text{Intermediate Term [Cc]:} \\
 &= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})} \\
 &= \sqrt{(2 \cdot \pi^2 \cdot 202713/205)} \\
 &= 139.712
 \end{aligned}$$

$$\begin{aligned}
 &\text{Slenderness ratio [KL/r]:} \\
 &= KL/r \\
 &= 1 \cdot 1045/27.3 \\
 &= 38.337
 \end{aligned}$$

$$\begin{aligned}
 &\text{Bending Moment [Rm]:} \\
 &= FI / (2 \cdot Bplen) \cdot e \cdot L / 2 \\
 &= 5940 / (2 \cdot 1880) \cdot 937 \cdot 1045/2 \\
 &= 773811.062 \text{ N-mm}
 \end{aligned}$$

$$\begin{aligned}
 &\text{Compressive Allowable, } KL/r < Cc \text{ (} 38.3 < 140 \text{) per AISC E2-1 [Sca]:} \\
 &= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3)) \\
 &= (1 - (38.3)^2 / (2 \cdot 140^2)) 205 / \\
 &\quad (5/3 + 3 \cdot (38.3) / (8 \cdot 140) - (38.3)^3 / (8 \cdot 140^3)) \\
 &= 111.6 \text{ MPa}
 \end{aligned}$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$\begin{aligned}
 &= Sc/Sca + (Rm \cdot C1 / I) / Sba \\
 &= 9.16/112 + (773811 \cdot 89/6579711) / 137 \\
 &= 0.159
 \end{aligned}$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	14.0	164.0	0.0	2296.0	0.0	0.658E+07
Web	937.0	14.0	0.0	13118.6	0.0	0.214E+06
Totals	...	...	...	15414.6	...	0.679E+07

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/15415$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 178/2$$

$$= 89.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 1600 - \cos(1082/2) (14 + 0 + 0) - 0$$

$$= 504.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(6790772/15415)}$$

$$= 20.989 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 504/21$$

$$= 24.012$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 5940/(2 \cdot 1880)$$

$$= 1.580 \text{ N/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 1.58 \cdot 937 \cdot 504$$

$$= 746412.812 \text{ N-mm}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 746413/76301$$

$$= 9.779 \text{ MPa}$$

Compressive Allowable, $KL/r < C_c$ ($24 < 140$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (24)^2 / (2 \cdot 140^2)) 205 / (5/3 + 3 \cdot (24) / (8 \cdot 140) - (24^3) / (8 \cdot 140^3))$$

$$= 116.7 \text{ MPa}$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (M_{base,c} \cdot C1/I) / S_{ba}$$

$$= 10.5/117 + (746413 \cdot 89/6790772) / 137$$

$$= 0.162$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Material Specification	SA-36	
Bolt Allowable Stress	Stba	114.00 MPa
Bolt Corrosion Allowance	Bca	3.0 mm
Distance from Bolts to Edge	Edgedis	50.0 mm
Nominal Bolt Diameter	Bnd	30.0000 mm
Thread Series	Series	TEMA Metric
BasePlate Allowable Stress	S	108.00 MPa
Area Available in a Single Bolt	BltArea	292.9056 mm ²



Saddle Load Q0 (Weight)	Q0	298605.8 N
Saddle Load QL (Wind/Seismic contribution)	QL	1397.7 N
Maximum Transverse Force	Ft	10448.9 N
Maximum Longitudinal Force	F1	5940.2 N
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [rb,l]:
 = $F1 / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 = $5940 / (293 \cdot 4)$
 = 5.1 MPa. Must be less than 68.4 MPa.

Shear Stress in a Single Bolt, Transverse Direction [rb,t]:
 = $Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 = $10449 / (293 \cdot 4)$
 = 8.9 MPa. Must be less than 68.4 MPa.

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
 = 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
 = $F1 / (\text{BoltShearAllowable} \cdot \text{Nbols})$
 = $5940 / (68.4 \cdot 4)$
 = 21.7136 mm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
 = $B \cdot Ft + \text{Sum of X Moments}$
 = $1600 \cdot 10449 + 0$
 = 16725018.00 N-mm

Eccentricity (e):
 = $Rmom / QO$
 = $16725018 / 298606$
 = 55.99 mm < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
 = 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
 = $\max[Bltarearl, Bltarears, Bltareart]$
 = $\max[0, 21.7, 0]$
 = 21.7136 mm²

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Note:
 Wear Pad Width (230.00) is less than $1.56 \cdot \sqrt{rm \cdot t}$
 and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:
 = $\min(b + 1.56 \cdot \sqrt{ Rm \cdot t }, 2a)$
 = $\min(178 + 1.56 \cdot \sqrt{ 1061 \cdot 22 }, 2 \cdot 1500)$
 = 416.3382 mm

Input and Calculated Values:

Vessel Mean Radius	Rm	1061.00	mm
Shell Thickness used in this Case	t	22.000	mm
Stiffened Vessel Length per 4.15.6	L	9800.00	mm
Distance from Saddle to Vessel tangent	a1 or a	1500.00	mm
Saddle Width	b1 or b	178.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Inside or Mean Depth of Head	Hi or hm	536.00	mm

Shell Allowable Stress used in Calculation 234.00 MPa
 Head Allowable Stress used in Calculation 234.00 MPa
 Circumferential Efficiency in Plane of Saddle 1.00
 Circumferential Efficiency at Mid-Span 1.00
 Distance from Saddle Base to Centerline B 1600.00 mm
 Coefficient of Friction mu 0.00

Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces 296709.62 N
 Pressure used in Analysis P 3.467 MPa

Horizontal Vessel Analysis Results:	Actual MPa	Allowable MPa
Long. Stress at Top of Midspan	80.47	234.00
Long. Stress at Bottom of Midspan	86.75	234.00
Long. Stress at Top of Saddles	93.49	234.00
Long. Stress at Bottom of Saddles	78.14	234.00
Tangential Shear in Shell	9.62	138.58
Circ. Stress at Horn of Saddle	56.70	350.99
Circ. Compressive Stress in Shell	2.46	234.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= Ftr(Ft/Num\ of\ Saddles + Z\ Force\ Load) \cdot B / E$
 $= 3 (20898/2 + 0) \cdot 1600/1874$
 $= 26762.3\ N$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(Fl,wind; Sum\ of\ X\ Forces) \cdot B / Ls$
 $= \max(5940, 0) \cdot 1600/6800$
 $= 1397.7\ N$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= Saddle\ Load + \max(Fwl, Fwt, Fsl, Fst)$
 $= 269947 + \max(1398, 26762, 0, 0)$
 $= 296709.6\ N$

Longitudinal Wind Force [Fl]:
 $= WindScalar \cdot WindPress(Platform\ Area + (\pi/4(OD \cdot WindDiaMult)^2))$
 $= 1.1 \cdot 1587 (4.72 + \pi/4(2.22 \cdot 1.2)^2)$
 $= 18001.596\ N$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight) 300126.38 N
 Transverse Shear Load 10448.90 N
 Longitudinal Shear Load 5940.22 N

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:
 $= \min(2(ShellID/2 + t + WearPadThickness)\sin(\theta/2), 2 \cdot Rm)$
 $= \min(2(2100/2 + 22 + 10)\sin(120/2), 2 \cdot 1061)$
 $= 1874.079\ mm$

The Computed K values from Table 4.15.1:

K1 = 0.1066 K2 = 1.1707 K3 = 0.8799 K4 = 0.4011
 K5 = 0.7603 K6 = 0.0529 K7 = 0.0529 K8 = 0.3405
 K9 = 0.2711 K10 = 0.0581 K1* = 0.1923

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:
 $= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L)) / (1 + (4hm)/3L)]$
 $= -296710 \cdot 1500 [1 - (1 - 1500/9800 + (1061^2 - 536^2)/$



$$\frac{(2 \cdot 1500 \cdot 9800)}{(1 + (4 \cdot 536)/(3 \cdot 9800))}]$$

$$= -81945984.0 \text{ N-mm}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L / 4 \cdot ((1 + 2(Rm^2 - hm^2)/(L^2)) / (1 + (4hm^2)/(3L)) - 4a/L)$$

$$= 296710 \cdot 9800 / 4 \cdot ((1 + 2(1061^2 - 536^2)/(9800^2)) / (1 + (4 \cdot 536)/(3 \cdot 9800)) - 4 \cdot 1500/9800)$$

$$= 244393712.0 \text{ N-mm}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot Rm / (2t) - M2 / (\pi \cdot Rm^2 t)$$

$$= 3.47 \cdot 1061 / (2 \cdot 22) - 244393712 / (\pi \cdot 1061^2 \cdot 22)$$

$$= 80.47 \text{ MPa}$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot Rm / (2t) + M2 / (\pi \cdot Rm^2 \cdot t)$$

$$= 3.47 \cdot 1061 / (2 \cdot 22) + 244393712 / (\pi \cdot 1061^2 \cdot 22)$$

$$= 86.75 \text{ MPa}$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot Rm / (2t) - M1 / (K1 \cdot \pi \cdot Rm^2 t)$$

$$= 3.47 \cdot 1061 / (2 \cdot 22) - 81945984 / (0.11 \cdot \pi \cdot 1061^2 \cdot 22)$$

$$= 93.49 \text{ MPa}$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot Rm / (2t) + M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t)$$

$$= 3.47 \cdot 1061 / (2 \cdot 22) + 81945984 / (0.19 \cdot \pi \cdot 1061^2 \cdot 22)$$

$$= 78.14 \text{ MPa}$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot hm/3))$$

$$= 296710 \cdot (9800 - 2 \cdot 1500) / (9800 + (4 \cdot 536/3))$$

$$= 191886.8 \text{ N}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (Rm \cdot t)$$

$$= 1.17 \cdot 191887 / (1061 \cdot 22)$$

$$= 9.62 \text{ MPa}$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(Rm \cdot t)}$$

$$= 0.78 \cdot \sqrt{(1061 \cdot 22)}$$

$$= 119.169 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K5 \cdot Q \cdot k / (t(b + X1 + X2))$$

$$= -0.76 \cdot 296710 \cdot 0.1 / (22(178 + 119 + 119))$$

$$= -2.46 \text{ MPa}$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8Rm$ (4.15.22) [σ_7]:

$$= -Q / (4 \cdot t(b + X1 + X2)) - 3 \cdot K7 \cdot Q / (2 \cdot t^2)$$

$$= -296710 / (4 \cdot 22(178 + 119 + 119)) - 3 \cdot 0.053 \cdot 296710 / (2 \cdot 22^2)$$

$$= -56.70 \text{ MPa}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(Rm \cdot t)}, 2a)$$

$$= \min(178 + 1.56 \cdot \sqrt{(1061 \cdot 22)}, 2 \cdot 1500)$$

$$= 416.34 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1880.0000	mm
Baseplate Thickness	Bpthk	14.0000	mm
Baseplate Width	Bpwid	204.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	14.0000	mm
Web Thickness	Webtk	14.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	205.0	MPa

Height of Web at Center Hw,c 492.0 mm
 Friction Coefficient mu 0.000

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(1050 \cdot 22)} + 230)^3) / 12$$

$$= 414472.531 \text{ mm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$$

$$= 414473 + 10276 (166 - 11)^2$$

$$= 246848000.000 \text{ mm}^4$$

	B mm	D mm	Y mm	A mm ²	AY mm ³	I + AD ⁽²⁾ mm ⁴
Shell	467.1	22.0	11.0	10276.2	113038.1	0.247E+09
Wearplate	230.0	10.0	27.0	2300.0	62100.0	0.444E+08
Web	14.0	504.0	284.0	7056.0	2003903.8	0.248E+09
BasePlate	204.0	14.0	543.0	2856.0	1550807.9	0.406E+09
Totals	...	...	...	22488.2	3729849.5	0.945E+09

Distance to Centroid [C1]:

$$= AY / A$$

$$= 3729849 / 22488$$

$$= 165.858 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 296710$$

$$= 60386.8477 \text{ N}$$

Tension Stress, St = (Fh/As) = 4.9453 MPa
 Allowed Stress, Sa = 0.6 * Yield Str = 122.9978 MPa

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\text{theta} / 2) / (\text{theta} / 2 \text{ in radians})$$

$$= 1600 - 1050 \cdot \sin(120 / 2) / 1.05$$

$$= 731.657 \text{ mm}$$

Bending Moment, M = Fh * d = 44200372.0000 N-mm

Bending Stress, Sb = (M * C1 / I) = 7.7524 MPa
 Allowed Stress, Sa = 2/3 * Yield Str = 136.6642 MPa

Minimum Thickness of Baseplate per Moss:

The maximum of 2/3*Yield and the Saddle Allowable Stress value was used to determine the baseplate thickness.

$$= \sqrt{3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{3(296710 + 3417)204 / (4 \cdot 1880 \cdot 138)}$$

$$= 13.305 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle} / 2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120/2) (1050 + 22 + 10)$$

$$= 1874.079 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nribs} - 1)$$

$$= 1874 / (3 - 1)$$

$$= 937.039 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 937 \cdot 204 / 2$$

$$= 95578.023 \text{ mm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 300126 / (1880 \cdot 204)$$

$$= 0.783 \text{ N/mm}^2$$

Axial Load [P]:

$$= \text{Ap} \cdot \text{Bp}$$

$$= 95578 \cdot 0.78$$

$$= 74795.281 \text{ N}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 2296 + 6559$$

$$= 8855.276 \text{ mm}^2$$

Compressive Stress [Sc]:

$$= P / \text{Ar}$$

$$= 74795 / 8855$$

$$= 8.447 \text{ MPa}$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	14.0	178.0	...	2492.0	...	0.658E+07

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 178 - 14$$

$$= 164.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= \text{AY} / \text{A}$$

$$= 0 / 8855$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 178 / 2$$

$$= 89.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(6579704 / 8855)}$$

$$= 27.259 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\text{theta}/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1600 - \cos(120/2) (1050 + 22 + 10) - 14$$

$$= 1045.000 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 202713 / 205)}$$

$$= 139.712$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 1045/27.3$$

$$= 38.337$$

Bending Moment [Rm]:

$$= FI / (2 \cdot Bplen) \cdot e \cdot L / 2$$

$$= 5940 / (2 \cdot 1880) \cdot 937 \cdot 1045/2$$

$$= 773811.062 \text{ N-mm}$$

Compressive Allowable, KL/r < Cc (38.3 < 140) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (38.3)^2 / (2 \cdot 140^2)) 205 / (5/3 + 3 \cdot (38.3) / (8 \cdot 140) - (38.3^3) / (8 \cdot 140^3))$$

$$= 111.6 \text{ MPa}$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 8.45/112 + (773811 \cdot 89/6579711) / 137$$

$$= 0.152$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	14.0	164.0	0.0	2296.0	0.0	0.658E+07
Web	937.0	14.0	0.0	13118.6	0.0	0.214E+06
Totals	...	...	...	15414.6	...	0.679E+07

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/15415$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 178/2$$

$$= 89.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - Bpthk$$

$$= 1600 - \cos(1082/2) (14 + 0 + 0) - 0$$

$$= 504.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ (\text{Total Inertia} / \text{Total Area}) }$$

$$= \sqrt{ (6790772/15415) }$$

$$= 20.989 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 504/21$$

$$= 24.012$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 5940 / (2 \cdot 1880)$$

$$= 1.580 \text{ N/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 1.58 \cdot 937 \cdot 504$$

$$= 746412.812 \text{ N-mm}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 746413/76301$$

$$= 9.779 \text{ MPa}$$

Compressive Allowable, $KL/r < C_c$ ($24 < 140$) per AISC E2-1 [Sca]:

$$= (1 - (Klr)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (Klr) / (8 \cdot Cc) - (Klr)^3 / (8 \cdot Cc^3))$$

$$= (1 - (24)^2 / (2 \cdot 140^2)) 205 /$$

$$(5/3 + 3 \cdot (24) / (8 \cdot 140) - (24^3) / (8 \cdot 140^3))$$

$$= 116.7 \text{ MPa}$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (Mbase,c \cdot C1/I) / Sba$$

$$= 9.7/117 + (746413 \cdot 89/6790772) / 137$$

$$= 0.155$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Material Specification		SA-36
Bolt Allowable Stress	Stba	114.00 MPa
Bolt Corrosion Allowance	Bca	3.0 mm
Distance from Bolts to Edge	Edgedis	50.0 mm
Nominal Bolt Diameter	Bnd	30.0000 mm
Thread Series	Series	TEMA Metric
BasePlate Allowable Stress	S	108.00 MPa
Area Available in a Single Bolt	BltArea	292.9056 mm ²
Saddle Load QO (Weight)	QO	273364.1 N
Saddle Load QL (Wind/Seismic contribution)	QL	1397.7 N
Maximum Transverse Force	Ft	10448.9 N
Maximum Longitudinal Force	Fl	5940.2 N
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [tb,l]:

$$= Fl / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 5940 / (293 \cdot 4)$$

$$= 5.1 \text{ MPa. Must be less than } 68.4 \text{ MPa.}$$

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:

$$= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 10449 / (293 \cdot 4)$$

$$= 8.9 \text{ MPa. Must be less than } 68.4 \text{ MPa.}$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (} QO > QL \text{ --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= Fl / (\text{BoltShearAllowable} \cdot \text{Nbolts})$$

$$= 5940 / (68.4 \cdot 4)$$

$$= 21.7136 \text{ mm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot Ft + \text{Sum of X Moments}$$

$$= 1600 \cdot 10449 + 0$$

$$= 16725018.00 \text{ N-mm}$$

Eccentricity (e):

$$= Rmom / QO$$

$$= 16725018 / 273364$$

$$= 61.16 \text{ mm} < Bplen/6 \text{ --> No Uplift in Transverse direction}$$

Bolt Area due to Transverse Load [Bltareart]:

$$= 0 \text{ (No Uplift)}$$

Required Area of a Single Bolt [Bltarear]:

$$= \max[Bltarearl, Bltarears, Bltareart]$$

$$= \max[0, 21.7, 0]$$

$$= 21.7136 \text{ mm}^2$$



Review notes about nozzle loadings and supports in the Warnings report.

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Input, Nozzle Desc: M124"

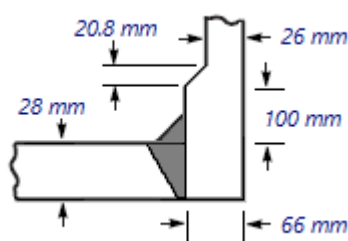
From: 10

Pressure for Reinforcement Calculations	P	2.6593	MPa
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	+N+IT
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Elliptical Head	D	2100.00	mm
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	28.0000	mm
Head Internal Corrosion Allowance	cas	1.5000	mm
Head External Corrosion Allowance	co	0.0000	mm
Distance from Head Centerline	L1	0.0000	mm
User Entered Minimum Design Metal Temperature		-40.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF1	+IT
Material UNS Number		K03009	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa
Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		24.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	26.0000	mm
Flange Material		SA-350 LF1	+IT
Hub Height of Integral Nozzle	h	100.0000	mm
Height of Beveled Transition	L'	20.7846	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		66.0000	mm
Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		22.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
This is a Manway, Coupling or Access Opening.			
Flange Class		300	
Flange Grade		GR 1.4	

The Pressure Design option was MAWP + static head (to the nozzle).



ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation 24.000 in.
 Actual Thickness Used in Calculation 1.024 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 2.66 \cdot 2103 \cdot 0.9 / (2 \cdot 138 \cdot 1 - 0.2 \cdot 2.66)$
 $= 18.2471 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot Ro / (Sn \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 2.66 \cdot 305 / (118 \cdot 1 + 0.4 \cdot 2.66)$
 $= 6.8077 \text{ mm}$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 40 / 0.5773$
 $= 69.2820 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

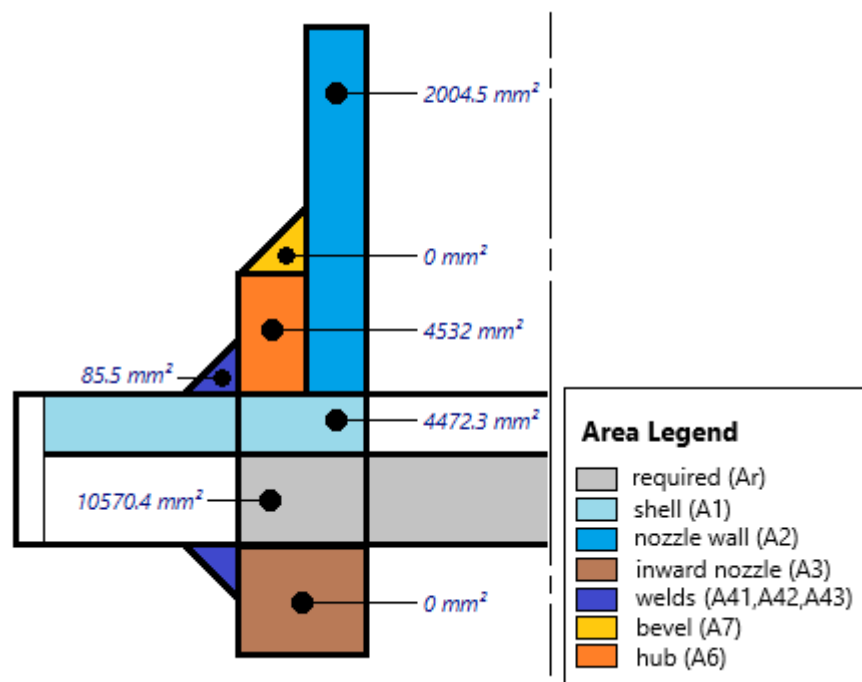
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	1121.2000	mm
Parallel to Vessel Wall, opening length	d	560.6000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	66.2500	mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 118/138)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 118/138)$
 $= 0.855$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.86, 1)$
 $= 0.855$



Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	10570.415	NA	NA
Area in Shell	A1	4472.310	NA	NA
Area in Nozzle Wall	A2	2004.530	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	85.509	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub	A6	4531.983	NA	NA
TOTAL AREA AVAILABLE	Atot	11094.333	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (561 \cdot 18.2 \cdot 1 + 2 \cdot 64.5 \cdot 18.2 \cdot 1 \cdot (1 - 0.86))$$

$$= 10570.415 \text{ mm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 561(1 \cdot 26.5 - 1 \cdot 18.2) - 2 \cdot 64.5$$

$$(1 \cdot 26.5 - 1 \cdot 18.2) \cdot (1 - 0.86)$$

$$= 4472.310 \text{ mm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp)(tn - trn)fr2$$

$$= (2 \cdot 66.3)(24.5 - 6.81)0.86$$

$$= 2004.530 \text{ mm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$$

$$= 10^2 \cdot 0.86 + (0)^2 \cdot 0.86$$

$$= 85.509 \text{ mm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lnp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot f_{r2}$$

$$= (2 \cdot \min(66.3, 300, 100)) \cdot (66 - 26) \cdot 0.86$$

$$= 4531.983 \text{ mm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement:

Wall Thickness for Internal/External pressures $t_a = 8.3077 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$= t_a$
 $= 8.3077 \text{ mm}$

Available Nozzle Neck Thickness = 26.0000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification $-29 \text{ }^{\circ}\text{C}$

Calculated Minimum Design Metal Temperature $-104 \text{ }^{\circ}\text{C}$

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification $-29 \text{ }^{\circ}\text{C}$

Gov. MDMT of the nozzle to shell joint welded assembly : $-29 \text{ }^{\circ}\text{C}$

Flange MDMT including Temperature reduction per UCS-66.1:

MDMT of ASME B16.5/47 flange per Matl. Specification $-29 \text{ }^{\circ}\text{C}$

Flange MDMT with Temp reduction per UCS-66(i)(2) $-50 \text{ }^{\circ}\text{C}$

Where the Stress Reduction Ratio per UCS-66(i)(2) is :

Design Pressure/Ambient Rating = $2.66/4.26 = 0.624$

Weld Size Calculations, Description: M124"

Intermediate Calc. for nozzle/shell Welds $T_{min} = 19.0000 \text{ mm}$

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 \cdot W_o \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A - A1 + 2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$$

$$= \max(0, (10570 - 4472 + 2 \cdot 64.5 \cdot 0.86 \cdot (1 \cdot 26.5 - 18.2)) \cdot 138)$$

$$= 967063.81 \text{ N}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2 + A4 - (W_i - C_{an} / .707)^2 \cdot f_{r2}) \cdot S_v$$

$$= (6537 + 85.5 - 0 \cdot 0.86) \cdot 138$$

$$= 913741.38 \text{ N}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$$

$$= (6537 + 0 + 85.5 + (2923)) \cdot 138$$

= 1317090.12 N

Weld Load [W3]:

$$= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f r_1)) \cdot S$$

$$= (6537 + 0 + 85.5 + 0 + (2923)) \cdot 138$$

$$= 1317090.12 \text{ N}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.14/2.0) \cdot 610 \cdot 10 \cdot 0.49 \cdot 118$$

$$= 553613. \text{ N}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n$$

$$= (3.14 \cdot 293) \cdot (66 - 1.5) \cdot 0.7 \cdot 118$$

$$= 4896128. \text{ N}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{g n v i} \cdot 0.74 \cdot S_{ng}$$

$$= (3.14/2.0) \cdot 610 \cdot (22 - 0) \cdot 0.74 \cdot 138$$

$$= 2151057. \text{ N}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (553613 + 4896128) = 5449740 \text{ N}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (553613 + 0 + 2151057 + 0) = 2704669 \text{ N}$$

Failure Path [Path 3-3]:

$$= (Sonw + Tngw + Sinw)$$

$$= (553613 + 2151057 + 0) = 2704669 \text{ N}$$

Summary of Failure Path Calculations:

Path 1-1 = 5449740 N	, must exceed W = 967063 N	or W1 = 913741 N
Path 2-2 = 2704669 N	, must exceed W = 967063 N	or W2 = 1317090 N
Path 3-3 = 2704669 N	, must exceed W = 967063 N	or W3 = 1317090 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 2.72 MPa

The Drop for this Nozzle is : 24.4458 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 352.4458 mm

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Input, Nozzle Desc: A16"

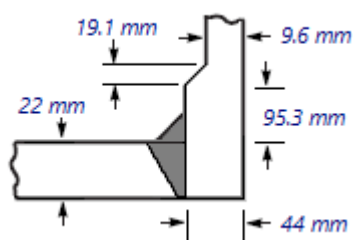
From: 20

Pressure for Reinforcement Calculations	P	2.6593	MPa
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	+N+IT
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	2100.00	mm
Shell Finished (Minimum) Thickness	t	22.0000	mm
Shell Internal Corrosion Allowance	cas	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		1550.00	mm
User Entered Minimum Design Metal Temperature		-40.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF1	+IT
Material UNS Number		K03009	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		6.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		80	
Flange Material		SA-350 LF1	+IT
Hub Height of Integral Nozzle	h	95.2682	mm
Height of Beveled Transition	L'	19.0683	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		44.0000	mm
Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		22.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		300	
Flange Grade		GR 1.4	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: A16"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 5.869 in.
 Actual Thickness Used in Calculation 0.378 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2.66 \cdot 1052 / (138 \cdot 1 - 0.6 \cdot 2.66)$
 $= 20.5000 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2.66 \cdot 76 / (118 \cdot 1 - 0.6 \cdot 2.66)$
 $= 1.7371 \text{ mm}$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 34.4 / 0.5773$
 $= 59.5805 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

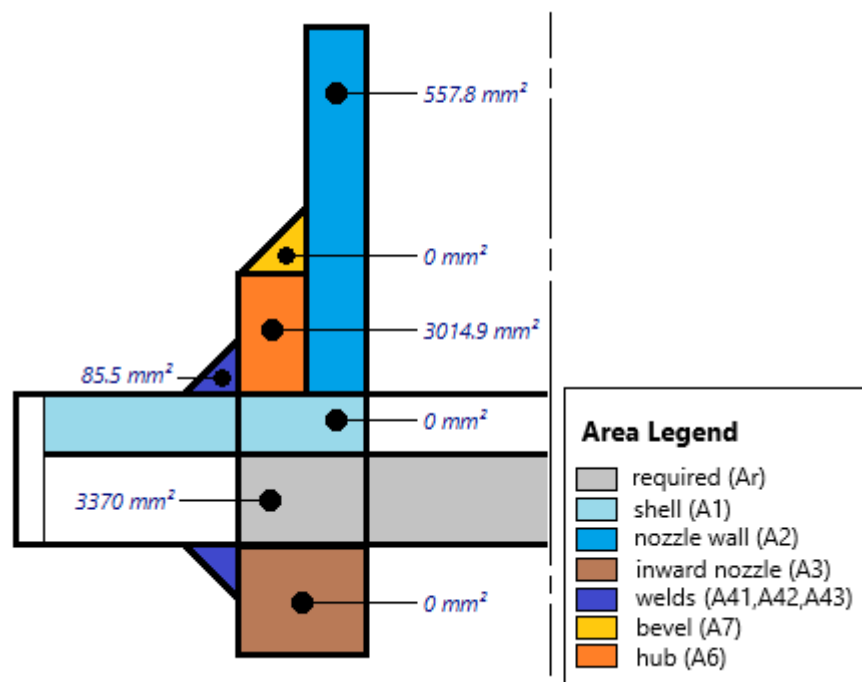
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	304.1452	mm
Parallel to Vessel Wall, opening length	d	152.0726	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	51.2500	mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118 / 138)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118 / 138)$
 $= 0.855$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.86, 1)$
 $= 0.855$



Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	3369.997	NA	NA
Area in Shell	A1	0.000	NA	NA
Area in Nozzle Wall	A2	557.794	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	85.509	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub	A6	3014.935	NA	NA
TOTAL AREA AVAILABLE	Atot	3658.237	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (152 \cdot 20.5 \cdot 1 + 2 \cdot 42.5 \cdot 20.5 \cdot 1 \cdot (1 - 0.86))$
 $= 3369.997 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 152(1 \cdot 20.5 - 1 \cdot 20.5) - 2 \cdot 42.5$
 $(1 \cdot 20.5 - 1 \cdot 20.5) \cdot (1 - 0.86)$
 $= 0.000 \text{ mm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn)fr2$
 $= (2 \cdot 51.2)(8.1 - 1.74)0.86$
 $= 557.794 \text{ mm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10^2 \cdot 0.86 + (0)^2 \cdot 0.86$
 $= 85.509 \text{ mm}^2$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lp}, h_o, H_{ubht})) \cdot (H_{ubt} - t_n) \cdot f_{r2}$$

$$= (2 \cdot \min(51.2, 200, 95.3)) \cdot (44 - 9.6) \cdot 0.86$$

$$= 3014.935 \text{ mm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $t_a = 3.2371 \text{ mm}$
 Wall Thickness per UG16(b), $tr_{16b} = 3.0000 \text{ mm}$
 Wall Thickness, shell/head, internal pressure $trb_1 = 22.0000 \text{ mm}$
 Wall Thickness $tb_1 = \max(trb_1, tr_{16b}) = 22.0000 \text{ mm}$
 Wall Thickness $tb_2 = \max(trb_2, tr_{16b}) = 3.0000 \text{ mm}$
 Wall Thickness per table UG-45 $tb_3 = 7.7200 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb_3, \max(tb_1, tb_2)]$$

$$= \min[7.72, \max(22, 3)]$$

$$= 7.7200 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(t_a, t_b)$$

$$= \max(3.24, 7.72)$$

$$= 7.7200 \text{ mm}$$

Available Nozzle Neck Thickness = 9.6012 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	43.9,	Allowable	:	118.0 MPa	Passed
Expansion	:	0.0,	Allowable	:	251.1 MPa	Passed
Occasional	:	11.8,	Allowable	:	156.9 MPa	Passed
Shear	:	21.7,	Allowable	:	82.6 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note:
This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification	-29 °C
Calculated Minimum Design Metal Temperature	-104 °C

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note:
This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification	-29 °C
Gov. MDMT of the nozzle to shell joint welded assembly :	-29 °C

Flange MDMT including Temperature reduction per UCS-66.1:

MDMT of ASME B16.5/47 flange per Matl. Specification	-29 °C
Flange MDMT with Temp reduction per UCS-66(i)(2)	-50 °C

Where the Stress Reduction Ratio per UCS-66(i)(2) is :
 Design Pressure/Ambient Rating = $2.66/4.26 = 0.624$

Weld Size Calculations, Description: A16"

Intermediate Calc. for nozzle/shell Welds	T_{min}	19.0000 mm
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Results Per UW-16.1:

Required Thickness	Actual Thickness
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Nozzle Weld 6.0000 = Min per Code 7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_r1 \cdot (E1 \cdot t-tr)) \cdot S_v)$
 $= \max(0, (3370 - 0 + 2 \cdot 42.5 \cdot 0.86 \cdot (1 \cdot 20.5 - 20.5)) \cdot 138)$
 $= 465011.69 \text{ N}$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A4-(W_i-Can/.707)^2 \cdot f_r2) \cdot S_v$
 $= (3573 + 85.5 - 0 \cdot 0.86) \cdot 138$
 $= 504784.72 \text{ N}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_r1)) \cdot S_v$
 $= (3573 + 0 + 85.5 + (1490)) \cdot 138$
 $= 710382.25 \text{ N}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_r1)) \cdot S$
 $= (3573 + 0 + 85.5 + 0 + (1490)) \cdot 138$
 $= 710382.25 \text{ N}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.14/2.0) \cdot 237 \cdot 10 \cdot 0.49 \cdot 118$
 $= 215299. \text{ N}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n$
 $= (3.14 \cdot 97.3) \cdot (44 - 1.5) \cdot 0.7 \cdot 118$
 $= 1072837. \text{ N}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot (T - cas) \cdot 0.74 \cdot S_{ng}$
 $= (3.14/2.0) \cdot 237 \cdot (22 - 1.5) \cdot 0.74 \cdot 138$
 $= 779509. \text{ N}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (SONW + SNW) = (215299 + 1072836) = 1288135 \text{ N}$

Failure Path [Path 2-2]:
 $= (Sonw + Tpgw + Tngw + Sinw)$
 $= (215299 + 0 + 779509 + 0) = 994808 \text{ N}$

Failure Path [Path 3-3]:
 $= (Sonw + Tngw + Sinw)$
 $= (215299 + 779509 + 0) = 994808 \text{ N}$

Summary of Failure Path Calculations:

Path 1-1 = 1288135 N	, must exceed W = 465011 N	or W1 = 504784 N
Path 2-2 = 994808 N	, must exceed W = 465011 N	or W2 = 710382 N
Path 3-3 = 994808 N	, must exceed W = 465011 N	or W3 = 710382 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 2.66 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 6.7123 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 228.7123 mm

Input Echo, WRC297 Item 1, Description: A16" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.5000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	2100.000	mm
Shell Thickness	22.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 138.00	MPa
Vessel Hot S.I. Allowable	Smh 138.00	MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	1.5000	mm
Nozzle Diameter	149.073	mm
Nozzle Thickness	44.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-350 LF1	
Nozzle Cold S.I. Allowable	SNmc 118.00	MPa
Nozzle Hot S.I. Allowable	SNmh 118.00	MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUStained loads.*

Design Internal Pressure	Dp	2.66	MPa
Radial Load	P	7199.67	N
Circumferential Shear	Vc	9599.58	N
Longitudinal Shear	Vl	9599.58	N
Circumferential Moment	Mc	0.288E+07	N-mm
Longitudinal Moment	Ml	0.375E+07	N-mm
Torsional Moment	Mt	0.432E+07	N-mm

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dn/Tn (5.388) is < 10
 check the Limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$
 $= 237.073 + 2 \cdot 1.65 \cdot \sqrt{(1061.75(22.0 - 1.5))}$
 $= 723.930 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2123.500	mm
Nozzle Outside Diameter	(d) =	237.073	mm
Vessel Thickness used	(T) =	20.500	mm
Nozzle Thickness used	(t) =	42.500	mm
T / t	=	0.482	
d / t	=	10.000	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.136$$

Nr/P	=	0.097
Mr/P	=	0.207

M0/P = 0.060
 N0/P = 0.126
 MrD/Mc = 0.305
 NrDL/Mc = 0.092
 M0d/Mc = 0.095
 N0DL/Mc = 0.046
 MrD/Ml = 0.179
 NrDL/Ml = 0.085
 M0D/Ml = 0.058
 N0DL/Ml = 0.125

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	-2	-2	-2	-2
Outplane Bending (P)	-6	6	-6	6
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-4	-4	4	4
Outplane Bending (ML)	-13	13	13	-13
Normal Pressure Stress	135	137	135	137
Outplane Stress Summary	108	150	144	133

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	-1	-1	-1	-1
Inplane Bending (P)	-21	21	-21	21
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-3	-3	3	3
Inplane Bending (ML)	-40	40	40	-40
Inplane Pressure Stress	67	67	67	67
Inplane Stress Summary	1	124	88	49

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	2	2	2	2
Shear Stress Summary	3	3	1	1

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	109	150	144	133

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	-2	-2	-2	-2
Outplane Bending (P)	-6	6	-6	6
Outplane Membrane (Mc)	-1	-1	1	1
Outplane Bending (Mc)	-16	16	16	-16
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0



Normal Pressure Stress	67	67	67	67
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Outplane Stress Summary	41	86	76	56
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	-1	-1	-1	-1
Inplane Bending (P)	-21	21	-21	21
Inplane Membrane (Mc)	-2	-2	2	2
Inplane Bending (Mc)	-52	52	52	-52
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	135	137	135	137

Inplane Stress Summary	56	207	167	107
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	2	2	2	2

Shear Stress Summary	1	1	3	3
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	56	207	167	107

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	-2	-2	-2	-2
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-4	-4	4	4
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	4	7	4	7

Hoop Stress Summary	-2	0	7	10
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	-3	3	-3	3
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-3	-3	3	3
Axial Bending (ML)	-7	7	7	-7
Axial Pressure Stress	2	2	2	2

Axial Stress Summary	-12	9	8	1
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
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Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	1	1	1	1

Shear Stress Summary	1	1	0	0
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress	12	9	8	10
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses in the	Left	Left	Right	Right
hoop direction)	Outside	Inside	Outside	Inside

Hoop Membrane (P)	-2	-2	-2	-2
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-1	-1	1	1
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	4	7	4	7

Hoop Stress Summary	0	4	3	6
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	-3	3	-3	3
Axial Membrane (Mc)	-2	-2	2	2
Axial Bending (Mc)	-10	10	10	-10
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	2	2	2	2

Axial Stress Summary	-14	13	11	-2
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	1	1	1	1

Shear Stress Summary	0	0	1	1
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress	15	13	11	9
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WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	135	137	135	137	135	137	135	137
Circ. Pl (SUS)	-6	-6	2	2	-3	-3	1	1
Circ. Q (SUS)	-19	19	7	-7	-73	73	31	-31

Long. Pm (SUS)	67	67	67	67	67	67	67	67
Long. Pl (SUS)	-4	-4	2	2	-3	-3	-1	-1
Long. Q (SUS)	-61	61	19	-19	-22	22	10	-10
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	135.0	137.0	135.0	137.0	135.0	137.0	135.0	137.0
Pm+Pl (SUS)	129.0	131.0	137.0	139.0	132.0	134.0	136.0	138.0
Pm+Pl+Q (Total)	110.1	150.3	144.0	132.0	59.1	207.0	167.1	107.2

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	137.00	138.00	Passed
Pm+Pl (SUS)	139.01	207.00	Passed
Pm+Pl+Q (TOTAL)	207.01	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	4	7	4	7	4	7	4	7
Circ. Pl (SUS)	-6	-6	2	2	-3	-3	-1	-1
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	2	2	2	2	2	2	2	2
Long. Pl (SUS)	-3	-3	3	3	-2	-2	2	2
Long. Q (SUS)	-10	10	4	-4	-13	13	7	-7
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	4.0	7.0	4.0	7.0	4.0	7.0	4.0	7.0
Pm+Pl (SUS)	2.0	2.0	6.0	9.0	1.0	4.0	4.0	6.0
Pm+Pl+Q (Total)	11.1	9.1	9.3	9.1	14.1	13.1	11.1	9.2

Nozzle Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	7.00	118.00	Passed
Pm+Pl (SUS)	9.00	177.00	Passed
Pm+Pl+Q (TOTAL)	14.14	354.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.



Input, Nozzle Desc: P12"

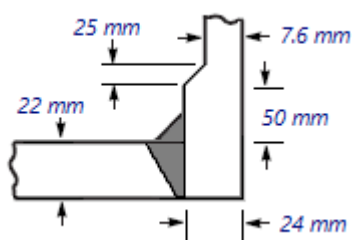
From: 20

Pressure for Reinforcement Calculations	P	2.6593	MPa
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	+N+IT
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	2100.00	mm
Shell Finished (Minimum) Thickness	t	22.0000	mm
Shell Internal Corrosion Allowance	cas	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		2550.00	mm
User Entered Minimum Design Metal Temperature		-40.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF1	+IT
Material UNS Number		K03009	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		160	
Flange Material		SA-350 LF1	+IT
Hub Height of Integral Nozzle	h	50.0000	mm
Height of Beveled Transition	L'	25.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		24.0000	mm
Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		22.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: P12"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 1.773 in.
 Actual Thickness Used in Calculation 0.301 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2.66 \cdot 1052 / (138 \cdot 1 - 0.6 \cdot 2.66)$
 $= 20.5000 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2.66 \cdot 24 / (118 \cdot 1 - 0.6 \cdot 2.66)$
 $= 0.5487 \text{ mm}$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 16.4 / 0.5773$
 $= 28.3270 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	101.3250	mm
Parallel to Vessel Wall	Rn+tn+t	50.6625	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	43.6905	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: P12".
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 2.0487	mm
Wall Thickness per UG16(b),	tr16b	= 3.0000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 22.0000	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 22.0000	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 3.0000	mm
Wall Thickness per table UG-45	tb3	= 4.9200	mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[4.92, \max(22, 3)]$
 $= 4.9200 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(2.05, 4.92)$
 $= 4.9200 \text{ mm}$

Available Nozzle Neck Thickness = 7.6454 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

*Note:
 This Material was specified as being an Impact Tested (Low Temperature) Material.*

Impact Test Temperature provided per Specification	-29 °C
Gov. MDMT of the nozzle to shell joint welded assembly :	-29 °C



Weld Size Calculations, Description: P12"

Intermediate Calc. for nozzle/shell welds T_{min} 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * W _o mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 1.0309 mm

The Cut Length for this Nozzle is, Drop + H_o + H + T : 223.0310 mm

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Input, Nozzle Desc: L72"

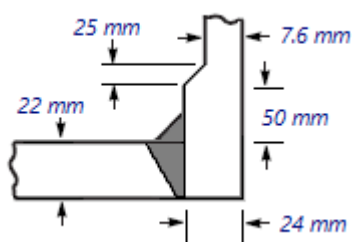
From: 20

Pressure for Reinforcement Calculations	P	2.6593	MPa
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	+N+IT
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	2100.00	mm
Shell Finished (Minimum) Thickness	t	22.0000	mm
Shell Internal Corrosion Allowance	cas	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		2750.00	mm
User Entered Minimum Design Metal Temperature		-40.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF1	+IT
Material UNS Number		K03009	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		30.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		160	
Flange Material		SA-350 LF1	+IT
Hub Height of Integral Nozzle	h	50.0000	mm
Height of Beveled Transition	L'	25.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		24.0000	mm
Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		22.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: L72"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 1.773 in.
 Actual Thickness Used in Calculation 0.301 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2.66 \cdot 1052 / (138 \cdot 1 - 0.6 \cdot 2.66)$
 $= 20.5000 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2.66 \cdot 24 / (118 \cdot 1 - 0.6 \cdot 2.66)$
 $= 0.5487 \text{ mm}$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 16.4 / 0.5773$
 $= 28.3270 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	101.3250	mm
Parallel to Vessel Wall	Rn+tn+t	50.6625	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	43.6905	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: L7.2".
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 2.0487	mm
Wall Thickness per UG16(b),	tr16b	= 3.0000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 22.0000	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 22.0000	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 3.0000	mm
Wall Thickness per table UG-45	tb3	= 4.9200	mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[4.92, \max(22, 3)]$
 $= 4.9200 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(2.05, 4.92)$
 $= 4.9200 \text{ mm}$

Available Nozzle Neck Thickness = 7.6454 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

*Note:
 This Material was specified as being an Impact Tested (Low Temperature) Material.*

Impact Test Temperature provided per Specification	-29 °C
Gov. MDMT of the nozzle to shell joint welded assembly :	-29 °C



Weld Size Calculations, Description: L72"

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 1.0309 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 223.0310 mm

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Input, Nozzle Desc: L82"

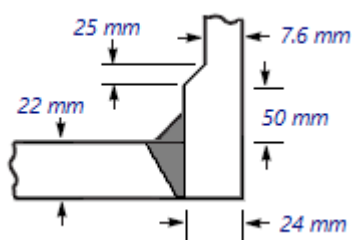
From: 20

Pressure for Reinforcement Calculations	P	2.6593	MPa
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	+N+IT
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	2100.00	mm
Shell Finished (Minimum) Thickness	t	22.0000	mm
Shell Internal Corrosion Allowance	cas	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		2750.00	mm
User Entered Minimum Design Metal Temperature		-40.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF1	+IT
Material UNS Number		K03009	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		330.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		160	
Flange Material		SA-350 LF1	+IT
Hub Height of Integral Nozzle	h	50.0000	mm
Height of Beveled Transition	L'	25.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		24.0000	mm
Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		22.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: L82"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 1.773 in.
 Actual Thickness Used in Calculation 0.301 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2.66 \cdot 1052 / (138 \cdot 1 - 0.6 \cdot 2.66)$
 $= 20.5000 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2.66 \cdot 24 / (118 \cdot 1 - 0.6 \cdot 2.66)$
 $= 0.5487 \text{ mm}$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 16.4 / 0.5773$
 $= 28.3270 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	101.3250	mm
Parallel to Vessel Wall	Rn+tn+t	50.6625	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	43.6905	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: L82".
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 2.0487	mm
Wall Thickness per UG16(b),	tr16b	= 3.0000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 22.0000	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 22.0000	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 3.0000	mm
Wall Thickness per table UG-45	tb3	= 4.9200	mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[4.92, \max(22, 3)]$
 $= 4.9200 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(2.05, 4.92)$
 $= 4.9200 \text{ mm}$

Available Nozzle Neck Thickness = 7.6454 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -29 °C



Weld Size Calculations, Description: L82"

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 1.0309 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 223.0310 mm

This nozzle or its diameter limit intersects weld seam # [3] on this element. The distance between the nozzle and seam is, 48.51 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 50.66 mm

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Input, Nozzle Desc: P22"

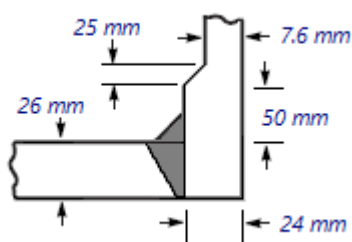
From: 30

Pressure for Reinforcement Calculations	P	2.6593	MPa
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	+N+IT
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	2100.00	mm
Shell Finished (Minimum) Thickness	t	26.0000	mm
Shell Internal Corrosion Allowance	cas	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		3550.00	mm
User Entered Minimum Design Metal Temperature		-40.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF1	+IT
Material UNS Number		K03009	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		160	
Flange Material		SA-350 LF1	+IT
Hub Height of Integral Nozzle	h	50.0000	mm
Height of Beveled Transition	L'	25.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		24.0000	mm
Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		22.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		300	
Flange Grade		GR 1.4	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: P22"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 1.773 in.
 Actual Thickness Used in Calculation 0.301 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2.66 \cdot 1052 / (138 \cdot 1 - 0.6 \cdot 2.66)$
 $= 20.5000 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2.66 \cdot 24 / (118 \cdot 1 - 0.6 \cdot 2.66)$
 $= 0.5487 \text{ mm}$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 16.4 / 0.5773$
 $= 28.3270 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	109.3250	mm
Parallel to Vessel Wall	Rn+tn+t	54.6625	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	43.6905	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: P22".
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 2.0487	mm
Wall Thickness per UG16(b),	tr16b	= 3.0000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 22.0000	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 22.0000	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 3.0000	mm
Wall Thickness per table UG-45	tb3	= 4.9200	mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[4.92, \max(22, 3)]$
 $= 4.9200 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(2.05, 4.92)$
 $= 4.9200 \text{ mm}$

Available Nozzle Neck Thickness = 7.6454 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification	-29 °C
Calculated Minimum Design Metal Temperature	-104 °C



Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Weld Size Calculations, Description: P22"

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 1.0309 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 227.0309 mm

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Input, Nozzle Desc: S112"

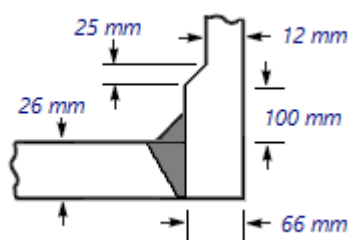
From: 30

Pressure for Reinforcement Calculations	P	2.6593	MPa
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	+N+IT
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	2100.00	mm
Shell Finished (Minimum) Thickness	t	26.0000	mm
Shell Internal Corrosion Allowance	cas	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		5050.00	mm
User Entered Minimum Design Metal Temperature		-40.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF1	+IT
Material UNS Number		K03009	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		12.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	12.0000	mm
Flange Material		SA-350 LF1	+IT
Hub Height of Integral Nozzle	h	100.0000	mm
Height of Beveled Transition	L'	25.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		66.0000	mm
Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		22.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		300	
Flange Grade		GR 1.4	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: S112"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 12.000 in.
Actual Thickness Used in Calculation 0.472 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $2.66 \cdot 1052 / (138 \cdot 1 - 0.6 \cdot 2.66)$
= 20.5000 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $2.66 \cdot 154 / (118 \cdot 1 - 0.6 \cdot 2.66)$
= 3.5159 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
= (Hub Thickness - Neck Thickness) / tan(30)
= 54/0.5773
= 93.5308 mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

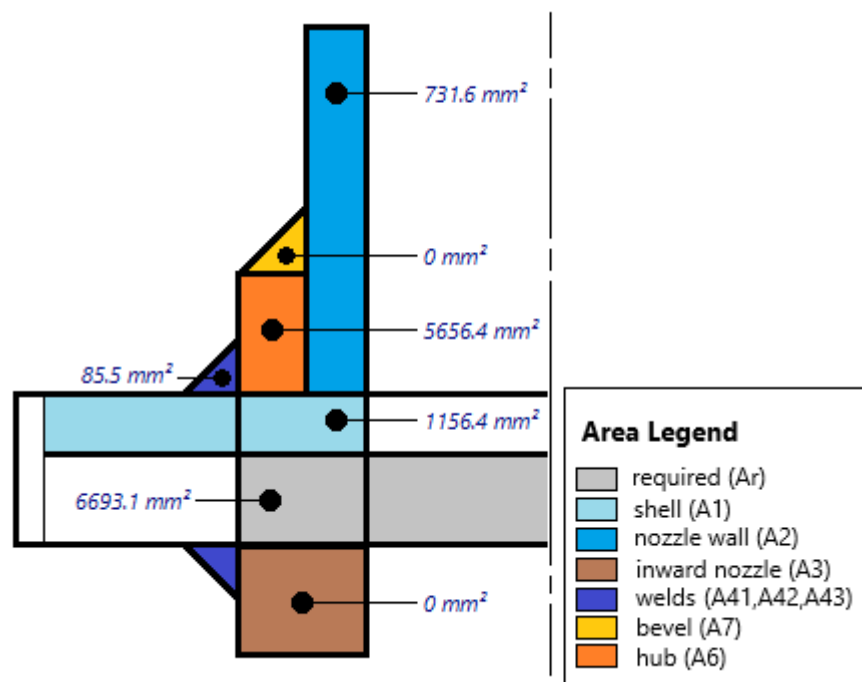
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	615.6000	mm
Parallel to Vessel Wall, opening length	d	307.8000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	61.2500	mm

Weld Strength Reduction Factor [fr1]:
= min(1, S_n/S_v)
= min(1, 118/138)
= 0.855

Weld Strength Reduction Factor [fr2]:
= min(1, S_n/S_v)
= min(1, 118/138)
= 0.855

Weld Strength Reduction Factor [fr3]:
= min(fr2, fr4)
= min(0.86, 1)
= 0.855



Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	6693.120	NA	NA
Area in Shell	A1	1156.426	NA	NA
Area in Nozzle Wall	A2	731.574	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	85.509	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub	A6	5656.407	NA	NA
TOTAL AREA AVAILABLE	Atot	7629.916	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$\begin{aligned}
 &= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)} \\
 &= (308 \cdot 20.5 \cdot 1 + 2 \cdot 64.5 \cdot 20.5 \cdot 1 \cdot (1 - 0.86)) \\
 &= 6693.120 \text{ mm}^2
 \end{aligned}$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$\begin{aligned}
 &= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1) \\
 &= 308(1 \cdot 24.5 - 1 \cdot 20.5) - 2 \cdot 64.5 \\
 &\quad (1 \cdot 24.5 - 1 \cdot 20.5) \cdot (1 - 0.86) \\
 &= 1156.426 \text{ mm}^2
 \end{aligned}$$

Area Available in Nozzle Projecting Outward [A2]:

$$\begin{aligned}
 &= (2 \cdot tlnp)(tn - trn)fr2 \\
 &= (2 \cdot 61.3)(10.5 - 3.52)0.86 \\
 &= 731.574 \text{ mm}^2
 \end{aligned}$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$\begin{aligned}
 &= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2 \\
 &= 10^2 \cdot 0.86 + (0)^2 \cdot 0.86 \\
 &= 85.509 \text{ mm}^2
 \end{aligned}$$



Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lp}, h_o, H_{ubht})) \cdot (H_{ubt} - t_n) \cdot f_{r2}$$

$$= (2 \cdot \min(61.3, 200, 100)) \cdot (66 - 12) \cdot 0.86$$

$$= 5656.407 \text{ mm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 5.0159 mm
Wall Thickness per UG16(b),	tr16b = 3.0000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 22.0000 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 22.0000 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 3.0000 mm
Wall Thickness per table UG-45	tb3 = 9.8312 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[9.83, \max(22, 3)]$$

$$= 9.8312 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(5.02, 9.83)$$

$$= 9.8312 \text{ mm}$$

Available Nozzle Neck Thickness = 12.0000 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 43.6,	Allowable	: 118.0 MPa	Passed
Expansion	: 0.0,	Allowable	: 251.4 MPa	Passed
Occasional	: 18.8,	Allowable	: 156.9 MPa	Passed
Shear	: 16.4,	Allowable	: 82.6 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification	-29 °C
Calculated Minimum Design Metal Temperature	-104 °C

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification	-29 °C
Gov. MDMT of the nozzle to shell joint welded assembly :	-29 °C

Flange MDMT including Temperature reduction per UCS-66.1:

MDMT of ASME B16.5/47 flange per Matl. Specification	-29 °C
Flange MDMT with Temp reduction per UCS-66(i)(2)	-50 °C

Where the Stress Reduction Ratio per UCS-66(i)(2) is :

$$\text{Design Pressure/Ambient Rating} = 2.66/4.26 = 0.624$$

Weld Size Calculations, Description: S112"

Intermediate Calc. for nozzle/shell Welds	Tmin	19.0000 mm
---	------	------------

Results Per UW-16.1:

Required Thickness	Actual Thickness
--------------------	------------------



Nozzle Weld 6.0000 = Min per Code 7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
= max(0, (A-A1+2•tn•fr1•(E1•t-tr))Sv)
= max(0, (6693 - 1156 + 2 • 64.5 • 0.86 •
(1 • 24.5 - 20.5))138)
= 824867.88 N

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
= (A2+A4-(Wi-Can/.707)²•fr2)•Sv
= (6388 + 85.5 - 0 • 0.86) • 138
= 893249.50 N

Weld Load [W2]:
= (A2 + A3 + A4 + (2 • tn • t • fr1)) • Sv
= (6388 + 0 + 85.5 + (2703)) • 138
= 1266156.62 N

Weld Load [W3]:
= (A2+A3+A4+A5+(2•tn•t•fr1))•S
= (6388 + 0 + 85.5 + 0 + (2703)) • 138
= 1266156.62 N

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
= (π/2) • Dlo • Wo • 0.49 • Snw
= (3.14/2.0) • 437 • 10 • 0.49 • 118
= 396683. N

Shear, Nozzle Wall [Snw]:
= (π • (Dlr + Dlo)/4) • (Thk - Can) • 0.7 • Sn
= (3.14 • 186) • (66 - 1.5) • 0.7 • 118
= 3115414. N

Tension, Shell Groove Weld [Tngw]:
= (π/2) • Dlo • Wgnvi • 0.74 • Sng
= (3.14/2.0) • 437 • (22 - 0) • 0.74 • 138
= 1541314. N

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
= (SONW + SNW) = (396683 + 3115413) = 3512096 N

Failure Path [Path 2-2]:
= (Sonw + Tpgw + Tngw + Sinw)
= (396683 + 0 + 1541314 + 0) = 1937997 N

Failure Path [Path 3-3]:
= (Sonw + Tngw + Sinw)
= (396683 + 1541314 + 0) = 1937997 N

Summary of Failure Path Calculations:

Path 1-1 = 3512096 N	, must exceed W = 824867 N	or W1 = 893249 N
Path 2-2 = 1937997 N	, must exceed W = 824867 N	or W2 = 1266156 N
Path 3-3 = 1937997 N	, must exceed W = 824867 N	or W3 = 1266156 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 2.85 MPa

The Drop for this Nozzle is : 22.9647 mm



The Cut Length for this Nozzle is, Drop + Ho + H + T : 248.9646 mm

Input Echo, WRC297 Item 1, Description: S112" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.5000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	2100.000	mm
Shell Thickness	26.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00 MPa
Vessel Hot S.I. Allowable	Smh	138.00 MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	1.5000	mm
Nozzle Diameter	304.800	mm
Nozzle Thickness	66.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-350 LF1	
Nozzle Cold S.I. Allowable	SNmc	118.00 MPa
Nozzle Hot S.I. Allowable	SNmh	118.00 MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSained loads.*

Design Internal Pressure	Dp	2.66	MPa
Radial Load	P	14399.42	N
Circumferential Shear	Vc	19199.08	N
Longitudinal Shear	Vl	19199.08	N
Circumferential Moment	Mc	0.115E+08	N-mm
Longitudinal Moment	Ml	0.150E+08	N-mm
Torsional Moment	Mt	0.173E+08	N-mm

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dn/Tn (6.618) is < 10
 check the limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 436.8 + 2 \cdot 1.65 \cdot \sqrt{(1063.75(26.0 - 1.5))}$$

$$= 969.542 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2127.500 mm
Nozzle Outside Diameter	(d) =	436.800 mm
Vessel Thickness used	(T) =	24.500 mm
Nozzle Thickness used	(t) =	64.500 mm
T / t	=	0.380
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.913$$

Nr/P	=	0.071
Mr/P	=	0.148
M0/P	=	0.043
N0/P	=	0.092



MrD/Mc = 0.288
 NrDL/Mc = 0.095
 M0d/Mc = 0.089
 N0DL/Mc = 0.071
 MrD/Ml = 0.116
 NrDL/Ml = 0.076
 M0D/Ml = 0.037
 N0DL/Ml = 0.117

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Outplane Membrane (P)	-2	-2	-2	-2
Outplane Bending (P)	-6	6	-6	6
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-6	-6	6	6
Outplane Bending (ML)	-12	12	12	-12
Normal Pressure Stress	112	115	112	115
Outplane Stress Summary	85	125	123	113

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Inplane Membrane (P)	-1	-1	-1	-1
Inplane Bending (P)	-21	21	-21	21
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-4	-4	4	4
Inplane Bending (ML)	-39	39	39	-39
Inplane Pressure Stress	56	56	56	56
Inplane Stress Summary	-10	111	77	40

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	2	2	2	2
Shear Stress Summary	3	3	1	1

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Two * Max Shear Stress	95	126	123	113

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	D1 Right Inside
Outplane Membrane (P)	-2	-2	-2	-2
Outplane Bending (P)	-6	6	-6	6
Outplane Membrane (Mc)	-3	-3	3	3
Outplane Bending (Mc)	-23	23	23	-23
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	56	56	56	56



Outplane Stress Summary 21 80 74 40

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE
 (Stresses parallel to
 circumferential plane)

	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside

Inplane Membrane (P)	-1	-1	-1	-1
Inplane Bending (P)	-21	21	-21	21
Inplane Membrane (Mc)	-4	-4	4	4
Inplane Bending (Mc)	-75	75	75	-75
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	112	115	112	115

Inplane Stress Summary 9 206 169 63

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE
 (Shear stress normal to
 circumferential plane)

	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	2	2	2	2

Shear Stress Summary 1 1 3 3

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE
 (Stress Intensities)

	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside

Two * Max Shear Stress 21 206 169 63

Nozzle Stresses (MPa):

LONGITUDINAL PLANE
 (Stresses in the
 hoop direction)

	Au	Al	Bu	Bl
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside

Hoop Membrane (P)	-2	-2	-2	-2
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-6	-6	6	6
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	6	9	6	9

Hoop Stress Summary -2 0 10 13

Nozzle Stresses (MPa):

LONGITUDINAL PLANE
 (Stresses Normal to
 pipe cross-section)

	Au	Al	Bu	Bl
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	-2	2	-2	2
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-2	-2	2	2
Axial Bending (ML)	-3	3	3	-3
Axial Pressure Stress	3	3	3	3

Axial Stress Summary -5 6 6 3

Nozzle Stresses (MPa):

LONGITUDINAL PLANE
 (Shear stress)

	Au	Al	Bu	Bl
	Outside	Inside	Outside	Inside

Shear due to (Vc) 0 0 0 0



Shear due to (V1)	0	0	0	0
Shear due to Torsion	0	0	0	0

Shear Stress Summary	1	1	0	0
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	6	6	10	13

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	-2	-2	-2	-2
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-3	-3	3	3
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	6	9	6	9
Hoop Stress Summary	0	4	7	10

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	-2	2	-2	2
Axial Membrane (Mc)	-1	-1	1	1
Axial Bending (Mc)	-9	9	9	-9
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	3	3	3	3
Axial Stress Summary	-10	12	11	-1

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (V1)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	0	0	1	1

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	11	12	11	12

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	112	115	112	115	112	115	112	115
Circ. Pl (SUS)	-8	-8	4	4	-5	-5	3	3
Circ. Q (SUS)	-18	18	6	-6	-96	96	54	-54
Long. Pm (SUS)	56	56	56	56	56	56	56	56



Long. P1 (SUS)	-5	-5	3	3	-5	-5	1	1
Long. Q (SUS)	-60	60	18	-18	-29	29	17	-17
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear P1 (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	112.0	115.0	112.0	115.0	112.0	115.0	112.0	115.0
Pm+P1 (SUS)	104.0	107.0	116.0	119.0	107.0	110.0	115.0	118.0
Pm+P1+Q (Total)	95.2	125.6	122.0	113.0	22.1	206.0	169.1	64.4

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	115.00	138.00	Passed
Pm+P1 (SUS)	119.02	207.00	Passed
Pm+P1+Q (TOTAL)	206.01	413.99	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	6	9	6	9	6	9	6	9
Circ. P1 (SUS)	-8	-8	4	4	-5	-5	1	1
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	3	3	3	3	3	3	3	3
Long. P1 (SUS)	-2	-2	2	2	-1	-1	1	1
Long. Q (SUS)	-5	5	1	-1	-11	11	7	-7
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear P1 (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	6.0	9.0	6.0	9.0	6.0	9.0	6.0	9.0
Pm+P1 (SUS)	3.0	1.0	10.0	13.0	2.0	4.0	7.0	10.0
Pm+P1+Q (Total)	4.0	6.0	10.0	13.0	10.0	13.0	11.0	13.0

Nozzle Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	9.00	118.00	Passed
Pm+P1 (SUS)	13.00	177.00	Passed
Pm+P1+Q (TOTAL)	13.00	354.00	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.



Input, Nozzle Desc: L32"

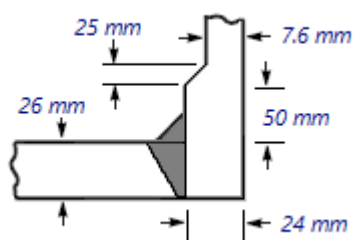
From: 30

Pressure for Reinforcement Calculations	P	2.6593	MPa
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	+N+IT
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	2100.00	mm
Shell Finished (Minimum) Thickness	t	26.0000	mm
Shell Internal Corrosion Allowance	cas	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		5050.00	mm
User Entered Minimum Design Metal Temperature		-40.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF1	+IT
Material UNS Number		K03009	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		30.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		160	
Flange Material		SA-105	+IT
Hub Height of Integral Nozzle	h	50.0000	mm
Height of Beveled Transition	L'	25.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		24.0000	mm
Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		22.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: L32"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 1.773 in.
 Actual Thickness Used in Calculation 0.301 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2.66 \cdot 1052 / (138 \cdot 1 - 0.6 \cdot 2.66)$
 $= 20.5000 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2.66 \cdot 24 / (118 \cdot 1 - 0.6 \cdot 2.66)$
 $= 0.5487 \text{ mm}$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 16.4 / 0.5773$
 $= 28.3270 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	109.3250	mm
Parallel to Vessel Wall	Rn+tn+t	54.6625	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	43.6905	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: L32".
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 2.0487	mm
Wall Thickness per UG16(b),	tr16b	= 3.0000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 22.0000	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 22.0000	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 3.0000	mm
Wall Thickness per table UG-45	tb3	= 4.9200	mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[4.92, \max(22, 3)]$
 $= 4.9200 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(2.05, 4.92)$
 $= 4.9200 \text{ mm}$

Available Nozzle Neck Thickness = 7.6454 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification	-29 °C
Calculated Minimum Design Metal Temperature	-104 °C



Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = 2.66/5.10 = 0.521

Weld Size Calculations, Description: L32"

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 1.0309 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 227.0309 mm

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Input, Nozzle Desc: L42"

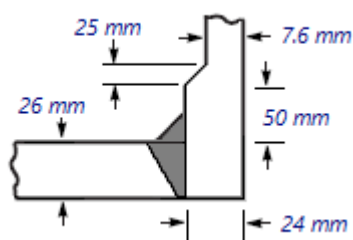
From: 30

Pressure for Reinforcement Calculations	P	2.6593	MPa
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	+N+IT
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	2100.00	mm
Shell Finished (Minimum) Thickness	t	26.0000	mm
Shell Internal Corrosion Allowance	cas	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		5050.00	mm
User Entered Minimum Design Metal Temperature		-40.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF1	+IT
Material UNS Number		K03009	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		330.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		160	
Flange Material		SA-105	+IT
Hub Height of Integral Nozzle	h	50.0000	mm
Height of Beveled Transition	L'	25.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		24.0000	mm
Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		22.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: L42"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 1.773 in.
Actual Thickness Used in Calculation 0.301 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $2.66 \cdot 1052 / (138 \cdot 1 - 0.6 \cdot 2.66)$
= 20.5000 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $2.66 \cdot 24 / (118 \cdot 1 - 0.6 \cdot 2.66)$
= 0.5487 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
= (Hub Thickness - Neck Thickness) / tan(30)
= 16.4/0.5773
= 28.3270 mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	109.3250	mm
Parallel to Vessel Wall	Rn+tn+t	54.6625	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	43.6905	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: L42".
This calculation is valid for nozzles that meet all the requirements of
paragraph UG-36. Please check the Code carefully, especially for nozzles
that are not isolated or do not meet Code spacing requirements. To force
the computation of areas for small nozzles go to Tools->Configuration
and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 2.0487	mm
Wall Thickness per UG16(b),	tr16b	= 3.0000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 22.0000	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 22.0000	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 3.0000	mm
Wall Thickness per table UG-45	tb3	= 4.9200	mm

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1, tb2)]
= min[4.92, max(22, 3)]
= 4.9200 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(2.05, 4.92)
= 4.9200 mm

Available Nozzle Neck Thickness = 7.6454 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification	-29 °C
Calculated Minimum Design Metal Temperature	-104 °C



Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C

Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

Design Pressure/Ambient Rating = 2.66/5.10 = 0.521

Weld Size Calculations, Description: L42"

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 1.0309 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 227.0309 mm

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Input, Nozzle Desc: L52"

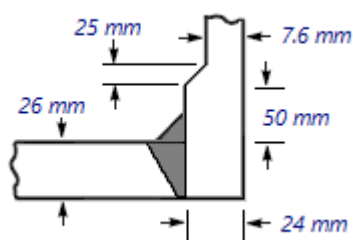
From: 30

Pressure for Reinforcement Calculations	P	2.6593	MPa
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	+N+IT
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	2100.00	mm
Shell Finished (Minimum) Thickness	t	26.0000	mm
Shell Internal Corrosion Allowance	cas	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		4050.00	mm
User Entered Minimum Design Metal Temperature		-40.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF1	+IT
Material UNS Number		K03009	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		30.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		160	
Flange Material		SA-105	+IT
Hub Height of Integral Nozzle	h	50.0000	mm
Height of Beveled Transition	L'	25.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		24.0000	mm
Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		22.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: L52"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 1.773 in.
Actual Thickness Used in Calculation 0.301 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $2.66 \cdot 1052 / (138 \cdot 1 - 0.6 \cdot 2.66)$
= 20.5000 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $2.66 \cdot 24 / (118 \cdot 1 - 0.6 \cdot 2.66)$
= 0.5487 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
= (Hub Thickness - Neck Thickness) / tan(30)
= 16.4/0.5773
= 28.3270 mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	109.3250	mm
Parallel to Vessel Wall	Rn+tn+t	54.6625	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	43.6905	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: L52".
This calculation is valid for nozzles that meet all the requirements of
paragraph UG-36. Please check the Code carefully, especially for nozzles
that are not isolated or do not meet Code spacing requirements. To force
the computation of areas for small nozzles go to Tools->Configuration
and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 2.0487	mm
Wall Thickness per UG16(b),	tr16b	= 3.0000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 22.0000	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 22.0000	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 3.0000	mm
Wall Thickness per table UG-45	tb3	= 4.9200	mm

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1,tb2)]
= min[4.92, max(22, 3)]
= 4.9200 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(2.05, 4.92)
= 4.9200 mm

Available Nozzle Neck Thickness = 7.6454 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification	-29 °C
Calculated Minimum Design Metal Temperature	-104 °C



Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C

Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

Design Pressure/Ambient Rating = $2.66/5.10 = 0.521$

Weld Size Calculations, Description: L52"

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 * W_o \text{ mm}$

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 1.0309 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 227.0309 mm

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Input, Nozzle Desc: L62"

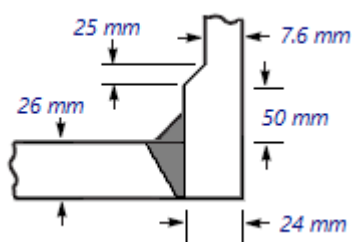
From: 30

Pressure for Reinforcement Calculations	P	2.6593	MPa
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	+N+IT
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	2100.00	mm
Shell Finished (Minimum) Thickness	t	26.0000	mm
Shell Internal Corrosion Allowance	cas	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		4050.00	mm
User Entered Minimum Design Metal Temperature		-40.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF1	+IT
Material UNS Number		K03009	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		330.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		160	
Flange Material		SA-105	+IT
Hub Height of Integral Nozzle	h	50.0000	mm
Height of Beveled Transition	L'	25.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		24.0000	mm
Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		22.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: L62"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 1.773 in.
 Actual Thickness Used in Calculation 0.301 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2.66 \cdot 1052 / (138 \cdot 1 - 0.6 \cdot 2.66)$
 $= 20.5000 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2.66 \cdot 24 / (118 \cdot 1 - 0.6 \cdot 2.66)$
 $= 0.5487 \text{ mm}$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 16.4 / 0.5773$
 $= 28.3270 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	109.3250	mm
Parallel to Vessel Wall	Rn+tn+t	54.6625	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	43.6905	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: L62".
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 2.0487	mm
Wall Thickness per UG16(b),	tr16b	= 3.0000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 22.0000	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 22.0000	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 3.0000	mm
Wall Thickness per table UG-45	tb3	= 4.9200	mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[4.92, \max(22, 3)]$
 $= 4.9200 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(2.05, 4.92)$
 $= 4.9200 \text{ mm}$

Available Nozzle Neck Thickness = 7.6454 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

*Note:
 This Material was specified as being an Impact Tested (Low Temperature) Material.*

Impact Test Temperature provided per Specification	-29 °C
Calculated Minimum Design Metal Temperature	-104 °C



Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = 2.66/5.10 = 0.521

Weld Size Calculations, Description: L62"

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 1.0309 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 227.0309 mm

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Input, Nozzle Desc: V13"

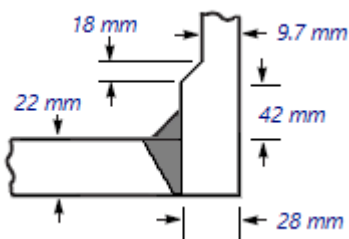
From: 40

Pressure for Reinforcement Calculations	P	2.6593	MPa
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	+N+IT
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	2100.00	mm
Shell Finished (Minimum) Thickness	t	22.0000	mm
Shell Internal Corrosion Allowance	cas	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		7550.00	mm
User Entered Minimum Design Metal Temperature		-40.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF1	+IT
Material UNS Number		K03009	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		3.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		160	
Flange Material		SA-350 LF1	+IT
Hub Height of Integral Nozzle	h	42.0000	mm
Height of Beveled Transition	L'	18.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		28.0000	mm
Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		22.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		600	
Flange Grade		GR 1.4	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: V13"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 2.734 in.
Actual Thickness Used in Calculation 0.383 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $2.66 \cdot 1052 / (138 \cdot 1 - 0.6 \cdot 2.66)$
= 20.5000 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $2.66 \cdot 36.2 / (118 \cdot 1 - 0.6 \cdot 2.66)$
= 0.8274 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
= (Hub Thickness - Neck Thickness) / $\tan(30)$
= $18.3 / 0.5773$
= 31.6367 mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

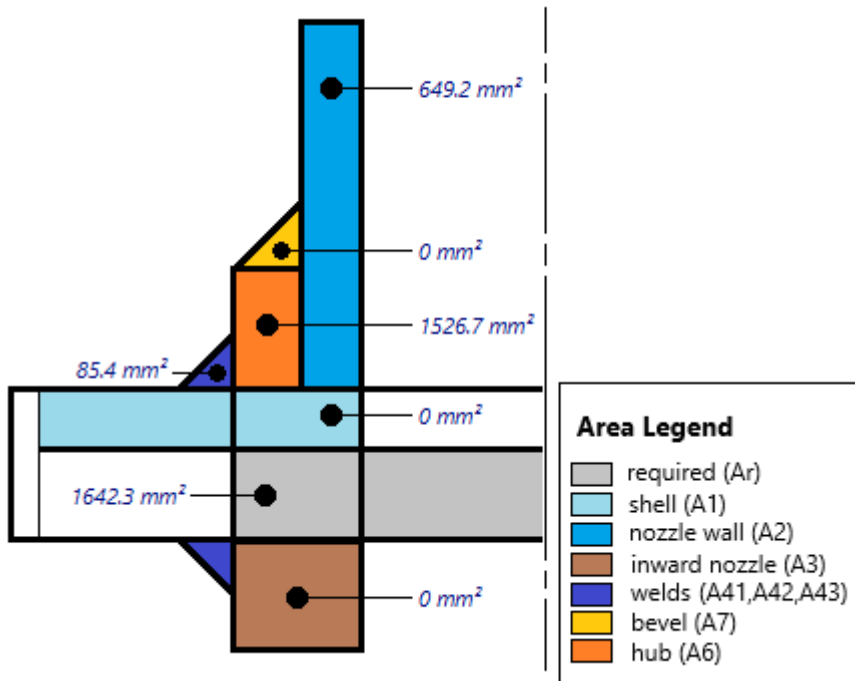
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	144.8618	mm
Parallel to Vessel Wall, opening length	d	72.4309	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	51.2500	mm

Weld Strength Reduction Factor [fr1]:
= $\min(1, S_n / S_v)$
= $\min(1, 118 / 138)$
= 0.855

Weld Strength Reduction Factor [fr2]:
= $\min(1, S_n / S_v)$
= $\min(1, 118 / 138)$
= 0.855

Weld Strength Reduction Factor [fr3]:
= $\min(fr2, fr4)$
= $\min(0.86, 1)$
= 0.855



Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	1642.280	NA	NA
Area in Shell	A1	0.000	NA	NA
Area in Nozzle Wall	A2	649.215	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	85.440	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub + Bevel	A6	1526.661	NA	NA
TOTAL AREA AVAILABLE	Atot	2261.315	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (72.4 \cdot 20.5 \cdot 1 + 2 \cdot 26.5 \cdot 20.5 \cdot 1 \cdot (1 - 0.86))$
 $= 1642.280 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 72.4(1 \cdot 20.5 - 1 \cdot 20.5) - 2 \cdot 26.5$
 $(1 \cdot 20.5 - 1 \cdot 20.5) \cdot (1 - 0.86)$
 $= 0.000 \text{ mm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn)fr2$
 $= (2 \cdot 51.2)(8.23 - 0.83)0.86$
 $= 649.215 \text{ mm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= (Wo^2 - \text{Area Lost}) \cdot fr2 + (Wi - can/0.707)^2 - \text{Area Lost} \cdot fr2$
 $= (10^2 - 0.081) \cdot 0.86 + (0^2 - 0) \cdot 0.86$
 $= 85.440 \text{ mm}^2$

Area Available in the Hub Section [A6]:
= (2 • min(Tlnp,ho,Hubht)) • (Hubtk - tn) • fr2
= (2 • min(51.2,200,42)) • (28 - 9.73) • 0.86)
= 1526.661 mm² Includes Bevel Area of 214.701 mm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures ta = 2.3274 mm
Wall Thickness per UG16(b), tr16b = 3.0000 mm
Wall Thickness, shell/head, internal pressure trb1 = 22.0000 mm
Wall Thickness tb1 = max(trb1, tr16b) = 22.0000 mm
Wall Thickness tb2 = max(trb2, tr16b) = 3.0000 mm
Wall Thickness per table UG-45 tb3 = 6.3000 mm

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1,tb2)]
= min[6.3, max(22, 3)]
= 6.3000 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(2.33, 6.3)
= 6.3000 mm

Available Nozzle Neck Thickness = 9.7345 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	37.6,	Allowable	:	118.0 MPa	Passed
Expansion	:	0.0,	Allowable	:	257.4 MPa	Passed
Occasional	:	5.3,	Allowable	:	156.9 MPa	Passed
Shear	:	23.2,	Allowable	:	82.6 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note:
This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification	-29 °C
Calculated Minimum Design Metal Temperature	-104 °C

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note:
This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification	-29 °C
Gov. MDMT of the nozzle to shell joint welded assembly :	-29 °C

Flange MDMT including Temperature reduction per UCS-66.1:

MDMT of ASME B16.5/47 flange per Matl. Specification	-29 °C
Flange MDMT with Temp reduction per UCS-66(i)(2)	-104 °C

Where the Stress Reduction Ratio per UCS-66(i)(2) is :
Design Pressure/Ambient Rating = 2.66/8.52 = 0.312

Weld Size Calculations, Description: V13"

Intermediate Calc. for nozzle/shell Welds	Tmin	19.0000 mm
---	------	------------

Results Per UW-16.1:

Required Thickness	Actual Thickness
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Nozzle Weld 6.0000 = Min per Code 7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
= max(0, (A-A1+2•tn•fr1•(E1•t-tr))•Sv)
= max(0, (1642 - 0 + 2 • 26.5 • 0.86 •
(1 • 20.5 - 20.5))138)
= 226611.31 N

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
= (A2+A4-(Wi-Can/.707)²•fr2)•Sv
= (2176 + 85.4 - 0 • 0.86) • 138
= 312029.34 N

Weld Load [W2]:
= (A2 + A3 + A4 + (2 • tn • t • fr1)) • Sv
= (2176 + 0 + 85.4 + (929)) • 138
= 440225.50 N

Weld Load [W3]:
= (A2+A3+A4+A5+(2•tn•t•fr1))•S
= (2176 + 0 + 85.4 + 0 + (929)) • 138
= 440225.50 N

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
= (π/2) • Dlo • Wo • 0.49 • Snw
= (3.14/2.0) • 125 • 10 • 0.49 • 118
= 113911. N

Shear, Nozzle Wall [Snw]:
= (π • (Dlr + Dlo)/4) • (Thk - Can) • 0.7 • Sn
= (3.14 • 49.5) • (28 - 1.5) • 0.7 • 118
= 340127. N

Tension, Shell Groove Weld [Tngw]:
= (π/2) • Dlo • (T - cas) • 0.74 • Sng
= (3.14/2.0) • 125 • (22 - 1.5) • 0.74 • 138
= 412424. N

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
= (SONW + SNW) = (113911 + 340127) = 454038 N

Failure Path [Path 2-2]:
= (Sonw + Tpgw + Tngw + Sinw)
= (113911 + 0 + 412424 + 0) = 526335 N

Failure Path [Path 3-3]:
= (Sonw + Tngw + Sinw)
= (113911 + 412424 + 0) = 526335 N

Summary of Failure Path Calculations:

Path 1-1 = 454037 N	, must exceed W = 226611 N	or W1 = 312029 N
Path 2-2 = 526335 N	, must exceed W = 226611 N	or W2 = 440225 N
Path 3-3 = 526335 N	, must exceed W = 226611 N	or W3 = 440225 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 2.66 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 1.8746 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 223.8746 mm

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 72.43 mm

Input Echo, WRC297 Item 1, Description: V13" :

Diameter Basis for Cylindrical Shell	ID		
Shell Internal Corrosion Allowance	1.5000	mm	
Shell External Corrosion Allowance	0.0000	mm	
Shell Diameter	2100.000	mm	
Shell Thickness	22.0000	mm	
Shell Stress Concentration Factor	1.000		
Vessel Material	SA-516 70		
Vessel Cold S.I. Allowable	Smc	138.00	MPa
Vessel Hot S.I. Allowable	Smh	138.00	MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter Basis for Nozzle	ID		
Nozzle Corrosion Allowance	1.5000	mm	
Nozzle Diameter	69.431	mm	
Nozzle Thickness	28.0000	mm	
Nozzle Stress Concentration Factor	1.000		
Nozzle Material	SA-350 LF1		
Nozzle Cold S.I. Allowable	SNmc	118.00	MPa
Nozzle Hot S.I. Allowable	SNmh	118.00	MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSained loads.*

Design Internal Pressure	Dp	2.66	MPa
Radial Load	P	3599.84	N
Circumferential Shear	Vc	4799.79	N
Longitudinal Shear	Vl	4799.79	N
Circumferential Moment	Mc	720253.69	N-mm
Longitudinal Moment	Ml	936329.88	N-mm
Torsional Moment	Mt	0.108E+07	N-mm

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dn/Tn (4.480) is < 10
 check the limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 125.431 + 2 \cdot 1.65 \cdot \sqrt{(1061.75(22.0 - 1.5))}$$

$$= 612.289 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2123.500 mm
Nozzle Outside Diameter	(d) =	125.431 mm
Vessel Thickness used	(T) =	20.500 mm
Nozzle Thickness used	(t) =	26.500 mm
T / t	=	0.774
d / t	=	10.000



$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.601$$

Nr/P	=	0.148
Mr/P	=	0.274
M0/P	=	0.088
N0/P	=	0.160
MrD/Mc	=	0.299
NrDL/Mc	=	0.116
M0d/Mc	=	0.108
N0DL/Mc	=	-0.059
MrD/M1	=	0.250
NrDL/M1	=	0.114
M0D/M1	=	0.090
N0DL/M1	=	0.047

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	-1	-1	-1	-1
Outplane Bending (P)	-4	4	-4	4
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	-9	9	9	-9
Normal Pressure Stress	135	137	135	137
Outplane Stress Summary	118	149	139	132

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	-1	-1	-1	-1
Inplane Bending (P)	-14	14	-14	14
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-2	-2	2	2
Inplane Bending (ML)	-26	26	26	-26
Inplane Pressure Stress	67	67	67	67
Inplane Stress Summary	23	104	80	55

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	2	2	2	2
Shear Stress Summary	3	3	0	0

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	118	149	139	132

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	-1	-1	-1	-1
Outplane Bending (P)	-4	4	-4	4



Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	-8	8	8	-8
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	67	67	67	67

Outplane Stress Summary	53	80	69	61
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Inplane Membrane (P)	-1	-1	-1	-1
Inplane Bending (P)	-14	14	-14	14
Inplane Membrane (Mc)	-1	-1	1	1
Inplane Bending (Mc)	-24	24	24	-24
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	135	137	135	137

Inplane Stress Summary	93	173	145	127
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	2	2	2	2

Shear Stress Summary	0	0	3	3
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities	Left	Left	Right	Right
	Outside	Inside	Outside	Inside

Two * Max Shear Stress	93	173	146	127
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses in the	Top	Top	Bottom	Bottom
hoop direction)	Outside	Inside	Outside	Inside

Hoop Membrane (P)	-1	-1	-1	-1
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	3	6	3	6

Hoop Stress Summary	0	4	2	5
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to	Top	Top	Bottom	Bottom
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	-6	6	-6	6
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-5	-5	5	5
Axial Bending (ML)	-12	12	12	-12
Axial Pressure Stress	1	1	1	1

Axial Stress Summary	-22	14	12	0
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	1	1	1	1
Shear Stress Summary	2	2	0	0

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	24	14	12	5

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	-1	-1	-1	-1
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	3	6	3	6
Hoop Stress Summary	2	5	1	4

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	-6	6	-6	6
Axial Membrane (Mc)	-4	-4	4	4
Axial Bending (Mc)	-11	11	11	-11
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	1	1	1	1
Axial Stress Summary	-20	14	11	0

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	1	1	1	1
Shear Stress Summary	0	0	2	2

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	23	15	11	6

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl



Circ. Pm (SUS)	135	137	135	137	135	137	135	137
Circ. Pl (SUS)	-1	-1	-1	-1	-2	-2	0	0
Circ. Q (SUS)	-13	13	5	-5	-38	38	10	-10
Long. Pm (SUS)	67	67	67	67	67	67	67	67
Long. Pl (SUS)	-3	-3	1	1	-1	-1	-1	-1
Long. Q (SUS)	-40	40	12	-12	-12	12	4	-4
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	135.0	137.0	135.0	137.0	135.0	137.0	135.0	137.0
Pm+Pl (SUS)	134.0	136.0	134.0	136.0	133.0	135.0	135.0	137.0
Pm+Pl+Q (Total)	121.1	149.2	139.0	131.0	95.0	173.0	145.1	127.1

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	137.00	138.00	Passed
Pm+Pl (SUS)	137.01	207.00	Passed
Pm+Pl+Q (TOTAL)	173.01	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	3	6	3	6	3	6	3	6
Circ. Pl (SUS)	-1	-1	-1	-1	-1	-1	-1	-1
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	1	1	1	1	1	1	1	1
Long. Pl (SUS)	-5	-5	5	5	-4	-4	4	4
Long. Q (SUS)	-18	18	6	-6	-17	17	5	-5
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	3.0	6.0	3.0	6.0	3.0	6.0	3.0	6.0
Pm+Pl (SUS)	6.0	9.0	6.0	6.0	5.0	8.0	5.0	5.0
Pm+Pl+Q (Total)	24.1	14.1	12.1	5.4	22.1	14.1	10.1	5.4

Nozzle Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	6.00	118.00	Passed
Pm+Pl (SUS)	9.00	177.00	Passed
Pm+Pl+Q (TOTAL)	24.08	354.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.



Input, Nozzle Desc: L12"

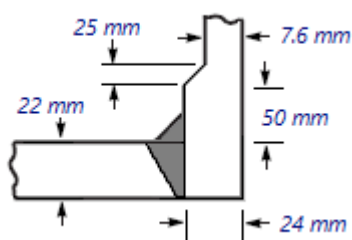
From: 40

Pressure for Reinforcement Calculations	P	2.6593	MPa
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	+N+IT
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	2100.00	mm
Shell Finished (Minimum) Thickness	t	22.0000	mm
Shell Internal Corrosion Allowance	cas	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		6550.00	mm
User Entered Minimum Design Metal Temperature		-40.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF1	+IT
Material UNS Number		K03009	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		30.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		160	
Flange Material		SA-350 LF1	+IT
Hub Height of Integral Nozzle	h	50.0000	mm
Height of Beveled Transition	L'	25.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		24.0000	mm
Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		22.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: L12"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 1.773 in.
 Actual Thickness Used in Calculation 0.301 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2.66 \cdot 1052 / (138 \cdot 1 - 0.6 \cdot 2.66)$
 $= 20.5000 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2.66 \cdot 24 / (118 \cdot 1 - 0.6 \cdot 2.66)$
 $= 0.5487 \text{ mm}$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 16.4 / 0.5773$
 $= 28.3270 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	101.3250	mm
Parallel to Vessel Wall	Rn+tn+t	50.6625	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	43.6905	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: L12".
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 2.0487	mm
Wall Thickness per UG16(b),	tr16b	= 3.0000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 22.0000	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 22.0000	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 3.0000	mm
Wall Thickness per table UG-45	tb3	= 4.9200	mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[4.92, \max(22, 3)]$
 $= 4.9200 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(2.05, 4.92)$
 $= 4.9200 \text{ mm}$

Available Nozzle Neck Thickness = 7.6454 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

*Note:
 This Material was specified as being an Impact Tested (Low Temperature) Material.*

Impact Test Temperature provided per Specification	-29 °C
Gov. MDMT of the nozzle to shell joint welded assembly :	-29 °C



Weld Size Calculations, Description: L12"

Intermediate Calc. for nozzle/shell welds T_{min} 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * W _o mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 1.0309 mm

The Cut Length for this Nozzle is, Drop + H_o + H + T : 223.0310 mm

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Input, Nozzle Desc: L22"

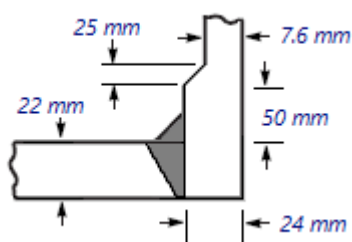
From: 40

Pressure for Reinforcement Calculations	P	2.6593	MPa
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	+N+IT
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	2100.00	mm
Shell Finished (Minimum) Thickness	t	22.0000	mm
Shell Internal Corrosion Allowance	cas	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		6550.00	mm
User Entered Minimum Design Metal Temperature		-40.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF1	+IT
Material UNS Number		K03009	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		330.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		160	
Flange Material		SA-350 LF1	+IT
Hub Height of Integral Nozzle	h	50.0000	mm
Height of Beveled Transition	L'	25.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		24.0000	mm
Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		22.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: L22"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 1.773 in.
Actual Thickness Used in Calculation 0.301 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $2.66 \cdot 1052 / (138 \cdot 1 - 0.6 \cdot 2.66)$
= 20.5000 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $2.66 \cdot 24 / (118 \cdot 1 - 0.6 \cdot 2.66)$
= 0.5487 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
= (Hub Thickness - Neck Thickness) / tan(30)
= 16.4/0.5773
= 28.3270 mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	101.3250	mm
Parallel to Vessel Wall	Rn+tn+t	50.6625	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	43.6905	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: L22".
This calculation is valid for nozzles that meet all the requirements of
paragraph UG-36. Please check the Code carefully, especially for nozzles
that are not isolated or do not meet Code spacing requirements. To force
the computation of areas for small nozzles go to Tools->Configuration
and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 2.0487	mm
Wall Thickness per UG16(b),	tr16b	= 3.0000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 22.0000	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 22.0000	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 3.0000	mm
Wall Thickness per table UG-45	tb3	= 4.9200	mm

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1,tb2)]
= min[4.92, max(22, 3)]
= 4.9200 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(2.05, 4.92)
= 4.9200 mm

Available Nozzle Neck Thickness = 7.6454 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -29 °C



Weld Size Calculations, Description: L22"

Intermediate Calc. for nozzle/shell welds T_{min} 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * W _o mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 1.0309 mm

The Cut Length for this Nozzle is, Drop + H_o + H + T : 223.0310 mm

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Input, Nozzle Desc: B26"

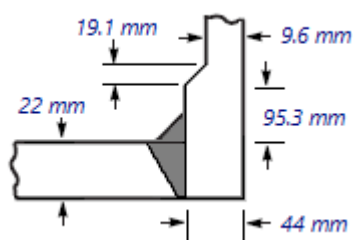
From: 50

Pressure for Reinforcement Calculations	P	2.6593	MPa
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	+N+IT
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	2100.00	mm
Shell Finished (Minimum) Thickness	t	22.0000	mm
Shell Internal Corrosion Allowance	cas	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		9250.00	mm
User Entered Minimum Design Metal Temperature		-40.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF1	+IT
Material UNS Number		K03009	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		6.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		80	
Flange Material		SA-350 LF1	+IT
Hub Height of Integral Nozzle	h	95.2682	mm
Height of Beveled Transition	L'	19.0683	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		44.0000	mm
Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		22.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		300	
Flange Grade		GR 1.4	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: B26"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 5.869 in.
 Actual Thickness Used in Calculation 0.378 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2.66 \cdot 1052 / (138 \cdot 1 - 0.6 \cdot 2.66)$
 $= 20.5000 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2.66 \cdot 76 / (118 \cdot 1 - 0.6 \cdot 2.66)$
 $= 1.7371 \text{ mm}$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 34.4 / 0.5773$
 $= 59.5805 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

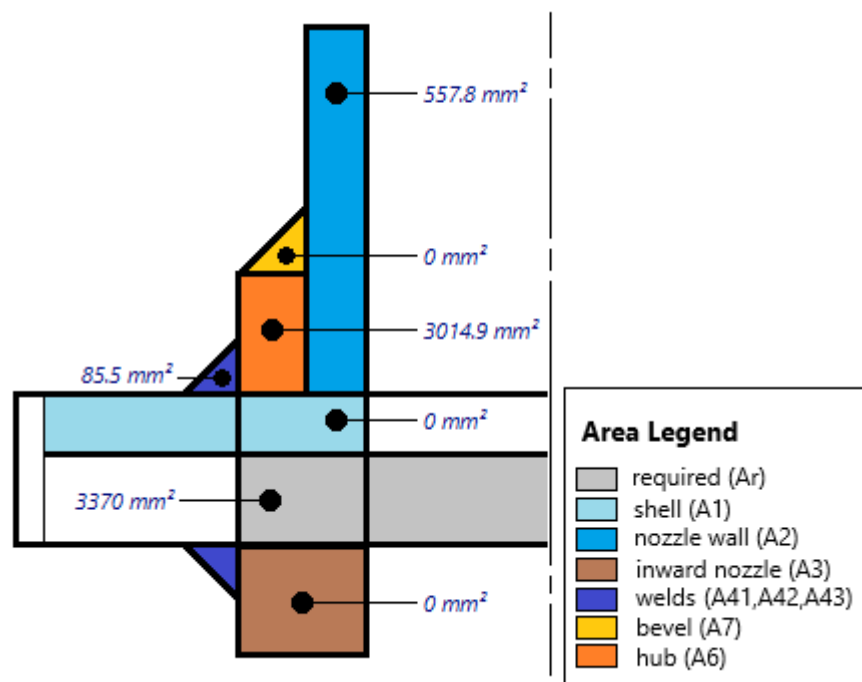
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	304.1452	mm
Parallel to Vessel Wall, opening length	d	152.0726	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	51.2500	mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118 / 138)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118 / 138)$
 $= 0.855$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.86, 1)$
 $= 0.855$



Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	3369.997	NA	NA
Area in Shell	A1	0.000	NA	NA
Area in Nozzle Wall	A2	557.794	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	85.509	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub	A6	3014.935	NA	NA
TOTAL AREA AVAILABLE	Atot	3658.237	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (152 \cdot 20.5 \cdot 1 + 2 \cdot 42.5 \cdot 20.5 \cdot 1 \cdot (1 - 0.86))$$

$$= 3369.997 \text{ mm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 152 (1 \cdot 20.5 - 1 \cdot 20.5) - 2 \cdot 42.5$$

$$(1 \cdot 20.5 - 1 \cdot 20.5) \cdot (1 - 0.86)$$

$$= 0.000 \text{ mm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp) (tn - trn) fr2$$

$$= (2 \cdot 51.2) (8.1 - 1.74) 0.86$$

$$= 557.794 \text{ mm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= Wo^2 \cdot fr2 + (Wi - can / 0.707)^2 \cdot fr2$$

$$= 10^2 \cdot 0.86 + (0)^2 \cdot 0.86$$

$$= 85.509 \text{ mm}^2$$

Area Available in the Hub Section [A6]:
= (2 • min(Tlnp,ho,Hubht)) • (Hubtk - tn) • fr2
= (2 • min(51.2,200,95.3)) • (44 - 9.6) • 0.86)
= 3014.935 mm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures ta = 3.2371 mm
Wall Thickness per UG16(b), tr16b = 3.0000 mm
Wall Thickness, shell/head, internal pressure trb1 = 22.0000 mm
Wall Thickness tb1 = max(trb1, tr16b) = 22.0000 mm
Wall Thickness tb2 = max(trb2, tr16b) = 3.0000 mm
Wall Thickness per table UG-45 tb3 = 7.7200 mm

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1,tb2)]
= min[7.72, max(22, 3)]
= 7.7200 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(3.24, 7.72)
= 7.7200 mm

Available Nozzle Neck Thickness = 9.6012 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	43.9,	Allowable	:	118.0 MPa	Passed
Expansion	:	0.0,	Allowable	:	251.1 MPa	Passed
Occasional	:	11.8,	Allowable	:	156.9 MPa	Passed
Shear	:	21.7,	Allowable	:	82.6 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note:
This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification	-29 °C
Calculated Minimum Design Metal Temperature	-104 °C

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note:
This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification	-29 °C
Gov. MDMT of the nozzle to shell joint welded assembly :	-29 °C

Flange MDMT including Temperature reduction per UCS-66.1:

MDMT of ASME B16.5/47 flange per Matl. Specification	-29 °C
Flange MDMT with Temp reduction per UCS-66(i)(2)	-50 °C

Where the Stress Reduction Ratio per UCS-66(i)(2) is :
Design Pressure/Ambient Rating = 2.66/4.26 = 0.624

Weld Size Calculations, Description: B26"

Intermediate Calc. for nozzle/shell Welds	Tmin	19.0000 mm
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Results Per UW-16.1:

Required Thickness	Actual Thickness
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Nozzle Weld 6.0000 = Min per Code 7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A-A1+2 \cdot tn \cdot fr1 \cdot (E1 \cdot t-tr)) \cdot Sv)$$

$$= \max(0, (3370 - 0 + 2 \cdot 42.5 \cdot 0.86 \cdot (1 \cdot 20.5 - 20.5)) \cdot 138)$$

$$= 465011.69 \text{ N}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv$$

$$= (3573 + 85.5 - 0 \cdot 0.86) \cdot 138$$

$$= 504784.72 \text{ N}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$$

$$= (3573 + 0 + 85.5 + (1490)) \cdot 138$$

$$= 710382.25 \text{ N}$$

Weld Load [W3]:

$$= (A2+A3+A4+A5+(2 \cdot tn \cdot t \cdot fr1)) \cdot S$$

$$= (3573 + 0 + 85.5 + 0 + (1490)) \cdot 138$$

$$= 710382.25 \text{ N}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$$

$$= (3.14/2.0) \cdot 237 \cdot 10 \cdot 0.49 \cdot 118$$

$$= 215299. \text{ N}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo)/4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.14 \cdot 97.3) \cdot (44 - 1.5) \cdot 0.7 \cdot 118$$

$$= 1072837. \text{ N}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$$

$$= (3.14/2.0) \cdot 237 \cdot (22 - 1.5) \cdot 0.74 \cdot 138$$

$$= 779509. \text{ N}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (215299 + 1072836) = 1288135 \text{ N}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (215299 + 0 + 779509 + 0) = 994808 \text{ N}$$

Failure Path [Path 3-3]:

$$= (Sonw + Tngw + Sinw)$$

$$= (215299 + 779509 + 0) = 994808 \text{ N}$$

Summary of Failure Path Calculations:

$$\text{Path 1-1} = 1288135 \text{ N}, \text{ must exceed } W = 465011 \text{ N} \quad \text{or } W1 = 504784 \text{ N}$$

$$\text{Path 2-2} = 994808 \text{ N}, \text{ must exceed } W = 465011 \text{ N} \quad \text{or } W2 = 710382 \text{ N}$$

$$\text{Path 3-3} = 994808 \text{ N}, \text{ must exceed } W = 465011 \text{ N} \quad \text{or } W3 = 710382 \text{ N}$$

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 2.66 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.



The Drop for this Nozzle is : 6.7123 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 228.7123 mm

Input Echo, WRC297 Item 1, Description: B26" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.5000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	2100.000	mm
Shell Thickness	22.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 138.00	MPa
Vessel Hot S.I. Allowable	Smh 138.00	MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	1.5000	mm
Nozzle Diameter	149.073	mm
Nozzle Thickness	44.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-350 LF1	
Nozzle Cold S.I. Allowable	SNmc 118.00	MPa
Nozzle Hot S.I. Allowable	SNmh 118.00	MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUstained loads.*

Design Internal Pressure	Dp	2.66	MPa
Radial Load	P	7199.67	N
Circumferential Shear	Vc	9599.58	N
Longitudinal Shear	Vl	9599.58	N
Circumferential Moment	Mc	0.288E+07	N-mm
Longitudinal Moment	Ml	0.375E+07	N-mm
Torsional Moment	Mt	0.432E+07	N-mm

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dn/Tn (5.388) is < 10
 check the Limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$
 $= 237.073 + 2 \cdot 1.65 \cdot \sqrt{(1061.75(22.0 - 1.5))}$
 $= 723.930 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2123.500	mm
Nozzle Outside Diameter	(d) =	237.073	mm
Vessel Thickness used	(T) =	20.500	mm
Nozzle Thickness used	(t) =	42.500	mm
T / t	=	0.482	
d / t	=	10.000	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.136$$

Nr/P	=	0.097
Mr/P	=	0.207



M0/P = 0.060
 N0/P = 0.126
 MrD/Mc = 0.305
 NrDL/Mc = 0.092
 M0d/Mc = 0.095
 N0DL/Mc = 0.046
 MrD/Ml = 0.179
 NrDL/Ml = 0.085
 M0D/Ml = 0.058
 N0DL/Ml = 0.125

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	-2	-2	-2	-2
Outplane Bending (P)	-6	6	-6	6
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-4	-4	4	4
Outplane Bending (ML)	-13	13	13	-13
Normal Pressure Stress	135	137	135	137
Outplane Stress Summary	108	150	144	133

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	-1	-1	-1	-1
Inplane Bending (P)	-21	21	-21	21
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-3	-3	3	3
Inplane Bending (ML)	-40	40	40	-40
Inplane Pressure Stress	67	67	67	67
Inplane Stress Summary	1	124	88	49

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	2	2	2	2
Shear Stress Summary	3	3	1	1

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	109	150	144	133

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	-2	-2	-2	-2
Outplane Bending (P)	-6	6	-6	6
Outplane Membrane (Mc)	-1	-1	1	1
Outplane Bending (Mc)	-16	16	16	-16
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0



Normal Pressure Stress	67	67	67	67
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Outplane Stress Summary	41	86	76	56
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Inplane Membrane (P)	-1	-1	-1	-1
Inplane Bending (P)	-21	21	-21	21
Inplane Membrane (Mc)	-2	-2	2	2
Inplane Bending (Mc)	-52	52	52	-52
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	135	137	135	137

Inplane Stress Summary	56	207	167	107
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	2	2	2	2

Shear Stress Summary	1	1	3	3
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Two * Max Shear Stress	56	207	167	107
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
---	----------------------	---------------------	-------------------------	------------------------

Hoop Membrane (P)	-2	-2	-2	-2
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-4	-4	4	4
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	4	7	4	7

Hoop Stress Summary	-2	0	7	10
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Axial Membrane (P)	0	0	0	0
Axial Bending (P)	-3	3	-3	3
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-3	-3	3	3
Axial Bending (ML)	-7	7	7	-7
Axial Pressure Stress	2	2	2	2

Axial Stress Summary	-12	9	8	1
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
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Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	1	1	1	1

Shear Stress Summary	1	1	0	0
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress	12	9	8	10
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses in the	Left	Left	Right	Right
hoop direction)	Outside	Inside	Outside	Inside

Hoop Membrane (P)	-2	-2	-2	-2
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-1	-1	1	1
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	4	7	4	7

Hoop Stress Summary	0	4	3	6
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	-3	3	-3	3
Axial Membrane (Mc)	-2	-2	2	2
Axial Bending (Mc)	-10	10	10	-10
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	2	2	2	2

Axial Stress Summary	-14	13	11	-2
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	1	1	1	1

Shear Stress Summary	0	0	1	1
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress	15	13	11	9
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WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	135	137	135	137	135	137	135	137
Circ. Pl (SUS)	-6	-6	2	2	-3	-3	1	1
Circ. Q (SUS)	-19	19	7	-7	-73	73	31	-31

Long. Pm (SUS)	67	67	67	67	67	67	67	67
Long. Pl (SUS)	-4	-4	2	2	-3	-3	-1	-1
Long. Q (SUS)	-61	61	19	-19	-22	22	10	-10
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	135.0	137.0	135.0	137.0	135.0	137.0	135.0	137.0
Pm+Pl (SUS)	129.0	131.0	137.0	139.0	132.0	134.0	136.0	138.0
Pm+Pl+Q (Total)	110.1	150.3	144.0	132.0	59.1	207.0	167.1	107.2

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	137.00	138.00	Passed
Pm+Pl (SUS)	139.01	207.00	Passed
Pm+Pl+Q (TOTAL)	207.01	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	4	7	4	7	4	7	4	7
Circ. Pl (SUS)	-6	-6	2	2	-3	-3	-1	-1
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	2	2	2	2	2	2	2	2
Long. Pl (SUS)	-3	-3	3	3	-2	-2	2	2
Long. Q (SUS)	-10	10	4	-4	-13	13	7	-7
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	4.0	7.0	4.0	7.0	4.0	7.0	4.0	7.0
Pm+Pl (SUS)	2.0	2.0	6.0	9.0	1.0	4.0	4.0	6.0
Pm+Pl+Q (Total)	11.1	9.1	9.3	9.1	14.1	13.1	11.1	9.2

Nozzle Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	7.00	118.00	Passed
Pm+Pl (SUS)	9.00	177.00	Passed
Pm+Pl+Q (TOTAL)	14.14	354.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.



Input, Nozzle Desc: B12"

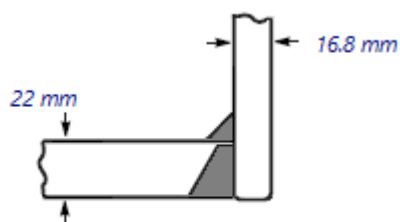
From: 50

Pressure for Reinforcement Calculations	P	2.6593	MPa
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	+N+IT
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	2100.00	mm
Shell Finished (Minimum) Thickness	t	22.0000	mm
Shell Internal Corrosion Allowance	cas	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		9050.00	mm
User Entered Minimum Design Metal Temperature		-40.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF1	+IT
Material UNS Number		K03009	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm
Flange Material		SA-350 LF1	+IT
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	196.8500	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	22.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		600	
Flange Grade		GR 1.4	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: B12"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	0.660	in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2.66 \cdot 1052 / (138 \cdot 1 - 0.6 \cdot 2.66)$
 $= 20.5000$ mm

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 26.9 \cdot (\exp([2.66 / (118 \cdot 1)] - 1))$
 $= 0.6131$ mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	125.3280	mm
Parallel to Vessel Wall	Rn+tn+tt	62.6640	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	38.1600	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: B12".
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 2.1131	mm
Wall Thickness per UG16(b),	tr16b	= 3.0000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 22.0000	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 22.0000	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 3.0000	mm
Wall Thickness per table UG-45	tb3	= 6.3000	mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[6.3, \max(22, 3)]$
 $= 6.3000$ mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(2.11, 6.3)$
 $= 6.3000$ mm

Available Nozzle Neck Thickness = 16.7640 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	13.2,	Allowable	:	118.0 MPa	Passed
Expansion	:	0.0,	Allowable	:	281.8 MPa	Passed
Occasional	:	1.8,	Allowable	:	156.9 MPa	Passed
Shear	:	10.4,	Allowable	:	82.6 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

*Note:
 This Material was specified as being an Impact Tested (Low Temperature) Material.*

Impact Test Temperature provided per Specification	-29 °C
Calculated Minimum Design Metal Temperature	-104 °C
Gov. MDMT of the nozzle to shell joint welded assembly :	-104 °C

Weld Size Calculations, Description: B12"

Intermediate Calc. for nozzle/shell Welds Tmin 15.2640 mm

Results Per UW-16.1:

Required Thickness Actual Thickness
 Nozzle Weld 6.0000 = Min per Code 7.0700 = $0.7 * W_o$ mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.8469 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 219.6970 mm

Input Echo, WRC297 Item 1, Description: B12" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.5000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	2100.000	mm
Shell Thickness	22.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 138.00	MPa
Vessel Hot S.I. Allowable	Smh 138.00	MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	1.5000	mm
Nozzle Diameter	50.800	mm
Nozzle Thickness	16.7640	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-350 LF1	
Nozzle Cold S.I. Allowable	SNmc 118.00	MPa
Nozzle Hot S.I. Allowable	SNmh 118.00	MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSstained loads.

Design Internal Pressure	Dp	2.66	MPa
Radial Load	P	2399.89	N
Circumferential Shear	Vc	3199.86	N
Longitudinal Shear	Vl	3199.86	N
Circumferential Moment	Mc	320112.56	N-mm
Longitudinal Moment	Ml	416146.78	N-mm
Torsional Moment	Mt	480169.53	N-mm

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, D_n/T_v , (3.833) is < 5
 check the Limitations of WRC bulletin 297.

Warning - The ratio, D_n/T_n (5.030) is < 10
 check the Limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$
 $= 84.328 + 2 \cdot 1.65 \cdot \sqrt{1061.75(22.0 - 1.5)}$
 $= 571.186 \text{ mm}$

WRC 297 Curve Access Parameters:



Vessel Mean Diameter (D) = 2123.500 mm
 Nozzle Outside Diameter (d) = 84.328 mm
 Vessel Thickness used (T) = 20.500 mm
 Nozzle Thickness used (t) = 15.264 mm
 T / t = 1.343
 d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.404$$

Nr/P = 0.202
 Mr/P = 0.277
 M0/P = 0.116
 N0/P = 0.179
 MrD/Mc = 0.270
 NrDL/Mc = 0.162
 M0d/Mc = 0.139
 N0DL/Mc = -0.111
 MrD/M1 = 0.254
 NrDL/M1 = 0.159
 M0D/M1 = 0.128
 N0DL/M1 = -0.038

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	-1	-1	-1	-1
Outplane Bending (P)	-3	3	-3	3
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	-9	9	9	-9
Normal Pressure Stress	135	137	135	137
Outplane Stress Summary	121	150	138	131

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	-1	-1	-1	-1
Inplane Bending (P)	-9	9	-9	9
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-1	-1	1	1
Inplane Bending (ML)	-17	17	17	-17
Inplane Pressure Stress	67	67	67	67
Inplane Stress Summary	37	91	76	59

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	2	2	2	2
Shear Stress Summary	3	3	0	0

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	121	150	138	131



Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	-1	-1	-1	-1
Outplane Bending (P)	-3	3	-3	3
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	-7	7	7	-7
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	67	67	67	67
Outplane Stress Summary	56	79	69	61

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	-1	-1	-1	-1
Inplane Bending (P)	-9	9	-9	9
Inplane Membrane (Mc)	-1	-1	1	1
Inplane Bending (Mc)	-14	14	14	-14
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	135	137	135	137
Inplane Stress Summary	108	159	140	132

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	2	2	2	2
Shear Stress Summary	0	0	3	3

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	108	159	140	133

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	-1	-1	-1	-1
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	4	7	4	7
Hoop Stress Summary	3	6	2	6

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	-10	10	-10	10



Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-8	-8	8	8
Axial Bending (ML)	-22	22	22	-22
Axial Pressure Stress	2	2	2	2

Axial Stress Summary	-40	26	21	-1
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	1	1	-1	-1
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	2	2	2	2
Shear Stress Summary	4	4	1	1

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	44	26	21	7

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	-1	-1	-1	-1
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	4	7	4	7
Hoop Stress Summary	4	7	2	5

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	-10	10	-10	10
Axial Membrane (Mc)	-6	-6	6	6
Axial Bending (Mc)	-18	18	18	-18
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	2	2	2	2
Axial Stress Summary	-34	24	15	0

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-1	-1	1	1
Shear due to Torsion	2	2	2	2
Shear Stress Summary	1	1	4	4

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	38	24	16	10



WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	135	137	135	137	135	137	135	137
Circ. Pl (SUS)	-1	-1	-1	-1	-2	-2	0	0
Circ. Q (SUS)	-12	12	6	-6	-23	23	5	-5
Long. Pm (SUS)	67	67	67	67	67	67	67	67
Long. Pl (SUS)	-2	-2	0	0	-1	-1	-1	-1
Long. Q (SUS)	-26	26	8	-8	-10	10	4	-4
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	135.0	137.0	135.0	137.0	135.0	137.0	135.0	137.0
Pm+Pl (SUS)	134.0	136.0	134.0	136.0	133.0	135.0	135.0	137.0
Pm+Pl+Q (Total)	122.1	148.2	140.0	130.0	110.0	158.0	140.1	132.1

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	137.00	138.00	Passed
Pm+Pl (SUS)	137.01	207.00	Passed
Pm+Pl+Q (TOTAL)	158.01	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	4	7	4	7	4	7	4	7
Circ. Pl (SUS)	-1	-1	-1	-1	-1	-1	-1	-1
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	2	2	2	2	2	2	2	2
Long. Pl (SUS)	-8	-8	8	8	-6	-6	6	6
Long. Q (SUS)	-32	32	12	-12	-28	28	8	-8
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	4.0	7.0	4.0	7.0	4.0	7.0	4.0	7.0
Pm+Pl (SUS)	9.2	12.2	10.1	10.2	7.3	10.2	8.2	8.4
Pm+Pl+Q (Total)	41.4	26.4	22.1	8.2	35.1	24.1	16.7	8.5

Nozzle Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	7.00	118.00	Passed

PV Elite 28 U2 Licensee: SL User

FileName : A150NRL14CM001_Let down vessel_7%

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Nozzle Calcs.: B1_2"

Nozl: 48 1:55pm Sep 18,2026



HEXAGON

Pm+Pl (SUS)	12.17	177.00	Passed
Pm+Pl+Q (TOTAL)	41.44	354.00	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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PV Elite[®] Pressure Vessel Design Notes

Project Information	
Project Number	
Client	
Date	
Designer	
Design Code	
Additional Standards	

Design Conditions		
Internal Design Pressure:		
External Design Pressure:		
Design Int. Temperature:		
Design Ext. Temperature:		
Corrosion Allowance:		

Vessel Geometry		
Vessel Type		
Diameter		
Tangent-to-Tangent Length		
Head Type		
Number of Segments		

Material Specifications	
Shell:	
Head:	
Nozzle:	

Nozzle/Opening Details	ID/OD	Size	Rating	Matl.	Reinf.

Miscellaneous		
Support Type		
Number of Supports		
Support Material		
Seismic Code		
Wind Code		
Insulation		
Pressure Test		
Radiography		
Fatigue Analysis		

Analysis Notes: